

Data Mining

Complete Numerical Problems & Solutions

CT 72502 (Elective I)

BEI / BEX · Year IV Part I

Institute of Engineering, Tribhuvan University

Every numerical the course sets, in one volume: **24 distinct methods** and **47 PYQ instances** drawn from **all 17 past papers (2070–2081 BS)**, plus a few extra practice problems where a likely pattern has no paper behind it yet. Each method is explained once, then every paper that used it is solved under the **question exactly as printed**, so a problem can be attempted from this PDF alone. **Every answer was independently recomputed, not copied from the source notes.**

Theory lives in the companion volume, **Data Mining — Complete Exam Notes**.

Ch	Topic	Hours	Papers	Marks	Share	Numerical	Theory	Num. %
1	Introduction	2	22/23	159	8.2%	0	159	0%
2	Data Preprocessing	6	23/23	295	15.2%	49	246	17%
3	Classification	12	23/23	420	21.7%	167	253	40%
4	Association Analysis	10	23/23	397	20.5%	157	240	40%
5	Cluster Analysis	9	23/23	366	18.9%	135	231	37%
6	Anomaly / Fraud Detection	3	23/23	158	8.2%	3	155	2%
7	Advanced Applications	3	22/23	141	7.3%	8	133	6%
Total		45		1936	100%	519	1417	27%

Share is of every mark the Detailed PYQ records. *Numerical* = a calculation, construction or trace of the kind the numerical companion solves; a question's whole marks go to one side. Bold numerical: at least half the chapter.

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Ch 2 – Data Preprocessing – Numerical Problems & Solutions

49 numerical marks, 2.5% of the archive, 17% of this chapter · 7 distinct numerical methods · 8 PYQ instances solved · 2 extra practice problems. Every answer independently recomputed, not copied from the source notes.

Methods in this chapter

- **M1** Normalisation: min-max, z-score, decimal scaling 78 Bh
- **M2** PCA: principal components + proportion of variance 76 Ch
- **M3** Distance measures + distance matrix (Euclidean, Manhattan, Supremum) 80 Bh, 76 Ch
- **M4** Cosine similarity 80 Bh, 80 Ba, 76 Ch
- **M5** Binary similarity: Jaccard, SMC, symmetric dissimilarity 80 Ba, 76 Ch
- **M6** Schema design + OLAP operation sequence 78 Bh, 76 Ch
- **M7** Central tendency, quartiles, IQR and outliers **(NEW)**, no PYQ

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-2 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
map the range	Min-max: slide min to new_min and stretch max to new_max; order and gaps kept. 1.1 1; 1.2 2; Practice B 1	dot product	Multiply component by component and add. 4.1 1; 4.2 1; 4.3 1
centre on mean	Z-score: subtract the mean, divide by σ ; the result counts standard deviations. 1.1 2; 1.2 2; 7.4 1	length	$\ d\ = \sqrt{\sum d_k^2}$ for each vector. 4.1 2; 4.2 2; 4.3 2
pick j from max	Decimal scaling: the smallest j with $\max v /10^j$ STRICTLY below 1. 1.1 3 ; 1.2 2	divide	$\cos = \text{dot} / (\text{product of lengths})$: the angle, magnitude cancelled. 4.1 3 ; 4.2 3 ; 4.3 3
equal-depth bins	Sort, then cut into bins holding the same COUNT (depth), not the same range. 1.2 1 ; Practice A 1	tally the pairs	Count $M_{11}, M_{10}, M_{01}, M_{00}$ position by position. 5.1 1 ; 5.2 2,3,4
bin mean	Replace every value by its bin's mean. 1.2 1 ; Practice A 1	split attributes	Decide which attributes are asymmetric (a shared 0 means nothing) and which symmetric. 5.2 1
bin boundary	Replace every value by the nearer of its bin's min and max. Practice A 2	Jaccard	$M_{11}/(M_{11} + M_{10} + M_{01})$: 0-0 matches ignored. 5.1 2; 5.2 2
equal-width bins	Cut the range into equal intervals: $W = (\max - \min)/N$; counts differ. Practice A 3	SMC	$(M_{11} + M_{00})/\text{all}$: 0-0 matches count. 5.1 3; 5.2 4
block split	A row and column of zeros decouple: solve each block alone. 2.1 1	symmetric d	$(M_{10} + M_{01})/\text{all} = 1 - \text{SMC}$. 5.2 3
characteristic eq.	$ \Sigma - \lambda I = 0$ gives the eigenvalues. 2.1 2	roll-up	Climb a dimension's hierarchy to a coarser level. 6.1 1,2,3; 6.2 1,2
trace check	The eigenvalues must add up to the trace (sum of the diagonal). 2.1 3	roll-up to all	A dimension the question never mentions is summed away. 6.1 4; 6.2 3
eigenvector	Solve $(\Sigma - \lambda I)v = 0$ from one row, then scale to unit length. 2.1 4	dice	Restrict two or more dimensions at once. 6.1 5
share of variance	$\lambda_i / \sum \lambda$: how much of the spread each component keeps. 2.1 5	slice	Restrict exactly one dimension. 6.2 4
differences	Subtract attribute by attribute first; every distance is built from these. 3.1 1; 3.2 1	cumulative freq.	Run the frequencies up to find the interval holding the $N/2$ -th value. 7.1 1 ; 7.2 1
Euclidean	$\sqrt{\sum d_k^2}$: the straight line ($r = 2$). Warm-up 1; 3.2 2	interpolate	Assume values spread evenly inside the interval and walk in proportionally. 7.1 2; 7.2 2
Manhattan	$\sum d_k $: along the grid ($r = 1$). Warm-up 2; 3.1 2	empirical rule	$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$. 7.3 1
supremum	$\max d_k $: the single largest difference ($r \rightarrow \infty$). Warm-up 3 ; 3.1 3	quartile position	Q_k sits at position $k(n+1)/4$; interpolate between neighbours. 7.4 2
fill by symmetry	A distance matrix has a zero diagonal and is symmetric: compute the lower triangle only. 3.1 2,3	1.5 IQR fences	Outside $Q_1 - 1.5 IQR$ or $Q_3 + 1.5 IQR$ is a suspected outlier. 7.4 3

M1. Normalisation

Method

- Min-max maps v into $[new_min, new_max]$, preserves relationships exactly:

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Z-score uses mean $\bar{A}$ and std deviation σ_A . Use when min/max unknown, or outliers would squash a min-max result:

$$v' = \frac{v - \bar{A}}{\sigma_A} \quad \sigma_A = \sqrt{\frac{1}{n} \sum (v_i - \bar{A})^2}$$

- Decimal scaling moves the decimal point:

$$v' = \frac{v}{10^j}, \quad j = \text{smallest int such that } \max(|v'|) < 1$$

Two traps. (1) Z-score uses the **mean**, not the min. (2) For decimal scaling, j must make the max *strictly* less than 1, so $\max |v| = 1000$ needs $j = 4$, not $j = 3$.

Problem 1.1 2+2+2

QUESTION AS PRINTED 2078 Bhadra · Q3 · [2+2+2]

Use the following methods to normalize the data: 200, 300, 400, 600 and 1000.

- Min-max normalization by setting min = 0 and max = 1
- Z-score normalization
- Normalization by decimal scaling

Given

$v = \{200, 300, 400, 600, 1000\}$, $n = 5$, $min_A = 200$, $max_A = 1000$.

(a) Min-max, $new_min = 0$, $new_max = 1$

$$v' = \frac{v - 200}{1000 - 200} = \frac{v - 200}{800}$$

v	v'
200	$0/800 = 0$
300	$100/800 = 0.125$
400	$200/800 = 0.25$
600	$400/800 = 0.5$
1000	$800/800 = 1$

(c) Decimal scaling

- $\max |v| = 1000$.
- $j = 3$: $1000/10^3 = 1$, not < 1 . **Rejected.**
- $j = 4$: $1000/10^4 = 0.1 < 1$. **Use $j = 4$.**

v	$v' = v/10000$
200	0.02
300	0.03
400	0.04
600	0.06
1000	0.1

Final answer

- Min-max: 0, 0.125, 0.25, 0.5, 1
- Z-score: -1.0607, -0.7071, -0.3536, +0.3536, +1.7678
- Decimal scaling: 0.02, 0.03, 0.04, 0.06, 0.1

(b) Z-score

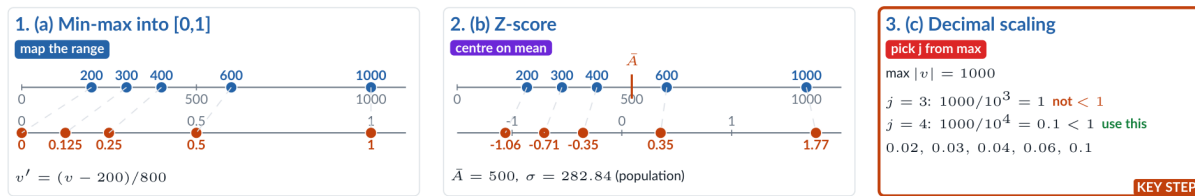
- Mean $\bar{A} = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$
- Deviations: -300, -200, -100, +100, +500
- $\sum (v - \bar{A})^2 = 90000 + 40000 + 10000 + 10000 + 250000 = 400000$
- $\sigma_A = \sqrt{400000/5} = \sqrt{80000} = 282.84$

v	$v' = (v - 500)/282.84$
200	$-300/282.84 = -1.0607$
300	$-200/282.84 = -0.7071$
400	$-100/282.84 = -0.3536$
600	$+100/282.84 = +0.3536$
1000	$+500/282.84 = +1.7678$

Note on σ verified/added

- Above uses the **population** std deviation, $\sqrt{\sum (v - \bar{A})^2 / n}$. This is the form Han & Kamber use, so it is the safe default.
- If the examiner wants the **sample** std deviation, divide by $n - 1$: $s = \sqrt{400000/4} = 316.23$, giving -0.9487, -0.6325, -0.3162, +0.3162, +1.5811.
- State which one you used.** Either is accepted if labelled.

Step by step: three ways to normalise the same five values



Problem 1.2 – all four transformations on one data set 6

QUESTION AS PRINTED 2079 Baishakh · Q2 · [6]

Suppose that the data for analysis include the attribute the frequency of stop words in documents. The values are given in increasing order: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- Use smoothing by bin means with a depth of 3.
- Use min-max normalization to transform the value 35 into the range from 0.0 to 1.0.
- Use z-score normalization to transform the value 35 where the standard deviation of the above frequency is 12.94.
- Use normalization by decimal scaling to transform the value 35.

$n = 27$, already sorted. **“Depth 3” means 3 values per bin**, not 3 bins — so $27/3 = 9$ bins. This is the single most common error on this question. Parts (b)–(d) each transform only the value 35, so no full table is needed there.

(a) Smoothing by bin means, depth 3

partition in order, replace every value by its bin mean

Bin	Values	Mean	Smoothed to
1	13, 15, 16	$44/3 = 14.67$	14.67, 14.67, 14.67
2	16, 19, 20	$55/3 = 18.33$	18.33, 18.33, 18.33
3	20, 21, 22	$63/3 = 21.00$	21, 21, 21
4	22, 25, 25	$72/3 = 24.00$	24, 24, 24
5	25, 25, 30	$80/3 = 26.67$	26.67, 26.67, 26.67
6	33, 33, 35	$101/3 = 33.67$	33.67, 33.67, 33.67
7	35, 35, 35	$105/3 = 35.00$	35, 35, 35
8	36, 40, 45	$121/3 = 40.33$	40.33, 40.33, 40.33
9	46, 52, 70	$168/3 = 56.00$	56, 56, 56

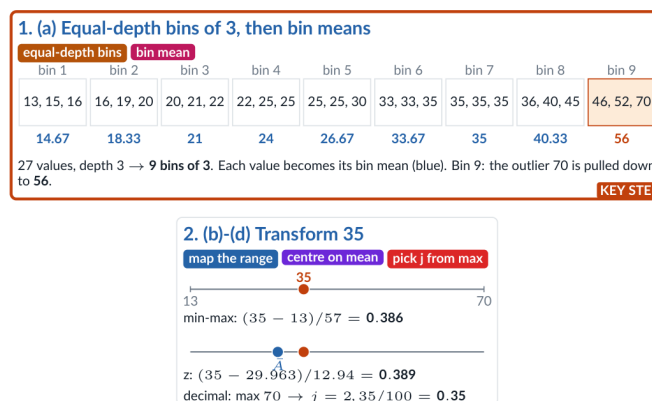
- Bin 9 shows why smoothing works:** the outlier 70 is pulled down to 56, and 46 pushed up. Noise is absorbed into the local average.
- This is **equal-depth** (equal-frequency) binning: same *count* per bin, not the same value range. Compare Practice A, which uses depth 4.

(b), (c), (d): transforming the single value 35

Method	Formula	Substitution	Result
Min-max into [0, 1]	$v' = \frac{v - \min}{\max - \min} (new_{\max} - new_{\min}) + new_{\min}$	$\frac{35 - 13}{70 - 13} = \frac{22}{57}$	0.386
Z-score	$v' = \frac{v - \bar{A}}{\sigma_A}, \bar{A} = \frac{809}{27} = 29.963$	$\frac{35 - 29.963}{12.94} = \frac{5.037}{12.94}$	0.389
Decimal scaling	$v' = v/10^j$, smallest j with $\max(v') < 1$	$\max = 70 \rightarrow j = 2, 35/100$	0.35

- σ is given as **12.94**, so do not recompute it — but you *must* compute the mean yourself. Sum the 27 values: 809.
- Decimal scaling:** $j = 2$, not 1. $j = 1$ would give $70/10 = 7 \not< 1$. Check the *maximum*, not the value being transformed.
- All three land near 0.35–0.39 here, which is coincidence: 35 sits just above the mean and just under the middle of the range.

Step by step: bins of three, then the value 35 three ways



M2. PCA: principal components and proportion of variance

Method

1. Start from the **covariance matrix** Σ (if given raw data: subtract the mean first, then compute Σ).
2. Solve the characteristic equation $|\Sigma - \lambda I| = 0$ for the eigenvalues λ .
3. Sort λ in **descending** order. λ_i = variance carried by component i .
4. For each λ_i solve $(\Sigma - \lambda_i I) \mathbf{v} = 0$ for the eigenvector, then normalise it to unit length.
5. Proportion of total variance explained by component i :

$$\frac{\lambda_i}{\sum_j \lambda_j}, \quad \sum_i \lambda_i = \text{trace}(\Sigma)$$

- **Check:** the eigenvalues must sum to the trace. Free and catches arithmetic slips.

Problem 2.1 4

QUESTION AS PRINTED 2076 Chaitra · Q1 · [4]

Find the principal components and the proportion of the total variance explained by each when the covariance matrix of the three random variables X_1 , X_2 and X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Step 1: characteristic equation

- The third row/column is decoupled, so Σ splits into a 2×2 block and a 1×1 block.
- From the 1×1 block: $(2 - \lambda) = 0 \Rightarrow \boxed{\lambda = 2}$
- From the 2×2 block:

$$(1 - \lambda)(5 - \lambda) - (-2)(-2) = 0$$

$$5 - 6\lambda + \lambda^2 - 4 = 0 \Rightarrow \lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 4}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

Step 2: eigenvalues, sorted

- $\lambda_1 = 3 + 2\sqrt{2} = \mathbf{5.8284}$
- $\lambda_2 = \mathbf{2}$
- $\lambda_3 = 3 - 2\sqrt{2} = \mathbf{0.1716}$
- **Check:** $5.8284 + 2 + 0.1716 = 8 = \text{trace}(\Sigma) = 1 + 5 + 2$. **Correct.**

Step 3: eigenvectors

- For $\lambda_1 = 5.8284$: row 1 gives $(1 - 5.8284)v_1 - 2v_2 = 0 \Rightarrow v_2 = -2.4142 v_1$; row 3 gives $v_3 = 0$.
 $\mathbf{e}_1 \propto (1, -2.4142, 0)$, length $\sqrt{1 + 5.8284} = 2.6131$
 $\mathbf{e}_1 = (0.3827, -0.9239, 0)$
- For $\lambda_2 = 2$: rows 1 and 2 force $v_1 = v_2 = 0$, v_3 free.
 $\mathbf{e}_2 = (0, 0, 1)$
- For $\lambda_3 = 0.1716$: row 1 gives $0.8284 v_1 = 2v_2 \Rightarrow v_2 = 0.4142 v_1$; $v_3 = 0$.
 $\mathbf{e}_3 \propto (1, 0.4142, 0)$, length 1.0824
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
- Check:** $\mathbf{e}_1 \cdot \mathbf{e}_3 = (0.3827)(0.9239) + (-0.9239)(0.3827) = 0$
 $\rightarrow$ orthogonal, as required for a symmetric matrix.

Final answer

- PC1: $\lambda_1 = 5.8284$, $\mathbf{e}_1 = (0.3827, -0.9239, 0)$, **72.86%**
- PC2: $\lambda_2 = 2$, $\mathbf{e}_2 = (0, 0, 1)$, **25.00%**
- PC3: $\lambda_3 = 0.1716$, $\mathbf{e}_3 = (0.9239, 0.3827, 0)$, **2.14%**

Step by step: eigenvalues, the check, the vectors, the variance kept

1. Spot the blocks

block split

1	-2	0
-2	5	0
0	0	2

2×2 block and a 1×1 block: solve each alone.

1×1 : $2 - \lambda = 0 \Rightarrow \lambda = 2$

2. Eigenvalues of the 2×2

characteristic eq.

$$(1 - \lambda)(5 - \lambda) - 4 = 0$$

$$\lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = 3 \pm 2\sqrt{2}$$

$\lambda_1 = 5.8284$, $\lambda_2 = 2$, $\lambda_3 = 0.1716$

KEY STEP

3. Trace check

trace check

5.83	2
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$5.8284 + 2 + 0.1716 = 8$
 trace = $1 + 5 + 2 = 8$ agrees

4. Eigenvectors

eigenvector

$\mathbf{e}_1 = (0.3827, -0.9239, 0)$
 $\mathbf{e}_3 = (0.9239, 0.3827, 0)$
 $\mathbf{e}_2 = (0, 0, 1)$, out of the page
 $\mathbf{e}_1 \cdot \mathbf{e}_3 = 0$: at right angles

5. Share of variance

share of variance

72.86%	25.00%
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PC1 + PC2 = **97.86%**; PC3 only 2.14%.
 Keep 2 components: 3 dims $\rightarrow$ 2.

M3. Distance measures and the distance matrix**Method**

- All three are Minkowski with a different r :

$$\text{dist}(p, q) = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{1/r}$$

- $r = 1$ **Manhattan**: $\sum |p_k - q_k|$
- $r = 2$ **Euclidean**: $\sqrt{\sum (p_k - q_k)^2}$
- $r \rightarrow \infty$ **Supremum** (L_∞ , Chebyshev): $\max_k |p_k - q_k|$
- A **distance matrix** is symmetric with a zero diagonal, so only the lower triangle needs computing.

Warm-up (Han & Kamber Example 2.19–2.20)

PROBLEM not a PYQ · Han & Kamber Example 2.19–2.20 verified/added

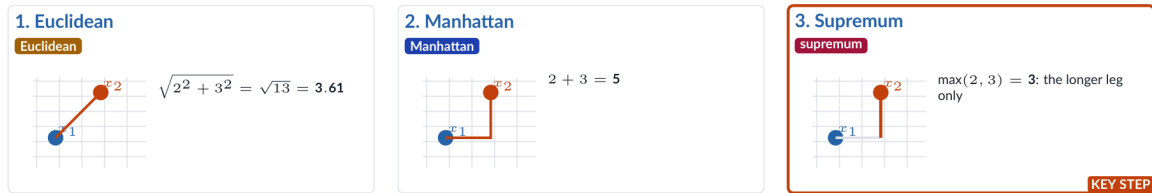
Given the two objects $x_1 = (1, 2)$ and $x_2 = (3, 5)$, compute the Euclidean, Manhattan and Supremum distances between them.

Smallest case that shows all three at once. Worth memorising as a check.

- $x_1 = (1, 2)$, $x_2 = (3, 5)$. Differences: 2 and 3.
- Euclidean = $\sqrt{2^2 + 3^2} = \sqrt{13} = \mathbf{3.61}$

- Manhattan = $2 + 3 = 5$
- Supremum = $\max(2, 3) = 3$
- **Ordering**: supremum $\leq$ Euclidean $\leq$ Manhattan. Always true. Use it to catch slips.

Step by step: the same two points, three distances



Problem 3.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (a) · [2]

Given the following points compute the distance matrix using the Manhattan and the Supremum distance.

Points	X	Y
P1	6	3
P2	2	2
P3	3	4

Manhattan (L_1)

- $d(P_1, P_2) = |6 - 2| + |3 - 2| = 4 + 1 = 5$
- $d(P_1, P_3) = |6 - 3| + |3 - 4| = 3 + 1 = 4$
- $d(P_2, P_3) = |2 - 3| + |2 - 4| = 1 + 2 = 3$

Supremum (L_∞)

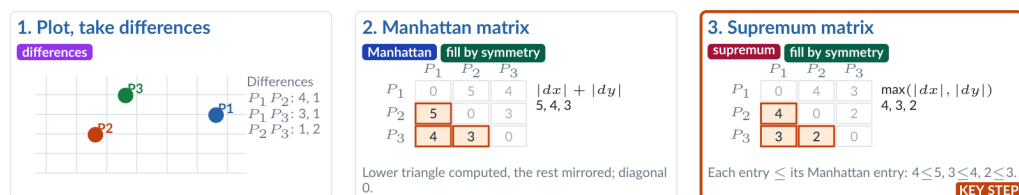
- $d(P_1, P_2) = \max(4, 1) = 4$
- $d(P_1, P_3) = \max(3, 1) = 3$
- $d(P_2, P_3) = \max(1, 2) = 2$

	P_1	P_2	P_3
P_1	0	5	4
P_2	5	0	3
P_3	4	3	0

	P_1	P_2	P_3
P_1	0	4	3
P_2	4	0	2
P_3	3	2	0

- **Sanity check**: supremum $\leq$ Euclidean $\leq$ Manhattan, always. Holds here ($4 \leq 5$, $3 \leq 4$, $2 \leq 3$).

Step by step: plot, then fill each matrix's lower triangle



Problem 3.2 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The cosine-similarity part of this same question is Problem 4.1 below.

Step-by-step

- $d(O_1, O_3)$: differences $(4 - 7, 56 - 58, 7 - 6, 10 - 9) = (-3, -2, 1, 1)$
 $= \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$
- $d(O_1, O_4)$: differences $(4 - 9, 56 - 55, 7 - 7, 10 - 12) = (-5, 1, 0, -2)$
 $= \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$

Final answer

$d(O_1, O_3) = 3.873$, $d(O_1, O_4) = 5.477$. Object 1 is closer to object 3 than to object 4.

Step by step: differences, then square, sum and root

1. Differences

differences

$O_1 - O_3$

Size	Wt	Col	Taste
-3	-2	1	1

$O_1 - O_4$

Size	Wt	Col	Taste
-5	1	0	-2

2. Square, sum, root

Euclidean

$d(O_1, O_3) = \sqrt{9 + 4 + 1 + 1} = \sqrt{15} = \mathbf{3.873}$

$d(O_1, O_4) = \sqrt{25 + 1 + 0 + 4} = \sqrt{30} = \mathbf{5.477}$

Object 1 is closer to 3 than to 4.

Square, add, root: the size gap of 5 dominates $d(O_1, O_4)$.

KEY STEP

M4. Cosine similarity**Method**

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}, \quad d_1 \cdot d_2 = \sum_k d_{1k} d_{2k}, \quad \|d\| = \sqrt{\sum_k d_k^2}$$

- Measures **angle only**, ignores magnitude → used for documents.
- Always lay out 3 lines: dot product, then each length, then divide.

Typo in your class notes verified/added : DBK's Lecture-2 cosine example prints $\|d_2\| = (6)^{0.5} = 2.245$. In fact $\sqrt{6} = \mathbf{2.449}$. His final answer 0.3150 is correct and only follows from $2.449 (5 / (6.481 \times 2.449) = 0.315)$, so 2.245 is a slip. Copy the method, not that number.

Problem 4.1 3

QUESTION AS PRINTED 2080 Bhadra · Q1 (b) · [3+3+3]

How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4.

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

The Euclidean-distance part of this same question is Problem 3.2 above.

Given

$O_2 = (3, 53, 8, 11)$, $O_4 = (9, 55, 7, 12)$.

Step-by-step

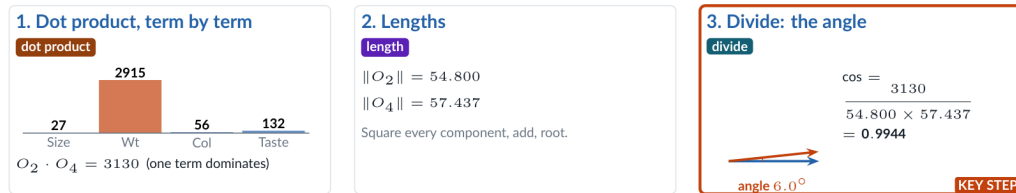
- Dot product: $3(9) + 53(55) + 8(7) + 11(12) = 27 + 2915 + 56 + 132 = 3130$
- $\|O_2\| = \sqrt{9 + 2809 + 64 + 121} = \sqrt{3003} = 54.800$
- $\|O_4\| = \sqrt{81 + 3025 + 49 + 144} = \sqrt{3299} = 57.437$
- $\cos = \frac{3130}{54.800 \times 57.437} = \frac{3130}{3147.5}$

Final answer

$\cos(O_2, O_4) = \mathbf{0.9944}$.

- Nearly 1 → the two objects point almost the same direction.
- **Why so high:** *Weight* (≈ 50) dwarfs the other attributes and dominates the dot product. **Normalise before using cosine** on mixed-scale attributes, or the answer means very little. verified/added

Step by step: cosine of objects 2 and 4 (the weight term dominates)



Problem 4.2 1

QUESTION AS PRINTED 2080 Baishakh · Q2 (b)(iv) · [1]

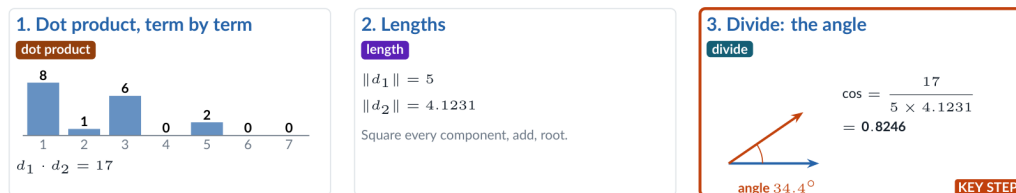
Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$. (the rest of Q2 b is Problem 5.2 below)

- Dot: $8 + 1 + 6 + 0 + 2 + 0 + 0 = 17$
- $\|d_1\| = \sqrt{16 + 1 + 4 + 0 + 4 + 0 + 0} = \sqrt{25} = 5$
- $\|d_2\| = \sqrt{4 + 1 + 9 + 0 + 1 + 1 + 1} = \sqrt{17} = 4.1231$
- $\cos = 17 / (5 \times 4.1231) = 17 / 20.6155$

Final answer

0.8246

Step by step: cosine of the two documents



Problem 4.3 1

QUESTION AS PRINTED 2076 Chaitra · Q2 (b) · [1]

Given the following two vectors compute the Cosine similarity between them.

$$D_1 = [4 \ 0 \ 2 \ 0 \ 1] \quad D_2 = [2 \ 0 \ 0 \ 2 \ 2]$$

- Dot: $8 + 0 + 0 + 0 + 2 = 10$
- $\|D_1\| = \sqrt{16 + 0 + 4 + 0 + 1} = \sqrt{21} = 4.5826$
- $\|D_2\| = \sqrt{4 + 0 + 0 + 4 + 4} = \sqrt{12} = 3.4641$
- $\cos = 10 / (4.5826 \times 3.4641) = 10 / 15.8745$

Final answer

0.6299

Step by step: cosine of the two vectors

1. Dot product, term by term

dot product



$D_1 \cdot D_2 = 10$ (one term dominates)

2. Lengths

length

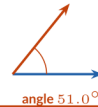
$\|D_1\| = 4.5826$
 $\|D_2\| = 3.4641$

Square every component, add, root.

3. Divide: the angle

divide

$\cos = \frac{10}{4.5826 \times 3.4641}$
 $= 0.6299$



angle 51.0° **KEY STEP**

M5. Binary similarity: Jaccard, SMC, symmetric dissimilarity

Method

- Encode each object as a 0/1 vector, then count over all attribute positions:
 - M_{11} : both 1 M_{00} : both 0
 - M_{10} : $p = 1, q = 0$ M_{01} : $p = 0, q = 1$
- SMC** (symmetric binary, 0-0 counts): $\frac{M_{11} + M_{00}}{M_{11} + M_{00} + M_{10} + M_{01}}$
- Jaccard** (asymmetric binary, 0-0 ignored): $\frac{M_{11}}{M_{11} + M_{10} + M_{01}}$
- Symmetric binary dissimilarity**: $d = \frac{M_{10} + M_{01}}{M_{11} + M_{00} + M_{10} + M_{01}} = 1 - SMC$
- Nominal (non-binary) dissimilarity**: $d = (p - m)/p$, m = matches, p = total attrs.

Problem 5.1 2

QUESTION AS PRINTED 2076 Chaitra · Q2 (c) · [2]

Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient.

$$P = [0 \ 0 \ 1 \ 1 \ 0 \ 1] \quad Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$$

Step-by-step: tabulate position by position

Position	1	2	3	4	5	6
P	0	0	1	1	0	1
Q	1	1	1	1	0	1
Category	M_{01}	M_{01}	M_{11}	M_{11}	M_{00}	M_{11}

- $M_{11} = 3, M_{01} = 2, M_{10} = 0, M_{00} = 1$ (total = 6 ✓)
- $J = \frac{3}{3 + 0 + 2} = \frac{3}{5}$
- $SMC = \frac{3 + 1}{6} = \frac{4}{6}$

Final answer

 $J = 0.6$, $SMC = 0.667$. SMC is higher because it credits the one 0-0 match that Jaccard discards.

Step by step: tally the pairs, then Jaccard and SMC

1. Tally position by position

tally the pairs

P	0	0	1	1	0	1
Q	1	1	1	1	0	1
pair	01	01	11	11	00	11

$M_{11}=3 \ M_{10}=0 \ M_{01}=2 \ M_{00}=1$

KEY STEP

2. Jaccard

Jaccard

$$J = \frac{M_{11}}{M_{11} + M_{10} + M_{01}} = \frac{3}{5} = 0.6$$

The single 0-0 (position 5) is left out.

3. SMC

SMC

$$SMC = \frac{M_{11} + M_{00}}{6} = \frac{4}{6} = 0.667$$

The 0-0 counts as agreement, so $SMC > J$.

Problem 5.2 2+2+2

QUESTION AS PRINTED 2080 Baishakh · Q2 (b) · [2+2+2+1]

Consider the following table:

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

- Calculated Jaccard Coefficient of $Sim_{jaccard}$ (Ram, Laxmi) for asymmetric binary attributes.
- Dissimilarity of symmetric binary attributes d (Laxmi, Shyam).
- Find the Simple matching coefficients of SMC (Ram, Shyam).
- Find the Cosine similarity between documents $d_1 = (4, 1, 2, 0, 2, 0, 0)$ and $d_2 = (2, 1, 3, 0, 1, 1, 1)$ — solved as Problem 4.2 above.

Step 0: split the attributes (the whole question turns on this)

- Asymmetric binary** = Test-1, Test-2, Fever, Cough. Encode $P = 1$, $N = 0$. A shared “N” is uninformative, which is exactly what “asymmetric” means.
- Symmetric** = Gender, Eyecolor, Haircolor. Both/all states are equally meaningful.
- Encoded: Ram = (1, 0, 1, 0), Laxmi = (1, 1, 0, 0), Shyam = (0, 1, 0, 1).

(i) Jaccard(Ram, Laxmi)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0
Cat.	M_{11}	M_{01}	M_{10}	M_{00}

$$- M_{11} = 1, M_{01} = 1, M_{10} = 1, M_{00} = 1$$

$$- J = \frac{1}{1 + 1 + 1} = \frac{1}{3}$$

Answer

$$J = 0.333$$

(iii) SMC(Ram, Shyam)

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1
Cat.	M_{10}	M_{01}	M_{10}	M_{01}

$$- M_{11} = 0, M_{00} = 0, M_{10} = 2, M_{01} = 2$$

$$- SMC = \frac{0 + 0}{4} = 0$$

Answer

$$SMC = 0 \text{ (they disagree on every test).}$$

(ii) d (Laxmi, Shyam), symmetric binary attributes

- Across these 3 people each of Gender / Eyecolor / Haircolor takes exactly **2 states**, and neither state is an “absence” → all 3 are **symmetric binary**.
- Encode: Gender M=1 · Eyecolor Black=1 · Haircolor Gray=1.

	Gender	Eye	Hair
Laxmi (F, Blue, Black)	0	0	0
Shyam (M, Blue, Gray)	1	0	1
Category	s	t	s

$$- q = 0, r = 0, s = 2, t = 1 \text{ (total } p = 3 \checkmark)$$

- Symmetric binary dissimilarity keeps t :

$$d = \frac{r + s}{q + r + s + t} = \frac{0 + 2}{3} = \frac{2}{3}$$

Answer

$$d = 0.667$$

Cross-check: for binary attrs the symmetric formula reduces to the nominal mismatch ratio $(p - m)/p = (3 - 1)/3 = 2/3$. Same answer by both routes, so the encoding choice is safe.

verified/added

Step by step: decide the attribute types first, then tally each pair

1. Split the attributes

split attributes

symmetric

Gender, Eye, Hair
both states count

asymmetric

Test-1, Test-2,
Fever, Cough
shared N = no info

P → 1, N → 0 on the tests.

KEY STEP

2. (i) Jaccard(Ram, Laxmi)

tally the pairs Jaccard

	T1	T2	Fev	Cou
Ram	1	0	1	0
Laxmi	1	1	0	0

pair 11 01 10 00

$M_{11}=1 \ M_{10}=1 \ M_{01}=1 \ M_{00}=1$

$J = 1/(1 + 1 + 1) = 0.333$

3. (ii) d (Laxmi, Shyam), symmetric

tally the pairs symmetric d

	Gen	Eye	Hair
Laxmi	0	0	0
Shyam	1	0	1

pair 01 00 01

$M_{11}=0 \ M_{10}=0 \ M_{01}=2 \ M_{00}=1$

$d = (0 + 2)/3 = 0.667$

4. (iii) SMC(Ram, Shyam)

tally the pairs SMC

	T1	T2	Fev	Cou
Ram	1	0	1	0
Shyam	0	1	0	1

pair 10 01 10 01

$M_{11}=0 \ M_{10}=2 \ M_{01}=2 \ M_{00}=0$

$SMC = (0 + 0)/4 = 0$

M6. Schema design + OLAP operation sequence

Method

1. Put the **measures** in a central **fact table**, plus one foreign key per dimension.
2. Draw one **dimension table** per dimension around it.
 - **Star** → leave dimension tables flat (de-normalised).
 - **Snowflake** → normalise the hierarchies into sub-tables (location → city → province).
3. For the OLAP sequence, work from the base cuboid and ask, per dimension:
 - Need a coarser level? → **roll-up**
 - Need a finer level? → **drill-down**
 - Restricting 1 dimension? → **slice**. Restricting ≥ 2 ? → **dice**
 - Dimension not wanted at all? → **roll-up to all** (dimension reduction)
4. n dimensions → 2^n cuboids in the lattice.

Problem 6.1 3+3

QUESTION AS PRINTED 2078 Bhadra · Q2 · [3+3]

Suppose that a data warehouse consists of the four dimensions data, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults or seniors, with each category having its own charge rate.

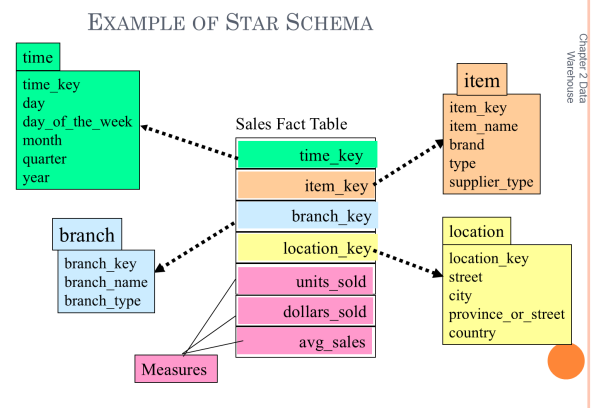
- a) Draw a star schema diagram for the data warehouse.
- b) Starting with the base cuboid [data, spectator, location, game], what specific OLAP operations should you perform in order to list the total charge paid by student spectators at Dashrath Stadium in 2021?

(a) Star schema

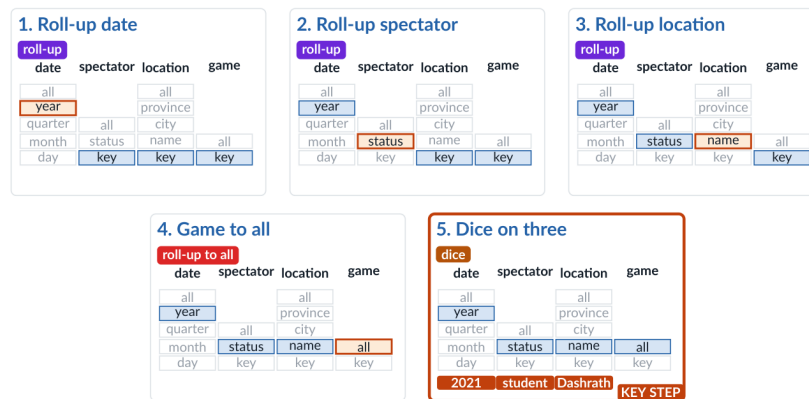
- **Fact table** Sales: date_key, spectator_key, location_key, game_key, count, charge.
- **Dimension tables** (flat, one hop from the fact table):
 - date(date_key, day, month, quarter, year)
 - spectator(spectator_key, name, status, phone, address) where status $\in$ {student, adult, senior}
 - location(location_key, name, street, city, province, country)
 - game(game_key, name, description, sport_type)
- 4 dimensions → $2^4 = 16$ cuboids in the lattice.

(b) OLAP operation sequence

1. **Roll-up** on *date* from date_key to year.
 2. **Roll-up** on *spectator* from spectator_key to status.
 3. **Roll-up** on *location* from location_key to location_name.
 4. **Roll-up** on *game* to all (the question does not split by game).
 5. **Dice** with year = 2021 and status = student and location_name = ``Dashrath Stadium''.
 6. Read off the charge measure (summed).
- **Dice, not slice** → 3 dimensions are being restricted at once.



Step by step: each dimension's level after every OLAP operation (orange = changed)



Problem 6.2 3+3

QUESTION AS PRINTED 2076 Chaitra · Q3 · [3+3]

Suppose that a data warehouse for a sales company consists of five dimensions: *time*, *location*, *supplier*, *brand*, and *product*, and two measures: *count* and *price*.

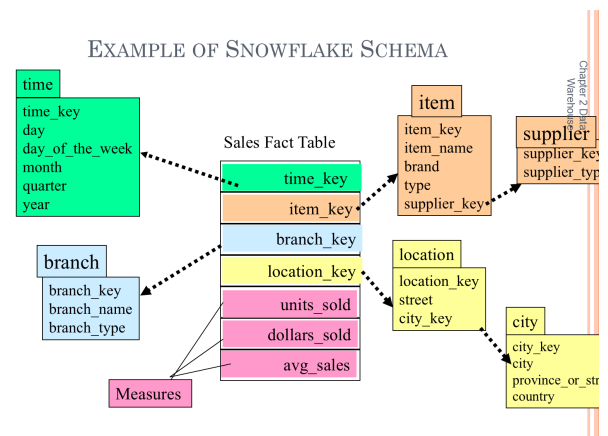
- Draw a *snowflake schema* diagram for the data warehouse.
- Starting with the base cuboid [*time*, *location*, *supplier*, *brand*, *product*], what specific OLAP operations should one perform in order to list the total count for a certain *brand* for each *state* per *year* (assume *location* has three levels: *country*, *state*, *city*; and assume *time* has three levels: *year*, *month*, *day*)?

(a) Snowflake schema

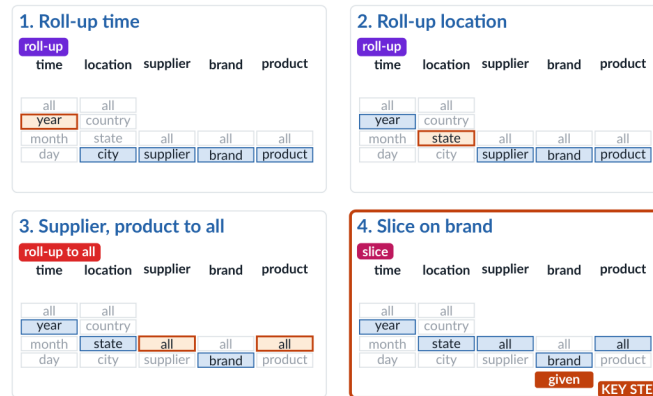
- Fact table** Sales: time_key, location_key, supplier_key, brand_key, product_key, count, price.
- Normalised hierarchies** (this is what makes it a snowflake, not a star):
 - time(time_key, day, month_key) → month(month_key, month, year)
 - location(location_key, city_key) → city(city_key, city, state_key) → state(state_key, state, country)
 - product(product_key, name, brand_key) → brand(branch_key, brand_name)
 - supplier(supplier_key, supplier_name, supplier_type)
- 5 dimensions → $2^5 = 32$ cuboids.

(b) OLAP operation sequence

- Roll-up on *time* from day to year.
 - Roll-up on *location* from city to state.
 - Roll-up on *supplier* to all (not wanted).
 - Roll-up on *product* to all: brand is its own dimension of this cube, so the product dimension is not needed.
 - Slice on brand = the given brand (1 dimension restricted).
 - Result: cuboid [year, state] holding total count.
- Do not roll product up to its brand level** as well: brand would then be held twice, once as a dimension and once as a level of product.



Step by step: each dimension's level after every OLAP operation (orange = changed)



M7. Central tendency, quartiles, IQR and outliers

NEW Not in any of the 23 papers. Added because Notes by PC/unit 2 Numerical.docx and Untitled spreadsheet.xlsx (2025-26) are built entirely around it. Theory in Ch 2, §2.5.

Method

- **Grouped median** = $L_1 + \left(\frac{N/2 - (\sum freq)_i}{freq_{median}} \right) \times width$. Find the median interval first: the one where cumulative frequency first reaches $N/2$.
- **Mode from mean and median** $\approx 3 median - 2 mean$ (unimodal, moderately skewed).
- **Quartiles**, interpolated convention: position of $Q_k = \frac{k(n+1)}{4}$, then interpolate between the neighbouring sorted values.
- **Outlier fences**: below $Q_1 - 1.5 IQR$ or above $Q_3 + 1.5 IQR$.

Problem 7.1 – grouped median, 100 employees 4

PROBLEM not a PYQ · PC's notes · [4]

Suppose the salaries of 100 employees of a company are grouped into \$10,000 intervals as shown in the table below. Find the median salary.

Salary (\$)	Freq	Cum freq
10,000–20,000	8	8
20,000–30,000	15	23
30,000–40,000	20	43
40,000–50,000	32	75
50,000–60,000	18	93
60,000–70,000	7	100

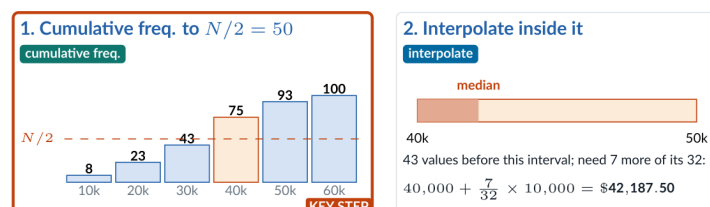
Solution

1. $N = 100 \rightarrow N/2 = 50$. Need the 50th observation.
2. Cumulative frequency passes 50 inside **40k–50k** (43 before it, 75 after) $\rightarrow$ that is the median interval.
3. $L_1 = 40,000$, $(\sum freq)_i = 43$,
 $freq_{median} = 75 - 43 = 32$, $width = 10,000$.
4. $median = 40,000 + \frac{50-43}{32} \times 10,000 = 40,000 + \frac{7}{32} \times 10,000$

Answer

$median = \$42,187.50$

Step by step: find the median interval, then interpolate



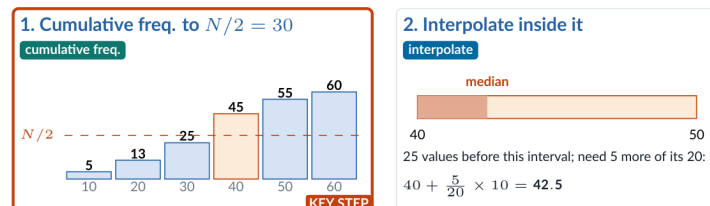
Problem 7.2 – same method, smaller table 3

PROBLEM not a PYQ · PC's notes · [3]

Salaries (in \$1000s) of 60 employees fall in the intervals 10–20, 20–30, 30–40, 40–50, 50–60 and 60–70 with frequencies 5, 8, 12, 20, 10 and 5 respectively. Find the median salary.

- $N = 5 + 8 + 12 + 20 + 10 + 5 = 60 \rightarrow N/2 = 30$.
- Cumulative before 40–50 = $5 + 8 + 12 = 25$; $freq_{median} = 20$, $L_1 = 40$, $width = 10$.
- $median = 40 + \frac{30-25}{20} \times 10 = 40 + 2.5 = 42.5$ (i.e. \$42,500).

Step by step: find the median interval, then interpolate



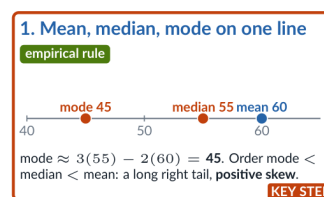
Problem 7.3 – mode from mean and median 2

PROBLEM not a PYQ · PC's notes · [2]

For a unimodal, moderately skewed distribution the mean is 60 and the median is 55. Estimate the mode, and state what the result says about the skew.

- $mean - mode \approx 3(mean - median) \Rightarrow mode \approx 3median - 2mean$.
- $mode \approx 3(55) - 2(60) = 165 - 120 = 45$.
- **Sanity check:** $mode < median < mean \rightarrow$ positively skewed. Consistent.

Step by step: the three centres on one line



Problem 7.4 – full descriptive summary of one dataset 2+2+2+2

PROBLEM not a PYQ · Han & Kamber Example 2.6 · [2+2+2+2]

Suppose that the data for analysis is the salaries (in \$1000s) of 12 employees, already sorted in increasing order:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

Find (a) the mean, median and mode; (b) the quartiles Q_1 and Q_3 and the IQR; (c) the five-number summary; (d) any outliers using the $1.5 \times IQR$ rule.

(a) Central tendency

- $\sum x = 696 \rightarrow \text{mean} = \frac{696}{12} = 58.$
- $n \text{ even} \rightarrow \text{median} = \frac{52+56}{2} = 54.$
- 52 and 70 each occur twice $\rightarrow$ **bimodal**, modes **52** and **70**.
- $\text{midrange} = \frac{30+110}{2} = 70.$
- $\text{mode} < \text{median} < \text{mean}$ is broken here by the bimodality; the single outlier 110 is what pulls the mean above the median.

(b) Quartiles, positional convention (Han & Kamber)

- Take the 3rd, 6th, 9th sorted values $\rightarrow Q_1 = 47, Q_2 = 52, Q_3 = 63.$
- $IQR = 63 - 47 = 16.$

(c) Quartiles, interpolated convention (PC's class notes)

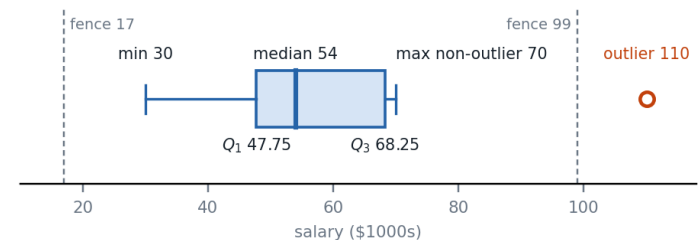
- $Q_1 \text{ at } \frac{1(13)}{4} = 3.25 \rightarrow 47 + 0.25(50 - 47) = 47.75$
- $Q_2 \text{ at } \frac{2(13)}{4} = 6.5 \rightarrow 52 + 0.5(56 - 52) = 54$
- $Q_3 \text{ at } \frac{3(13)}{4} = 9.75 \rightarrow 63 + 0.75(70 - 63) = 68.25$
- $IQR = 68.25 - 47.75 = 20.5$

(d) Five-number summary and outliers

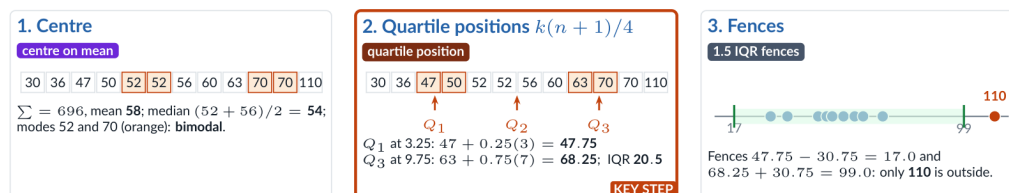
- Five-number summary (interpolated): **30, 47.75, 54, 68.25, 110.**
- Fences: $47.75 - 1.5(20.5) = 17.0$ and $68.25 + 1.5(20.5) = 99.0.$
- $110 > 99 \rightarrow$ **110** is a suspected outlier. Nothing below 17.
- **Cross-check with the other convention:** fences $47 - 24 = 23$ and $63 + 24 = 87.$ **110 is still the only outlier** $\rightarrow$ the verdict does not depend on the convention, so either is safe in the exam as long as you name it.

(e) Boxplot

- Box from Q_1 to Q_3 , line at 54, whiskers to 30 and to the largest non-outlier (70), and **110 plotted as a lone point.**



Drawn to scale by `boxfig.py` from the values above. The whisker stops at 70, not at the fence.

Step by step: centre, quartiles, fences**Extra practice (patterns not yet fully covered)****Practice A – binning for noise smoothing**

Binning is examined in prose 72 Ka, 76 Ash but never as a numerical. **This is DBK's own worked example** (Lecture-1, "Binning Methods for Data Smoothing"), so it is the likeliest form any future numerical would take. Answers below independently recomputed and they match his slide exactly.

PROBLEM not a PYQ · DBK Lecture-1 worked example verified/added

The following price data has already been sorted:

4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

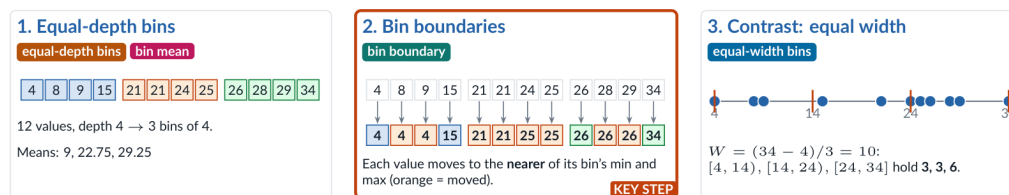
Partition the data into **equal-depth** bins of depth 4, then smooth the noise by (i) bin means and (ii) bin boundaries.

Solution

- 12 values, depth 4 $\rightarrow$ 3 bins.

- Bin 1: 4, 8, 9, 15 Bin 2: 21, 21, 24, 25 Bin 3: 26, 28, 29, 34
- (i) **Smoothing by bin means**
 - Bin 1 mean = $36/4 = 9 \rightarrow 9, 9, 9, 9$
 - Bin 2 mean = $91/4 = 22.75 \approx 23 \rightarrow 23, 23, 23, 23$
 - Bin 3 mean = $117/4 = 29.25 \approx 29 \rightarrow 29, 29, 29, 29$
- (ii) **Smoothing by bin boundaries** (each value $\rightarrow$ nearer of min/max)
 - Bin 1 (min 4, max 15): 4, 8, 9, 15 $\rightarrow 4, 4, 4, 15$ (8 and 9 are nearer 4)
 - Bin 2 (min 21, max 25): 21, 21, 24, 25 $\rightarrow 21, 21, 25, 25$
 - Bin 3 (min 26, max 34): 26, 28, 29, 34 $\rightarrow 26, 26, 26, 34$
- **Equal-width for contrast:** range = $34 - 4 = 30$, $N = 3 \rightarrow W = (B - A)/N = 10$. Intervals $[4, 14)$, $[14, 24)$, $[24, 34]$ $\rightarrow$ counts 3, 3, 6. **Unequal counts: that is the difference from equal-depth.**

Step by step: equal-depth bins, boundaries, and equal width for contrast



Practice B — min-max into a range other than [0,1]

Every PYQ so far uses $[0, 1]$. The figures below are Han & Kamber 3rd ed Example 3.4, so they are the canonical ones. **verified/added**

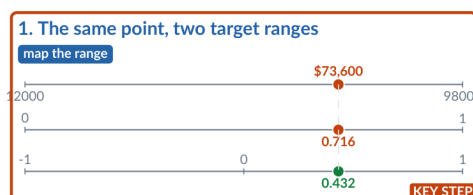
PROBLEM not a PYQ · Han & Kamber 3rd ed Example 3.4 **verified/added**

Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000 respectively. Normalise **\$73,600** by min-max normalization, first to the range $[0, 1]$ and then to the range $[-1, 1]$.

Solution

- To $[0, 1]$: $v' = \frac{73600 - 12000}{98000 - 12000}(1 - 0) + 0 = \frac{61600}{86000} = \mathbf{0.716}$
- To $[-1, 1]$: $v' = \frac{61600}{86000}(1 - (-1)) + (-1) = 0.716(2) - 1 = \mathbf{0.432}$
- **Check:** the value sits 71.6% of the way up the range, so in $[-1, 1]$ it must land slightly above the midpoint 0. It does.

Step by step: one income, mapped into two ranges



Ch 3 – Classification – Numerical Problems & Solutions

167 numerical marks, 8.6% of the archive, 40% of this chapter · 6 distinct methods · 12 PYQ instances solved · every answer independently recomputed.
Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

- **M1** Decision tree by information gain (ID3) 79 Bh, 78 Bh, 76 Ch
- **M2** Gini index 75 Ch, 71 Ch
- **M3** Naive Bayes classification 72 Ch, 79 Bh
- **M4** k -Nearest Neighbour 80 Bh
- **M5** Confusion matrix metrics **8 papers**
- **M6** ANN gradients (backpropagation) 76 Ch

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-3 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
entropy	Info = $-\sum p \log_2 p$ of the class counts at a node. 1.4 1	k nearest	Sort the distances and keep the k smallest. 4.1 2
gain	Info before minus weighted Info after the split; the largest wins. 1.4 2	majority vote	The class most common among the k nearest is the prediction. 4.1 3
pure branch	A partition holding one class has Info (or Gini) 0 and becomes a leaf. 1.4 3; Gini 3	read the axes	Find which axis is Actual before naming TP, FN, FP, TN. Papers flip it. 5.1 1; 5.2 1; 5.3/5.4 1; 5.5 1; 5.6 1; 5.7 1; 5.9 1; 5.10 1; 5.8 2
Gini	$1 - \sum p^2$: the impurity Gini-based trees minimise. Gini 1	tally	Compare actual and predicted position by position and count each outcome. 5.8 1
binary split	Gini splits in two: group the values into two subsets, try each grouping. Gini 2, 3	diagonal	Accuracy uses the diagonal (TP + TN) over everything. 5.1 2; 5.2 2; 5.3/5.4 2; 5.5 2; 5.9 2; 5.10 4; 5.8 3
prior	$P(C)$ = class count / all tuples. 3.1 1; 3.2 1	off-diagonal	Error rate uses the off-diagonal (FP + FN) over everything. 5.6 2; 5.9 3
class count	$P(x_k C)$ = tuples of class C with that value / tuples of class C . 3.1 2; 3.2 2	actual total	TPR, specificity and FPR divide by an ACTUAL class total, P or N . 5.1 3, 5; 5.2 3, 4; 5.3/5.4 3, 4; 5.5 3, 4; 5.6 3, 4, 5; 5.7 2, 3; 5.9 4, 5; 5.10 2, 3; 5.8 5, 6
Laplace	A zero count wipes out the product: add 1 to each count, k to the total. 3.2 3	predicted total	Precision divides by the PREDICTED-positive total, TP + FP. 5.1 4; 5.2 5; 5.3/5.4 5; 5.5 5; 5.7 4; 5.9 6; 5.8 4
Gaussian	A continuous attribute: fit μ , σ per class, read the density at x . 3.2 4	weighted sum	A neuron adds its weighted inputs, $net = \sum w_i x_i$. 6.1 1
product	Naive independence: multiply the per-attribute probabilities. 3.1 3; 3.2 5	sigmoid	$\sigma(z) = 1/(1 + e^{-z})$, and $\sigma' = \sigma(1 - \sigma)$. 6.1 1, 3
largest wins	Compare the candidates; the largest score is the answer. Gini 4; 3.1 4; 3.2 5	chain rule	Multiply the local derivatives along the one path from the loss to the weight. 6.1 2, 3
distance	Euclidean distance from the query to every training tuple. 4.1 1		

M1. Decision tree by information gain (ID3)

Two things to know before you start.

- **Problems 1.2 and 1.4 are different tables.** Same ten-row transport layout, same class counts, but four rows differ. The root is Travel cost in both, yet the runner-up is not: Income level (0.6955) in 1.2, a dead tie between Car ownership and Income level (0.5345) in 1.4. Never copy one answer into the other.
- **Two of the four tables have three classes** (1.2 and 1.4: Bus, Car, Train; 1.3: high, moderate, low). *Info* then has three terms and can exceed 1; the maximum for three classes is $\log_2 3 = 1.585$.

The recipe

- **Step 1.** Count the classes in the whole set D , then $Info(D) = -\sum_i p_i \log_2 p_i$, one term per class.
- **Step 2.** For each attribute A , split D by its values and fill *one row per value* of the table below: the class counts, that branch's $Info(D_j)$, and the branch weighted by its share of D .
- **Step 3.** Sum the weighted column. That is $Info_A(D)$, the **residual** entropy after splitting on A .
- **Step 4.** $Gain(A) = Info(D) - Info_A(D)$. Compare the gains; the largest wins and becomes the node's test.

- **Step 5.** Recurse on each branch that is not already pure ($Info = 0$), using only that branch's rows and only the attributes not yet used on its path.

The table idiom, used unchanged in every problem below

Attribute	Value	Class 1	Class 2	$ D_j $	$Info(D_j)$	$\frac{ D_j }{ D } Info(D_j)$
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Time-savers: a pure branch contributes 0; a 1:1 branch has $Info$ exactly 1; branches with the same class ratio share one $Info$, so compute it once. Keep 4 decimals throughout.

Every entropy this section needs

$\log_2 x = \frac{\ln x}{\ln 2}$. Ratios are class counts in any order: 2:1 covers (2 Yes, 1 No) and (1 Yes, 2 No) alike. $Info = 0$ whenever only one class is present.

Split	1:1	2:1	3:1	3:2	4:1	5:1	4:3	6:1	9:5
$Info$	1.0000	0.9183	0.8113	0.9710	0.7219	0.6500	0.9852	0.5917	0.9403

Split	2:2:1	3:1:1	3:2:1	3:2:2	4:2:1	4:3:3	6:3:2	6:5:3
$Info$	1.5219	1.3710	1.4591	1.5567	1.3788	1.5710	1.4354	1.5306

Problem 1.1 10

QUESTION AS PRINTED 2079 Bhadra · Q3 · [10]

Draw decision tree for the given data using ID3 algorithm.

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Step 1 Entropy of the whole set

14 rows: 9 Yes, 5 No.

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.9403}$$

Steps 2 and 3 Split on each attribute in turn

Attribute	Value	Yes	No	$ D_j $	$Info(D_j)$	$\frac{ D_j }{ D } Info(D_j)$
Age	Youth	2	3	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
	Middle_Aged	4	0	4	0	0
	Senior	3	2	5	0.9710	$\frac{5}{14}(0.9710) = 0.3468$
$Info_{Age}(D) = 0.6936$						Gain = 0.9403 - 0.6936 = 0.2467
Income	High	2	2	4	1.0000	$\frac{4}{14}(1.0000) = 0.2857$
	Medium	4	2	6	0.9183	$\frac{6}{14}(0.9183) = 0.3936$
	Low	3	1	4	0.8113	$\frac{4}{14}(0.8113) = 0.2318$
$Info_{Income}(D) = 0.9111$						Gain = 0.9403 - 0.9111 = 0.0292
Student	No	3	4	7	0.9852	$\frac{7}{14}(0.9852) = 0.4926$
	Yes	6	1	7	0.5917	$\frac{7}{14}(0.5917) = 0.2959$
$Info_{Student}(D) = 0.7885$						Gain = 0.9403 - 0.7885 = 0.1518
Credit_Rating	Fair	6	2	8	0.8113	$\frac{8}{14}(0.8113) = 0.4636$
	Excellent	3	3	6	1.0000	$\frac{6}{14}(1.0000) = 0.4286$
$Info_{Credit_Rating}(D) = 0.8922$						Gain = 0.9403 - 0.8922 = 0.0481

Step 4 Compare

Attribute	Gain
Age	0.2467
Student	0.1518
Credit_Rating	0.0481
Income	0.0292

Answer. The root attribute is **Age**, with the highest information gain, **0.2467**.

Why Age wins

- Age is the only attribute with a **pure branch**: all four *Middle_Aged* rows (3, 7, 12, 13) are Yes, so it removes more uncertainty than any other.
- Income removes almost none: its three branches are nearly as mixed as the parent.
- Middle_Aged* becomes a **leaf** at once. *Youth* (2:3) and *Senior* (3:2) still have $Info = 0.9710$, so each needs its own split: Step 5.

Step 5 Recurse: Steps 1 to 4 again, inside each impure branch

Rows numbered 1 to 14 top to bottom. Each branch uses only its own rows and only the attributes Age has not used.

Age = Youth rows 1, 2, 8, 9, 11 2 Yes, 3 No $Info = 0.9710$

Attribute	Value (rows)	Yes	No	Info	Gain
Student	No (1,2,8)	0	3	0	0.9710
	Yes (9,11)	2	0	0	
Income	High (1,2)	0	2	0	0.5710
	Medium (8,11)	1	1	1.0000	
	Low (9)	1	0	0	
Credit_Rating	Fair (1,8,9)	1	2	0.9183	0.0200
	Excellent (2,11)	1	1	1.0000	

Student: residual 0, gain $0.9710 - 0.0000 = 0.9710$. Income: residual $\frac{2}{5}(1.0000) = 0.4000$, gain $0.9710 - 0.4000 = 0.5710$. Credit_Rating: residual $\frac{3}{5}(0.9183) + \frac{2}{5}(1.0000) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. **Split on Student.**

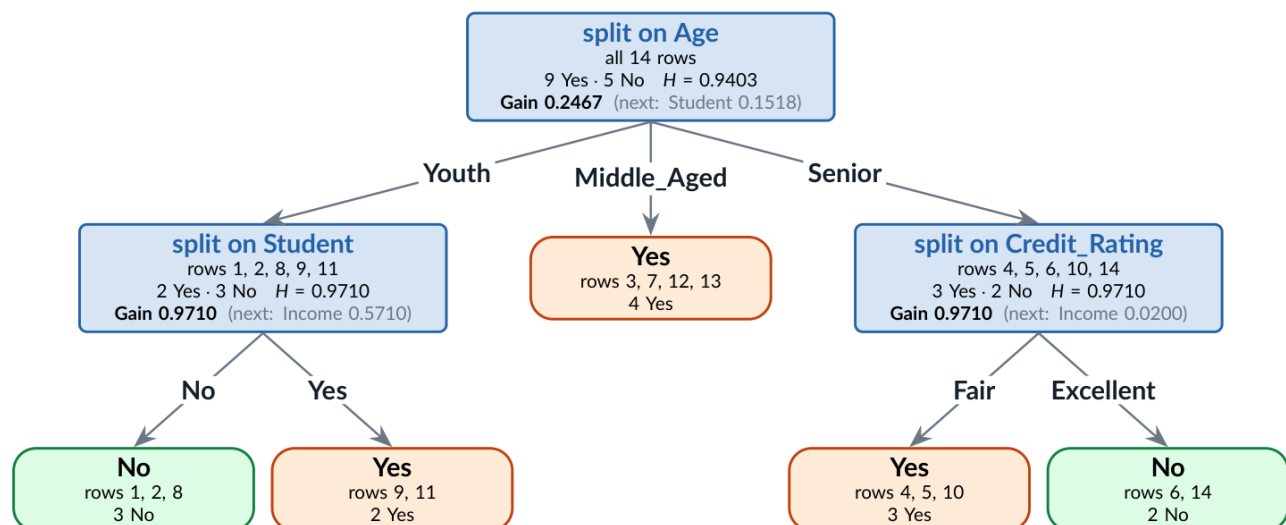
Age = Senior rows 4, 5, 6, 10, 14 3 Yes, 2 No $Info = 0.9710$

Attribute	Value (rows)	Yes	No	Info	Gain
Credit_Rating	Fair (4,5,10)	3	0	0	0.9710
	Excellent (6,14)	0	2	0	
Income	Medium (4,10,14)	2	1	0.9183	0.0200
	Low (5,6)	1	1	1.0000	
Student	No (4,14)	1	1	1.0000	0.0200
	Yes (5,6,10)	2	1	0.9183	

Credit_Rating: residual 0, gain $0.9710 - 0.0000 = 0.9710$. Income: residual $\frac{3}{5}(0.9183) + \frac{2}{5}(1.0000) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. Student: residual $\frac{2}{5}(1.0000) + \frac{3}{5}(0.9183) = 0.9510$, gain $0.9710 - 0.9510 = 0.0200$. **Split on Credit_Rating.**

The tree, 2 levels of tests and 5 leaves, one rule per leaf: Age = Youth, Student = No $\rightarrow$ No; Age = Youth, Student = Yes $\rightarrow$ Yes; Age = Middle_Aged $\rightarrow$ Yes; Age = Senior, Credit_Rating = Fair $\rightarrow$ Yes; Age = Senior, Credit_Rating = Excellent $\rightarrow$ No. Income never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: buys_computer (rows numbered 1 to 14 top to bottom)



5 leaves, depth 2; classifies all 14 training tuples correctly. The same tree as Han & Kamber 3rd ed Fig 8.2, whose Example 8.1 prints the same gains (to its own truncation: 0.246, 0.151).

Problem 1.2 8

QUESTION AS PRINTED 2078 Bhadra · Q4 · [8]

Construct a decision tree for the following data set using information gain.

Predict the class label for a data point with values <Female, 2, standard, high>

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Step 1 Entropy of the whole set three classes, so three terms

10 rows: 4 Bus, 3 Car, 3 Train.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.5710}$$

Steps 2 and 3 Split on each attribute in turn class columns (Bus, Car, Train)

Attribute	Value	Bus	Car	Train	$ D_j $	$Info(D_j)$	$\frac{ D_j }{10} Info(D_j)$
Gender	Male	3	1	1	5	1.3710	$\frac{5}{10}(1.3710) = 0.6855$
	Female	1	2	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
$Info_{Gender}(D) = 1.4465$							Gain = $1.5710 - 1.4465 = 0.1245$
Car ownership	0	2	0	1	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
	1	2	1	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	2	0	2	0	2	0	0
$Info_{Car\ ownership}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$
Travel cost	Cheap	4	0	1	5	0.7219	$\frac{5}{10}(0.7219) = 0.3609$
	Expensive	0	3	0	3	0	0
	Standard	0	0	2	2	0	0
$Info_{Travel\ cost}(D) = 0.3609$							Gain = $1.5710 - 0.3609 = \mathbf{1.2101}$
Income level	Low	2	0	0	2	0	0
	Medium	2	1	3	6	1.4591	$\frac{6}{10}(1.4591) = 0.8755$
	High	0	2	0	2	0	0
$Info_{Income\ level}(D) = 0.8755$							Gain = $1.5710 - 0.8755 = 0.6955$

Step 4 Compare

Attribute	Gain
Travel cost	1.2101
Income level	0.6955
Car ownership	0.5345
Gender	0.1245

Why Travel cost wins

- Two of its three branches are already **pure**: Expensive is all Car, Standard all Train. No other attribute leaves so little mixed.
- Gender barely helps: Male (3,1,1) and Female (1,2,2) are each almost as mixed as the whole set.
- Only the Cheap branch needs splitting** (4 Bus, 1 Train): Step 5.

Answer. The root attribute is **Travel cost**, with the highest information gain, **1.2101**.

Step 5 Recurse on the Cheap branch, then on what it leaves mixed

Rows numbered 1 to 10 top to bottom. Only Bus and Train occur in this branch, so the Car column is dropped.

Travel cost = Cheap rows 1, 2, 3, 4, 8 4 Bus, 1 Train

$Info = 0.7219$

Attribute	Value (rows)	Bus	Train	Info	Gain
Gender	Male (1,2,4)	3	0	0	0.3219
	Female (3,8)	1	1	1.0000	
Car ownership	0 (1,3)	2	0	0	0.1709
	1 (2,4,8)	2	1	0.9183	
Income level	Low (1,3)	2	0	0	0.1709
	Medium (2,4,8)	2	1	0.9183	

Gender: residual $\frac{2}{5}(1.0000) = 0.4000$, gain $0.7219 - 0.4000 = 0.3219$.

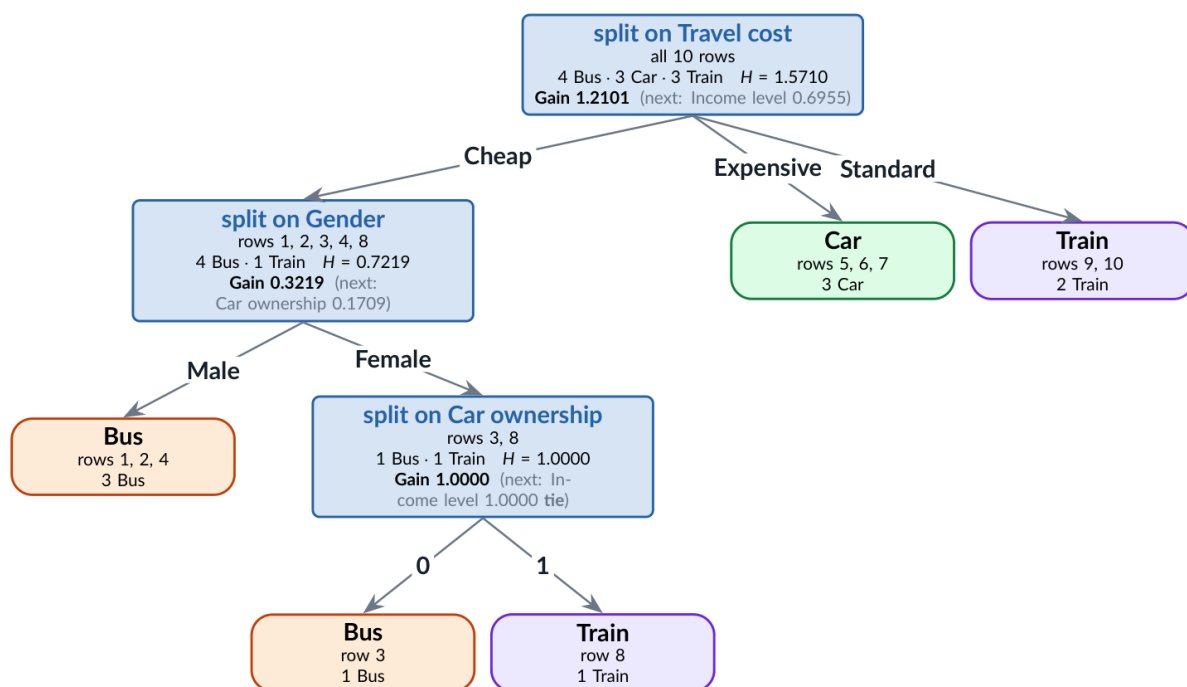
Car ownership: residual $\frac{3}{5}(0.9183) = 0.5510$, gain

$0.7219 - 0.5510 = 0.1709$. Income level: residual $\frac{3}{5}(0.9183) = 0.5510$,

gain $0.7219 - 0.5510 = 0.1709$. **Split on Gender.**

The tree, 3 levels of tests and 5 leaves, one rule per leaf: Travel cost = Cheap, Gender = Male $\rightarrow$ Bus; Travel cost = Cheap, Gender = Female, Car ownership = 0 $\rightarrow$ Bus; Travel cost = Cheap, Gender = Female, Car ownership = 1 $\rightarrow$ Train; Travel cost = Expensive $\rightarrow$ Car; Travel cost = Standard $\rightarrow$ Train. Income level never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: transport mode (rows numbered 1 to 10 top to bottom)



5 leaves, depth 3; classifies all 10 training tuples correctly.

Prediction for <Female, 2, standard, high>

- Read the root first: Travel cost = **standard**.
- That branch is **pure Train**, so the walk stops immediately.
- **No other attribute is consulted** – Gender, car ownership and income are never reached. **Answer: Train.**
- **This is the point of the question:** a decision tree only tests the attributes on the path it takes, not all of them.

Problem 1.3 2+9+3

QUESTION AS PRINTED 2082 Bhadra · Q3 · [2+9+3]

When do we use decision tree? Construct a complete decision tree using ID-3 algorithm for following dataset. How do you evaluate the performance of this model?

SN	Credit history	Debt	Collateral	Income	Credit Risk
1	bad	high	none	\$0 to \$15K	high
2	unknown	high	none	\$15 to \$35K	high
3	unknown	low	none	\$15 to \$35K	moderate
4	unknown	low	none	\$0 to \$15K	high
5	unknown	low	none	over \$35K	low
6	unknown	low	adequate	over \$35K	low
7	bad	low	none	\$0 to \$15K	high
8	bad	low	adequate	over \$35K	moderate
9	good	low	none	over \$35K	low
10	good	high	adequate	over \$35K	low
11	good	high	none	\$0 to \$15K	high
12	good	high	none	\$15 to \$35K	moderate
13	good	high	none	over \$35K	low
14	bad	high	none	\$15 to \$35K	high

Three classes, not two, so *Info* has three terms. Rows are the table's SN.

Step 1 Entropy of the whole set

14 rows: 6 high, 3 moderate, 5 low.

$$Info(D) = -\frac{6}{14} \log_2 \frac{6}{14} - \frac{3}{14} \log_2 \frac{3}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.5239 + 0.4762 + 0.5305 = \mathbf{1.5306}$$

Steps 2 and 3 Split on each attribute in turn class columns (high, moderate, low)

Attribute	Value	high	mod	low	$ D_j $	$Info(D_j)$	$\frac{ D_j }{14} Info(D_j)$
Credit history	bad	3	1	0	4	0.8113	$\frac{4}{14}(0.8113) = 0.2318$
	unknown	2	1	2	5	1.5219	$\frac{5}{14}(1.5219) = 0.5435$
	good	1	1	3	5	1.3710	$\frac{5}{14}(1.3710) = 0.4896$
$Info_{Credit\ history}(D) = 1.2649$							Gain = 1.5306 – 1.2649 = 0.2657
Debt	high	4	1	2	7	1.3788	$\frac{7}{14}(1.3788) = 0.6894$
	low	2	2	3	7	1.5567	$\frac{7}{14}(1.5567) = 0.7783$
$Info_{Debt}(D) = 1.4677$							Gain = 1.5306 – 1.4677 = 0.0629
Collateral	none	6	2	3	11	1.4354	$\frac{11}{14}(1.4354) = 1.1278$
	adequate	0	1	2	3	0.9183	$\frac{3}{14}(0.9183) = 0.1968$
$Info_{Collateral}(D) = 1.3246$							Gain = 1.5306 – 1.3246 = 0.2060
Income	\$0–15K	4	0	0	4	0	0
	\$15–35K	2	2	0	4	1.0000	$\frac{4}{14}(1.0000) = 0.2857$
	over \$35K	0	1	5	6	0.6500	$\frac{6}{14}(0.6500) = 0.2786$
$Info_{Income}(D) = 0.5643$							Gain = 1.5306 – 0.5643 = 0.9663

Step 4 Compare

Attribute	Gain
Income	0.9663
Credit history	0.2657
Collateral	0.2060
Debt	0.0629

Why Income wins, by a wide margin

- Its \$0–15K branch is **pure**: rows 1, 4, 7, 11 are all high risk.
- Its over \$35K branch is nearly pure (1 moderate, 5 low), $Info = 0.6500$.
- Debt splits 7/7 into two branches about as mixed as the parent, so it gains almost nothing.

Answer. The root attribute is **Income**, with the highest information gain, **0.9663**.

Step 5 Recurse on the two impure income bands

Only the classes present in a branch get a column.

Income = \$15–35K rows 2, 3, 12, 14 2 high, 2 moderate

$Info = 1.0000$

Attribute	Value (rows)	high	mod	Info	Gain
Credit history	unknown (2,3)	1	1	1.0000	
	good (12)	0	1	0	
	bad (14)	1	0	0	0.5000
Debt	high (2,12,14)	2	1	0.9183	
	low (3)	0	1	0	0.3113
Collateral	none (2,3,12,14)	2	2	1.0000	0.0000

Credit history: residual $\frac{2}{4}(1.0000) = 0.5000$, gain

$1.0000 - 0.5000 = 0.5000$. Debt: residual $\frac{3}{4}(0.9183) = 0.6887$, gain

$1.0000 - 0.6887 = 0.3113$. Collateral: residual 1.0000, gain

$1.0000 - 1.0000 = 0.0000$. **Split on Credit history.**

Income = \$15–35K, Credit history = unknown rows 2, 3 1

high, 1 moderate $Info = 1.0000$

Attribute	Value (rows)	high	mod	Info	Gain
Debt	high (2)	1	0	0	
	low (3)	0	1	0	1.0000
Collateral	none (2,3)	1	1	1.0000	0.0000

Debt: residual 0, gain $1.0000 - 0.0000 = 1.0000$. Collateral: residual

1.0000 , gain $1.0000 - 1.0000 = 0.0000$. **Split on Debt.**

Income = over \$35K rows 5, 6, 8, 9, 10, 13 1 moderate, 5 low

$Info = 0.6500$

Attribute	Value (rows)	mod	low	Info	Gain
Credit history	unknown (5,6)	0	2	0	
	bad (8)	1	0	0	
	good (9,10,13)	0	3	0	0.6500
Collateral	none (5,9,13)	0	3	0	
Debt	adequate (6,8,10)	1	2	0.9183	0.1908
	low (5,6,8,9)	1	3	0.8113	
	high (10,13)	0	2	0	0.1091

Credit history: residual 0, gain $0.6500 - 0.0000 = 0.6500$. Collateral:

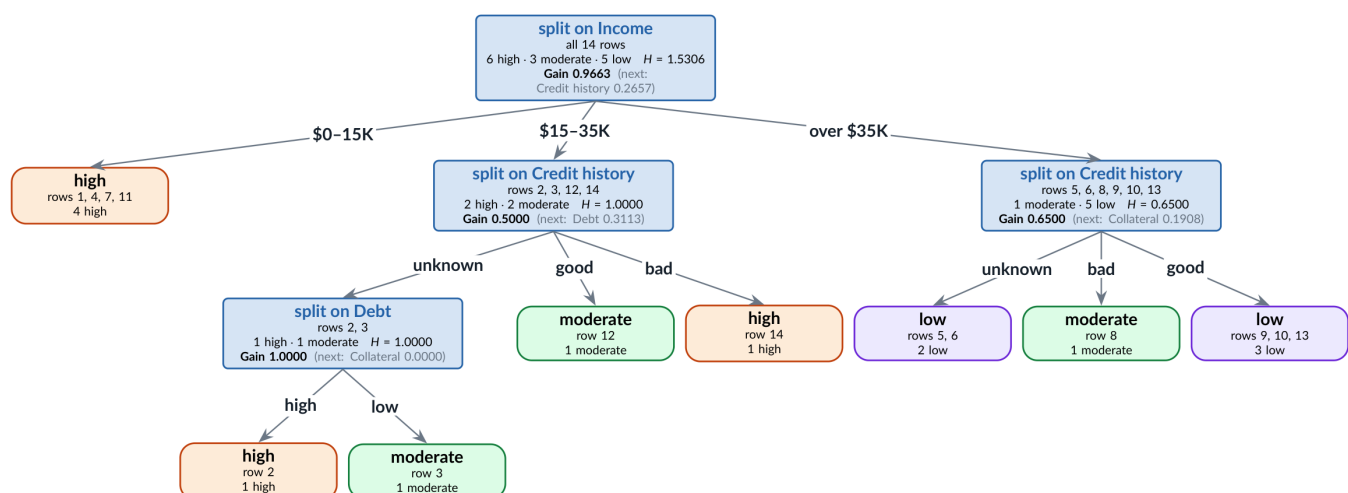
residual $\frac{3}{6}(0.9183) = 0.4592$, gain $0.6500 - 0.4592 = 0.1908$. Debt:

residual $\frac{4}{6}(0.8113) = 0.5409$, gain $0.6500 - 0.5409 = 0.1091$. **Split on**

Credit history.

The tree, 3 levels of tests and 8 leaves, one rule per leaf: Income = \$0–15K → high; Income = \$15–35K, Credit history = unknown, Debt = high → high; Income = \$15–35K, Credit history = unknown, Debt = low → moderate; Income = \$15–35K, Credit history = good → moderate; Income = \$15–35K, Credit history = bad → high; Income = over \$35K, Credit history = unknown → low; Income = over \$35K, Credit history = bad → moderate; Income = over \$35K, Credit history = good → low. Collateral never appears: **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

The finished tree: credit risk (rows are the table's SN)



Reading the tree

- 8 leaves; classifies all 14 tuples correctly.
- **Income** alone decides the \$0–15K band (all high risk).
- **Credit history** is tested twice, once under each other income band: an attribute used on one path can be reused on another.
- Only the \$15–35K, unknown-history tuples need a third test, **Debt**.

The 3-mark part: evaluating the model

- **Do not evaluate on the training set** – this tree scores 100 % there by construction, which measures nothing.
- Hold out data, or better **stratified 10-fold cross-validation** (§3.7).
- Build a **confusion matrix** on the held-out tuples, then report **accuracy**, and per class **precision, recall** and F_1 . With three classes, report them **per class** – a single accuracy hides which risk band is being missed.
- Mention **pruning** and a validation set, since a fully grown ID3 tree overfits.

Problem 1.4 – root node only 2+8

QUESTION AS PRINTED 2081 Bhadra · Q5 · [2+8]

State an algorithm for constructing a decision tree. Using ID3 identify the root node for the following data set.

Gender	Car Ownership	Travel Cost	Income Level	Transport Mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	High	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

A different 10-row table from Problem 1.2, despite looking almost identical: four rows differ. **Only the root is asked for**, so stop after Step 4; do not grow the whole tree.

Step 1 Entropy of the whole set

10 rows: 4 Bus, 3 Train, 3 Car.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.5710}$$

Steps 2 and 3 Split on each attribute in turn class columns (Bus, Train, Car)

Attribute	Value	Bus	Train	Car	$ D_j $	$Info(D_j)$	$\frac{ D_j }{10} Info(D_j)$
Gender	Male	3	1	1	5	1.3710	$\frac{5}{10}(1.3710) = 0.6855$
	Female	1	2	2	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
$Info_{Gender}(D) = 1.4465$							Gain = $1.5710 - 1.4465 = 0.1245$
Car ownership	0	2	1	0	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
	1	2	2	1	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	2	0	0	2	2	0	0
$Info_{Car ownership}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$
Travel cost	Cheap	4	1	0	5	0.7219	$\frac{5}{10}(0.7219) = 0.3609$
	Standard	0	2	0	2	0	0
	Expensive	0	0	3	3	0	0
$Info_{Travel cost}(D) = 0.3609$							Gain = $1.5710 - 0.3609 = 1.2101$
Income level	Low	2	0	0	2	0	0
	Medium	2	2	1	5	1.5219	$\frac{5}{10}(1.5219) = 0.7610$
	High	0	1	2	3	0.9183	$\frac{3}{10}(0.9183) = 0.2755$
$Info_{Income level}(D) = 1.0365$							Gain = $1.5710 - 1.0365 = 0.5345$

Step 4 Compare

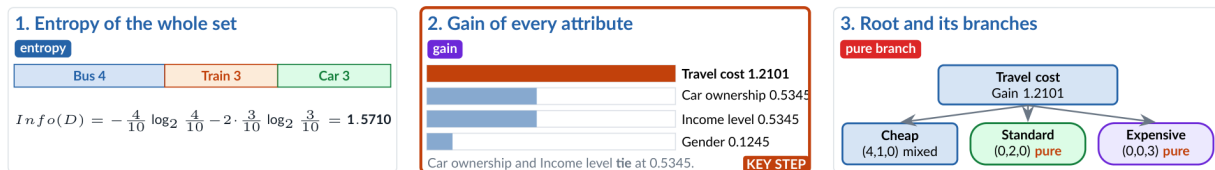
Attribute	Gain
Travel cost	1.2101
Car ownership	0.5345 tie with Income level
Income level	0.5345 tie with Car ownership
Gender	0.1245

Answer. The root attribute is **Travel cost**, with the highest information gain, **1.2101**.

Why Travel cost wins

- Two of its three branches are already **pure** (Standard all Train, Expensive all Car), so it removes most of the uncertainty in one test. **Say that**; do not just quote the number.
- Its gain is more than double the next best.
- Car ownership and Income level tie exactly.** Note it: had the question asked for the second level, you would have to break that tie explicitly.

Step by step: the root node, and why it wins



M2. Gini index

Method

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- Gini uses BINARY splits only.** For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

QUESTION AS PRINTED 2075 Chaitra · Q3 · [6+4]

Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree?

QUESTION AS PRINTED 2071 Chaitra · Q3 · [10]

What is a decision tree? Explain Gini Index with suitable example.

Neither paper prints a dataset – the “suitable example” is yours to supply. The 14-tuple *buys_computer* set below is the safe one to reuse. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$
- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

Step by step: Gini on two candidate binary splits

1. Gini of the whole set

Gini

Yes 9	No 5
-------	------

$$1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

2. Income

binary split

{low, medium}: 10 tuples

7	3	0.420
---	---	-------

{high}: 4 tuples

2	2	0.500
---	---	-------

$$Gini_A = 0.443, \Delta = 0.016$$

3. Age

binary split pure branch

{youth, senior}: 10 tuples

5	5	0.500
---	---	-------

{middle_aged}: 4 tuples

4	0 pure
---	--------

$$Gini_A = 0.357, \Delta = 0.102$$

KEY STEP

4. Largest drop wins

largest wins

Income Δ 0.016Age Δ 0.102

Age is chosen, as by information gain.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
 - **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

QUESTION AS PRINTED 2072 Chaitra · Q3 · [10]

Predict Class label using naive Bayesian classifier for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$ using the following data set.

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}.$

- Matches Han & Kamber 3rd ed Example 8.4.

Step by step: Naive Bayes for $X = (\text{youth, medium, yes, fair})$

1. Priors

prior

Yes 9	No 5
-------	------

$$P(\text{Yes}) = 9/14 = 0.643$$

$$P(\text{No}) = 5/14 = 0.357$$

2. Count within each class

class count

	Yes	No
age = youth	2/9	3/5
income = med	4/9	2/5
student = yes	6/9	1/5
credit = fair	6/9	2/5

3. Multiply

product

$$P(X|\text{Yes}) = 0.222 \cdot 0.444 \cdot 0.667 \cdot 0.667$$

$$= 0.0439$$

$$P(X|\text{No}) = 0.600 \cdot 0.400 \cdot 0.200 \cdot 0.400$$

$$= 0.0192$$

Attributes treated as independent given the class.

4. Times the prior, compare

largest wins

Yes: 0.028
No: 0.007

 $\times P(C)$, then compare. $P(X)$ cancels.

buys_computer = Yes

KEY STEP

Problem 3.2 6

QUESTION AS PRINTED 2079 Bhadra · Q4 · [6]

Suppose you have a test record " $X = (\text{Home Owner} = \text{No}, \text{Marital Status} = \text{Married}, \text{Income} = \$120\text{K})$ ". Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given

Defaulted: **No** = 7, **Yes** = 3. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** — effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = 1.64 \times 10^{-3}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx 4.8 \times 10^{-11}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

Step by step: a zero count, the Laplace fix and a Gaussian attribute

1. Priors

prior

No 7	Yes 3
------	-------

$P(\text{No}) = 0.7$
 $P(\text{Yes}) = 0.3$

2. Categorical counts

class count

HO = No |No |Yes
 4/7 3/3

Married |No |Yes
 4/7 0/3

A single zero makes $P(X|\text{Yes}) = 0$ whatever else holds.

3. Laplace fix

Laplace

$P(\text{Married}|\text{Yes}) = \frac{0+1}{3+3} = 0.167$
 $P(\text{HO=No}|\text{Yes}) = \frac{3+1}{3+3} = 0.8$

+1 per count, +k per total (k = values the attribute takes: 3, 2).

4. Income: a Gaussian per class

Gaussian

5. Combine, compare

product largest wins

No: $0.571 \cdot 0.571 \cdot 0.00719 \cdot 0.7$
 $= 1.64 \times 10^{-3}$

Yes: $0.8 \cdot 0.167 \cdot 1.2 \times 10^{-9} \cdot 0.3$
 $\approx 4.8 \times 10^{-11}$

Defaulted = No, larger by about 10^7 .

M4. k -Nearest Neighbour

Method

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** – that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (b) · [3+6]

When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for $K = 3$ with Euclidean distance matrix.

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

Column names are

the paper's own spellings.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

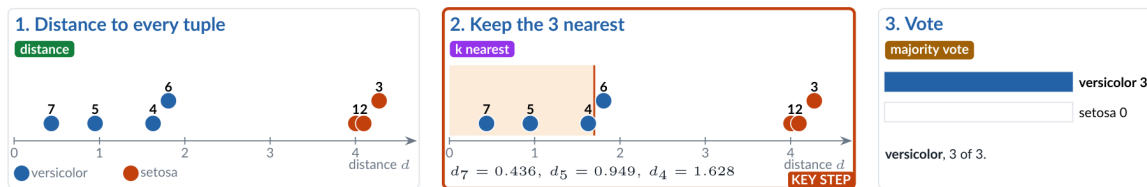
- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** → versicolor, versicolor, versicolor.

Final answer

versicolor (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

Step by step: 3-NN on the flower



M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. **verified/added**

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1 – accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

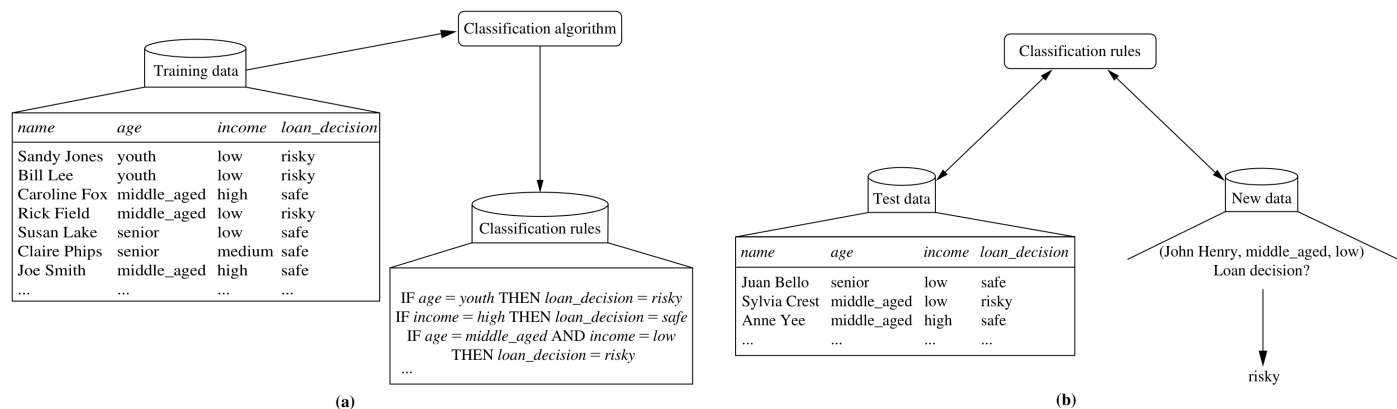
QUESTION AS PRINTED 2074 Chaitra · Q3 · [2+3+5]

Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model.

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

“Positive” is the paper’s own spelling.

The [2] block diagram



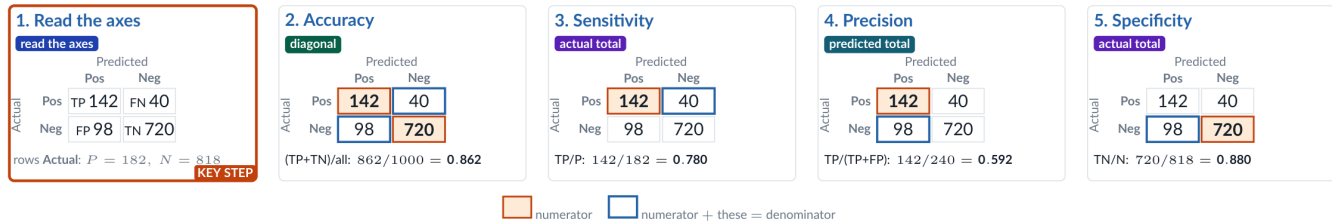
(a) learning: training data → algorithm → model; (b) classification: model checked on test data, then applied to new data. The precision–recall part is Ch 3 theory, §3.7. H&K Fig 8.1.

Step 1: read the four cells

rows are **Actual here**, so $TP = 142$, $FN = 40$, $FP = 98$, $TN = 720 \rightarrow P = 182$, $N = 818$, total = 1000.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
Sensitivity = TPR	TP/P	$142/182$	0.780
Precision	$TP/(TP + FP)$	$142/240$	0.592
Specificity	TN/N	$720/818$	0.880

Step by step: read the axes, then each metric's cells (rows Actual)



Problem 5.2 4

QUESTION AS PRINTED 2078 Bhadra · Q6 · [4]

Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier.

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

Same four numbers as 5.1, but the rows are Predicted here.

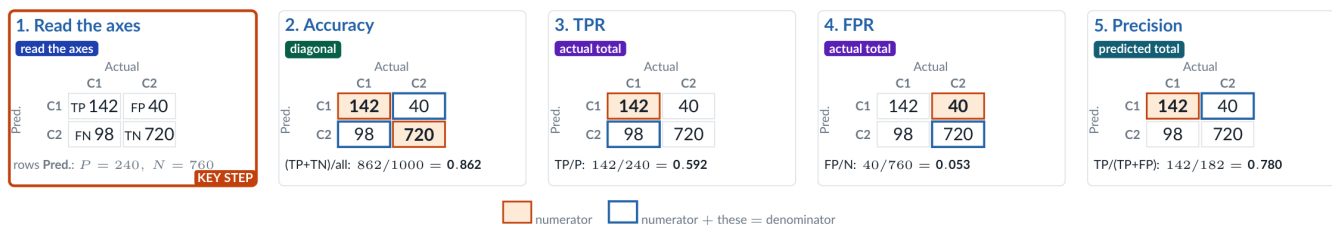
Step 1: read the four cells

rows are Predicted here, so $TP = 142$, $FP = 40$, $FN = 98$, $TN = 720 \rightarrow P = 142 + 98 = 240$, $N = 40 + 720 = 760$.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(142 + 720)/1000$	0.862
TPR	TP/P	$142/240$	0.592
FPR	FP/N	$40/760$	0.053
Precision	$TP/(TP + FP)$	$142/182$	0.780

- Accuracy is **unchanged from 5.1** – the diagonal is the same either way round.
- **Compare with 5.1**: sensitivity and precision have exchanged values. This is the orientation trap, live.

Step by step: read the axes, then each metric's cells (rows Predicted)



Problems 5.3 and 5.4 10

QUESTION AS PRINTED 2075 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

Calculate: Accuracy, Sensitivity, Specificity and Precision.

QUESTION AS PRINTED 2072 Chaitra · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall

Same four numbers, opposite orientation – that is the whole point of solving them side by side.

Both at once, so the orientation trap is visible

- 75 Ash, rows are **Actual**: $TP = 21$, $FN = 6$, $FP = 7$, $TN = 41$; $P = 27$, $N = 48$.
- 72 Ch, rows are **Predicted**: $TP = 21$, $FP = 6$, $FN = 7$, $TN = 41$; $P = 28$, $N = 47$.
- Both total 75.

Metric	Formula	75 Ash (rows Actual)		72 Ch (rows Predicted)	
		Subst.	Value	Subst.	Value
Accuracy	$(TP+TN)/(P+N)$	62/75	0.827	62/75	0.827
Sensitivity = recall	TP/P	21/27	0.778	21/28	0.750
Specificity	TN/N	41/48	0.854	41/47	0.872
Precision	$TP/(TP+FP)$	21/28	0.750	21/27	0.778

- **Accuracy is identical, every other metric moves.** Sensitivity and precision simply swap. Read the row/column labels before touching the arithmetic.

Step by step: read the axes, then each metric's cells (the same four numbers, both ways round)

1. Read the axes
read the axes

		Predicted				Actual	
		C1	C2			C1	C2
Actual	75 Ash	TP 21	FN 6	Pred.	72 Ch	TP 21	FP 6
		C2	FP 7			C2	TN 41

rows Actual: $P = 27$, $N = 48$ rows Pred: $P = 28$, $N = 47$
KEY STEP

2. Accuracy
diagonal

		Predicted				Actual	
		C1	C2			C1	C2
Actual	75 Ash	21	6	Pred.	72 Ch	21	6
		C2	7			C2	41

$(TP+TN)/all: 62/75 = 0.827$ $62/75 = 0.827$

3. Sensitivity = recall
actual total

		Predicted				Actual	
		C1	C2			C1	C2
Actual	75 Ash	21	6	Pred.	72 Ch	21	6
		C2	7			C2	41

$TP/P: 21/27 = 0.778$ $21/28 = 0.750$

4. Specificity
actual total

		Predicted				Actual	
		C1	C2			C1	C2
Actual	75 Ash	21	6	Pred.	72 Ch	21	6
		C2	7			C2	41

$TN/N: 41/48 = 0.854$ $41/47 = 0.872$

5. Precision
predicted total

		Predicted				Actual	
		C1	C2			C1	C2
Actual	75 Ash	21	6	Pred.	72 Ch	21	6
		C2	7			C2	41

$TP/(TP+FP): 21/28 = 0.750$ $21/27 = 0.778$

numerator
 numerator + these = denominator

Problem 5.5 10

QUESTION AS PRINTED 2074 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

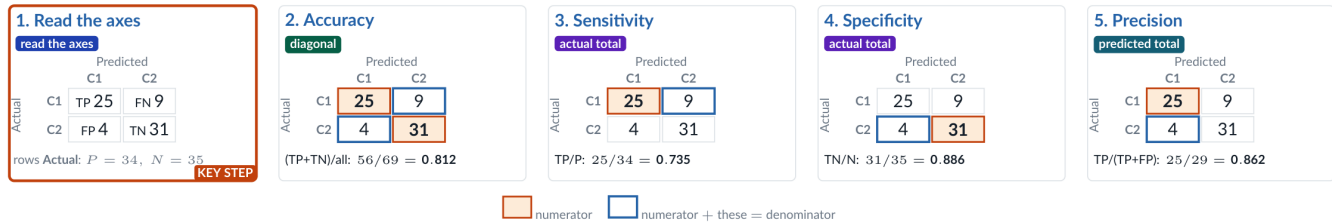
Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision

Step 1: read the four cells

rows are Actual, so $TP = 25$, $FN = 9$, $FP = 4$, $TN = 31 \rightarrow P = 34$, $N = 35$, total 69.

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	56/69	0.812
Sensitivity	TP/P	25/34	0.735
Specificity	TN/N	31/35	0.886
Precision	$TP/(TP + FP)$	25/29	0.862

Step by step: read the axes, then each metric's cells (rows Actual)



Problem 5.6 3+5

QUESTION AS PRINTED 2080 Baishakh · Q3 · [3+5]

In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity.

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

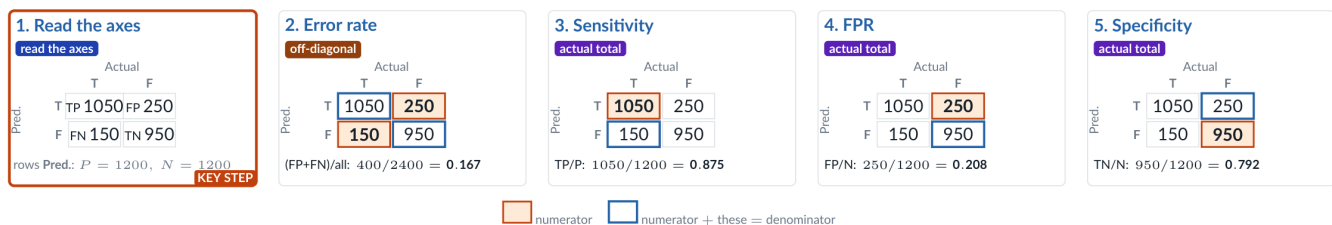
Step 1: read the four cells

rows are Predicted here, so $TP = 1050$, $FP = 250$, $FN = 150$, $TN = 950 \rightarrow P = 1200$, $N = 1200$, total 2400.

Metric	Formula	Substitution	Value
Classification error	$(FP + FN)/(P + N)$	400/2400	0.167
Sensitivity	TP/P	1050/1200	0.875
False alarm rate (FPR)	FP/N	250/1200	0.208
Specificity	TN/N	950/1200	0.792

– Check: $FPR + specificity = 0.208 + 0.792 = 1 \checkmark$

Step by step: read the axes, then each metric's cells (rows Predicted)



Problem 5.7 3+6

QUESTION AS PRINTED 2073 Shrawan · Q4 · [1+3+6]

Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix.

	A	B
A	20	5
B	10	40

Classify, A = Yes, B = No

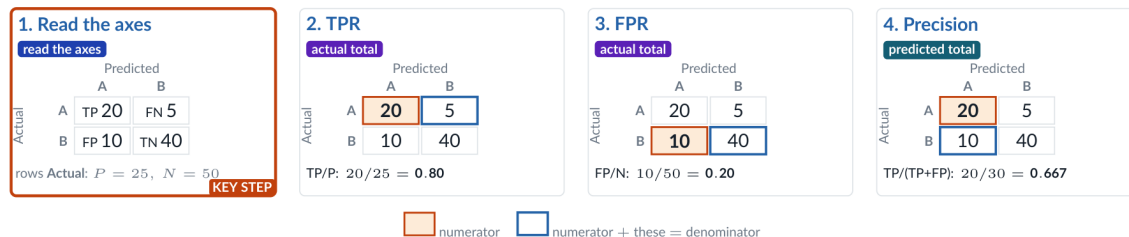
Step 1: read the four cells

A = Yes is the positive class, rows are Actual, so $TP = 20$, $FN = 5$, $FP = 10$, $TN = 40 \rightarrow P = 25$, $N = 50$.

Metric	Formula	Substitution	Value
TPR	TP/P	$20/25$	0.80
FPR	FP/N	$10/50$	0.20
Precision	$TP/(TP + FP)$	$20/30$	0.667

- The ROC point for this classifier is (0.20, 0.80), comfortably above the diagonal.

Step by step: read the axes, then each metric's cells (A = Yes, rows Actual)



Problem 5.8 10

QUESTION AS PRINTED 2070 Chaitra · Q3 · [10]

An input sequence “A A B B B A A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B B A A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A A B B A A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding.

- i) Accuracy ii) Precision iii) True Positive Rate iv) False Positive Rate

Step 1: tabulate Y position by position

(take A as the positive class)

Position	1	2	3	4	5	6	7	8	9	10
Actual	A	A	B	B	B	A	A	A	B	B
Y	A	A	A	A	B	B	A	A	A	B
Category	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Step 2: the two confusion matrices, then the metrics

- X is identical to the actual sequence: $TP = 5$, $TN = 5$, $FP = 0$, $FN = 0$.
- Y from the table above: $TP = 4$, $FN = 1$, $FP = 3$, $TN = 2$; $P = 5$, $N = 5$ (total 10 ✓).

Metric	Formula	Classifier X Subst.	Value	Classifier Y Subst.	Value
Accuracy	$(TP + TN)/(P + N)$	10/10	1.00	6/10	0.60
Precision	$TP/(TP + FP)$	5/5	1.00	4/7	0.571
TPR	TP/P	5/5	1.00	4/5	0.80
FPR	FP/N	0/5	0	3/5	0.60

- X is a perfect classifier** — its ROC point is the top-left corner (0, 1).
- Reading:** Y has decent recall (0.80) but poor precision (0.571) — it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point (0.60, 0.80) is only just above the diagonal, so it is barely better than guessing.

Step by step: tally, build both matrices, then each metric

1. Tally Y (A = positive)

tally

Actual \ Y	A	A	B	B	B	A	A	A	B	B
Y is	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Y: $TP = 4$, $FN = 1$, $FP = 3$, $TN = 2$.
X matches Actual at every position: $TP = 5$, $TN = 5$.

KEY STEP

2. Read the axes

read the axes

Classifier X

Predicted	
A	B
Actual A	TP 5 FN 0
Actual B	FP 0 TN 5

rows Actual: $P = 5$, $N = 5$

Classifier Y

Predicted	
A	B
Actual A	TP 4 FN 1
Actual B	FP 3 TN 2

rows Actual: $P = 5$, $N = 5$

3. Accuracy

diagonal

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

(TP+TN)/all: $10/10 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$6/10 = 0.60$

4. Precision

predicted total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

TP/(TP+FP): $5/5 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$4/7 = 0.571$

5. TPR

actual total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

TP/P: $5/5 = 1.00$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$4/5 = 0.80$

6. FPR

actual total

Classifier X

Predicted	
A	B
Actual A	5 0
Actual B	0 5

FP/N: $0/5 = 0$

Classifier Y

Predicted	
A	B
Actual A	4 1
Actual B	3 2

$3/5 = 0.60$

numerator numerator + these = denominator

Problem 5.9 4

QUESTION AS PRINTED 2079 Baishakh · Q6 · [4]

Given the following confusion matrix, determine Accuracy, Error rate, Sensitivity, Specificity, Precision, Recall of the classifier model.

$n = 165$	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

This paper labels the four cells for you, so there is no orientation trap – just read them off. $P = 105$, $N = 60$, total 165. ✓

Metric	Formula	Substitution	Value
Accuracy	$(TP + TN)/(P + N)$	$(100 + 50)/165$	0.909
Error rate	$(FP + FN)/(P + N)$	$(10 + 5)/165$	0.091
Sensitivity	TP/P	$100/105$	0.952
Specificity	TN/N	$50/60$	0.833
Precision	$TP/(TP + FP)$	$100/110$	0.909
Recall	= sensitivity	$100/105$	0.952

- **Checks:** accuracy + error rate = 1 ✓
- **Recall and sensitivity are the same thing.** The question lists both, so write both rows and say they are equal – do not assume it is a trick.
- Precision equalling accuracy here is a coincidence of the numbers, not a rule.

Step by step: read the axes, then each metric's cells (positive = YES, the second row and column)

1. Read the axes

read the axes

Predicted	
NO	YES
Actual NO	TN 50 FP 10
Actual YES	FN 5 TP 100

rows Actual: $P = 105$, $N = 60$

KEY STEP

2. Accuracy

diagonal

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

(TP+TN)/all: $150/165 = 0.909$

3. Error rate

off-diagonal

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

(FP+FN)/all: $15/165 = 0.091$

4. Sensitivity = recall

actual total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TP/P: $100/105 = 0.952$

5. Specificity

actual total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TN/N: $50/60 = 0.833$

6. Precision

predicted total

Predicted	
NO	YES
Actual NO	50 10
Actual YES	5 100

TP/(TP+FP): $100/110 = 0.909$

numerator numerator + these = denominator

Problem 5.10 – the mislabelled matrix 2+6

QUESTION AS PRINTED 2073 Chaitra · Q4 · [2+6]

What is ID3 algorithm? Calculate TPR, FPR and Accuracy for given confusion matrix.

	Predicted +	Predicted –
Predicted +	100	40
Predicted –	60	300

The paper labels BOTH axes “Predicted” – reproduced exactly as printed. The row stubs should read *Actual*. Say so in the answer, take rows as *Actual*, and state the assumption. That sentence is worth more than the arithmetic.

Reading rows as Actual (the intended matrix)

$TP = 100, FN = 40, FP = 60, TN = 300 \rightarrow P = 140, N = 360, \text{total } 500.$

Metric	Formula	Substitution	Value
TPR	TP/P	$100/140$	0.714
FPR	FP/N	$60/360$	0.167
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

If instead the rows really are Predicted

$TP = 100, FP = 40, FN = 60, TN = 300 \rightarrow P = 160, N = 340.$

Metric	Formula	Substitution	Value
TPR	TP/P	$100/160$	0.625
FPR	FP/N	$40/340$	0.118
Accuracy	$(TP + TN)/(P + N)$	$400/500$	0.800

- **Accuracy is 0.800 either way** – the diagonal does not move when you transpose. TPR and FPR do.
- **The safe exam answer:** give the first table, and add one line noting the paper's label defect and what the numbers would become under the other reading. Full marks either way, and it shows you understand the matrix rather than a memorised formula.

Step by step: read the axes, then each metric's cells (both readings of the mislabelled matrix)

1. Read the axes
read the axes

	+ Predicted	–			+ Predicted	–	
Actual	+	TP 100	FN 40	Pred.	+	TP 100	FP 40
	–	FP 60	TN 300		–	FN 60	TN 300

rows Actual: $P = 140, N = 360$ rows Predicted: $P = 160, N = 340$ KEY STEP

2. TPR
actual total

	+ Predicted	–			+ Predicted	–	
Actual	+	100	40	Pred.	+	100	40
	–	60	300		–	60	300

TP/P: $100/140 = 0.714$ 100/160 = 0.625

3. FPR
actual total

	+ Predicted	–			+ Predicted	–	
Actual	+	100	40	Pred.	+	100	40
	–	60	300		–	60	300

FP/N: $60/360 = 0.167$ 40/340 = 0.118

4. Accuracy
diagonal

	+ Predicted	–			+ Predicted	–	
Actual	+	100	40	Pred.	+	100	40
	–	60	300		–	60	300

(TP+TN)/all: $400/500 = 0.800$ 400/500 = 0.800

numerator
 numerator + these = denominator

M6. ANN gradients (backpropagation)

Method

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

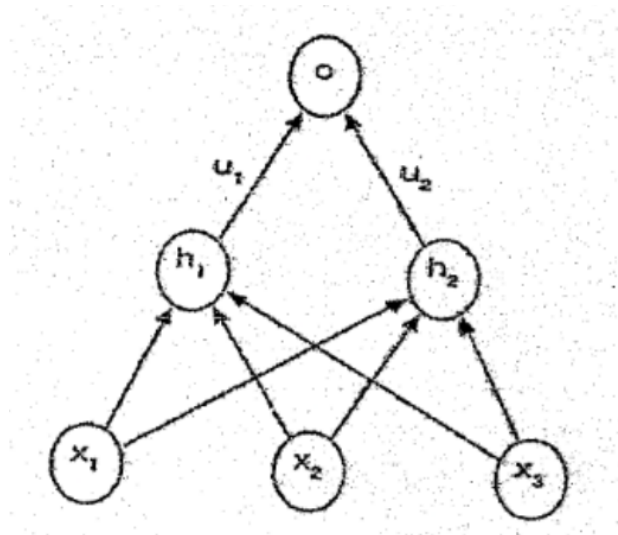
QUESTION AS PRINTED 2076 Chaitra · Q6 · [2+3+4]

Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs x_1, x_2 and x_3 connected to a hidden layer consisting of two nodes h_1 and h_2 . The weight of the edge connecting x_i to h_j is w_{ji} . The two hidden nodes are connected to the output node o . The weight of the edge connecting the hidden node h_i to the output node o is u_i . The activation functions at hidden and output layers is set to sigmoid function defined as follows:

$$\sigma(\theta) = \frac{1}{1 + e^{-\theta}}$$

Using the target output t , the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- Using the symbols given above, compute the activation at h_1 .
- Compute the gradient of the loss with respect to the output o .
- Compute the gradient of the loss with respect to the weight w_{12} .

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + e^{-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3)}}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

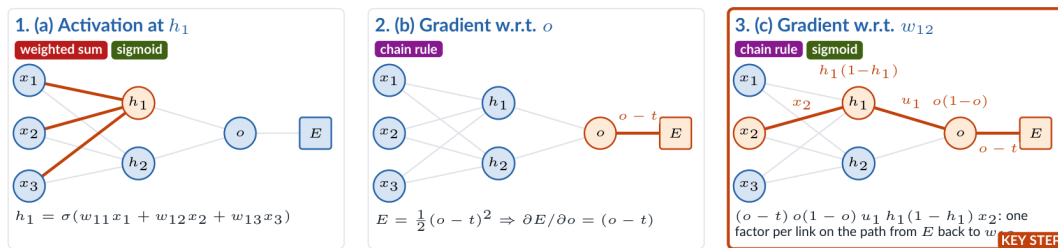
- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.
- Write $net_o = u_1h_1 + u_2h_2$ and $net_{h_1} = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2$$

- **Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- Weight update: $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Step by step: forward to h_1 , then back along one path



Ch 4 – Association Analysis – Numerical Problems & Solutions

157 numerical marks, 8.1% of the archive, 40% of this chapter · 5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- M1 Apriori: candidate generation, pruning, frequent itemsets
- M2 Rule generation from frequent itemsets
- M3 FP-tree construction and FP-Growth mining
- M4 Pattern evaluation: lift and the correlation measures
- M5 Subsequence enumeration

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-4 numerical below is built from

Move	What it says (where: problem, step; orange = key step)	Move	What it says (where: problem, step; orange = key step)
threshold to count	Turn min support into a COUNT first: $\lceil \text{fraction} \times N \rceil$. 1.3 1; 1.4/1.5 1; 1.6 1; 1.7 1; 1.8 1; 1.9 1; 1.10 1	node-links	Follow the item's node-links to every node carrying it. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
count and keep	Count each candidate over ALL transactions; keep those $\geq$ the min count. 1.1 1,2,3,4; 1.2 1,2,3; 1.3 2,3; 1.4/1.5 2,3,4,5; 1.6 2,3; 1.7 2,3,4; 1.8 2,3,4; 1.9 2,3; 1.10 2,3	prefix paths	Each node's whole root path, tagged with THAT node's count: the pattern base. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
join	Join two frequent $(k-1)$ -itemsets sharing their first $k-2$ items. 1.1 2,3,4,5; 1.2 2,3,4; 1.3 3,4; 1.4/1.5 3,4,5; 1.6 3; 1.7 3,4,5; 1.8 3,4,5; 1.9 3,4; 1.10 3,4	conditional tree	Add up the base per item, drop those below the min count, append the item. 3.1 1,2,3,4,5; 3.3 1,2,3,4,5
prune	Drop a candidate if any $(k-1)$ -subset is not frequent. No scan needed. 1.1 3; 1.4/1.5 4,5; 1.7 4; 1.8 4	lift	$P(A \cup B) / (P(A)P(B))$: below 1 is negative correlation, whatever the confidence. 4.1 3
stop	No candidates (or none survive): the run ends. 1.1 5; 1.2 4; 1.3 4; 1.4/1.5 5; 1.6 3; 1.7 5; 1.8 5; 1.9 4; 1.10 4	chi-square	$\sum (obs - exp)^2 / exp$ over the contingency table. 4.1 4
confidence	$\sigma(l) / \sigma(s)$: same numerator for every rule from l . 1.1 6; 1.4 1; 1.5 2; 1.6 4; 1.7 6; 1.8 6; 1.9 5; 1.10 5; M2 1; 3.1 1; 4.1 1,2	item splits	Enumerate how many items come from each element, then choose them. 5.1 1
threshold line	A rule is strong when its confidence is $\geq$ min_conf (equal counts). 1.1 6; 1.4 1; 1.5 2; 1.6 4; 1.7 6; 1.8 6; 1.9 5; 1.10 5; M2 1; 3.1 1	remove duplicates	The same sequence reached by two splits counts once. 5.1 2
F-list	Frequent items sorted by descending count, ties broken by a stated rule. –	keep element order	Subsets of each element, in the original element order, never merged. 5.2 1
reorder	Rewrite each transaction in F-list order, dropping infrequent items. –	common core	Remove one edge from each parent so the rest is identical: the core. 6.1 1
share prefix	A transaction that starts like an existing path increments its counts. 3.1 2,3,4,5; 3.2 2,3,5; 3.3 2,4,5,6; 3.4 2,4,5,6	attach edges	Add each parent's unique edge to the core at every legal vertex. 6.1 2,3
new branch	Where the prefix stops matching, a new child (or root child) is added. 3.1 1,2,4; 3.2 1,2,3,4; 3.3 1,3,4,5; 3.4 1,2,3,4,5	symmetry	Placements that map onto each other by a symmetry are one candidate. 6.1 2,3

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $min_sup_count = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq \text{min_sup_count} \rightarrow L_1$.
 2. Join L_{k-1} with itself: combine 2 itemsets only if their first $k - 2$ items match.
 3. Prune: discard a candidate if *any* of its $(k - 1)$ -subsets is missing from L_{k-1} .
 4. Count survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- Show the prune step explicitly – it is where the method marks are.

Problem 1.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [2+7]

What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%.

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

T106 genuinely prints "F, A, C, I, B, A, P" – **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: $C_1 \rightarrow L_1$

count all 18 distinct items over the 7 transactions, min count = 4

It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep	It.	Cnt	Keep
A	6	✓	F	5	✓	K	1	×	O	2	×
B	5	✓	G	1	×	L	2	×	P	6	✓
C	5	✓	H	1	×	M	4	✓	S	2	×
D	2	×	I	3	×	N	2	×	V	3	×
E	1	×	J	1	×						

$L_1 = \{A(6), B(5), C(5), F(5), M(4), P(6)\}$ – 6 items survive of 18.

Step 2: $C_2 \rightarrow L_2$

all $\binom{6}{2} = 15$ pairs from L_1 , counted against D

Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
AB	4	✓	AP	5	✓	BP	4	✓	CP	4	✓
AC	5	✓	BC	3	×	CF	4	✓	FM	3	×
AF	4	✓	BF	3	×	CM	3	×	FP	5	✓
AM	4	✓	BM	2	×				MP	4	✓

$L_2 = \{AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)\}$

Step 3: $C_3 \rightarrow L_3$

join L_2 pairs sharing their first item, then **prune before counting**

Cand.	Subset test (all 2-subsets must be in L_2)	Transactions	Count	Keep
ABC	$BC \notin L_2$	pruned, never counted	—	×
ABF	$BF \notin L_2$	pruned, never counted	—	×
ABM	$BM \notin L_2$	pruned, never counted	—	×
ABP	$AB \checkmark AP \checkmark BP \checkmark \rightarrow$ survives pruning	T101, T105, T106	3	×
ACF	$AC \checkmark AF \checkmark CF \checkmark$	T100, T101, T104, T106	4	✓
ACM	$CM \notin L_2$	pruned, never counted	—	×
ACP	$AC \checkmark AP \checkmark CP \checkmark$	T100, T101, T104, T106	4	✓
AFM	$FM \notin L_2$	pruned, never counted	—	×
AFP	$AF \checkmark AP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓
AMP	$AM \checkmark AP \checkmark MP \checkmark$	T100, T101, T104, T105	4	✓
CFP	$CF \checkmark CP \checkmark FP \checkmark$	T100, T101, T104, T106	4	✓

$L_3 = \{\text{ACF}(4), \text{ACP}(4), \text{AFP}(4), \text{AMP}(4), \text{CFP}(4)\}$

- **Five candidates died at the prune step without a single database scan** – that saving is the whole point of the Apriori property, so show the column.
- ABP is the instructive one: it *passes* pruning and still fails on the count. **Pruning is sufficient to reject, never sufficient to accept.**

Step 4: $C_4 \rightarrow L_4$

join L_3 triples sharing their first two items

Cand.	Joined from	Subset test	Count	Keep
ACFP	ACF + ACP	$ACF \checkmark ACP \checkmark AFP \checkmark CFP \checkmark$	4 (T100, T101, T104, T106)	✓

$L_4 = \{\text{ACFP}(4)\}$; no two 4-itemsets remain to join $\rightarrow C_5 = \emptyset \rightarrow$ **stop**.

Frequent itemsets

Maximal frequent itemset = **{A, C, F, P} with support $4/7 = 57\%$.**

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Step 5: the rules (the question gives confidence = 80 %, so this half is asked)

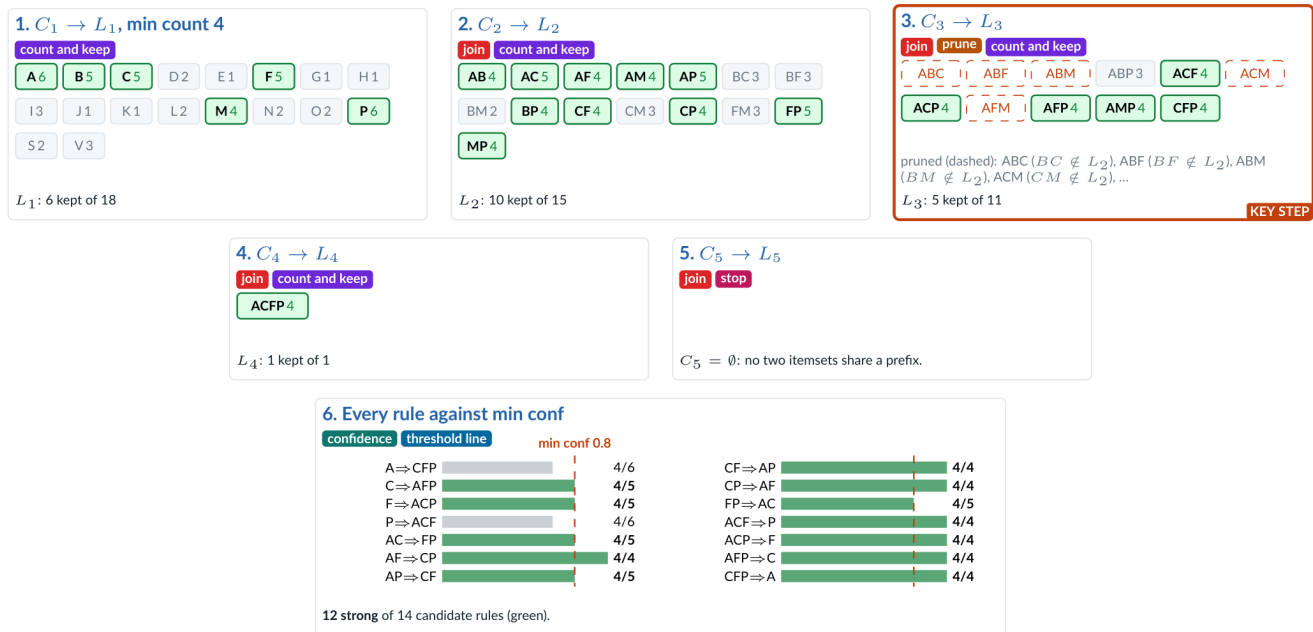
- From the maximal itemset $l = \{A, C, F, P\}$, $\sigma(l) = 4$. For every non-empty proper subset s , output $s \Rightarrow (l - s)$ with $conf = \frac{\sigma(l)}{\sigma(s)} = \frac{4}{\sigma(s)}$.
- $2^4 - 2 = 14$ candidate rules. $conf \geq 0.80$ **needs** $\sigma(s) \leq 5$, so read the answer straight off the antecedent's count.

Rule	$\sigma(s)$	Confidence
$C \Rightarrow AFP$	5	$4/5 = 80\% \checkmark$
$F \Rightarrow ACP$	5	$4/5 = 80\% \checkmark$
$AC \Rightarrow FP$	5	$4/5 = 80\% \checkmark$
$AP \Rightarrow CF$	5	$4/5 = 80\% \checkmark$
$FP \Rightarrow AC$	5	$4/5 = 80\% \checkmark$
$AF \Rightarrow CP$	4	$4/4 = 100\% \checkmark$
$CF \Rightarrow AP$	4	$4/4 = 100\% \checkmark$
$CP \Rightarrow AF$	4	$4/4 = 100\% \checkmark$
$ACF \Rightarrow P$	4	$4/4 = 100\% \checkmark$
$ACP \Rightarrow F$	4	$4/4 = 100\% \checkmark$
$AFP \Rightarrow C$	4	$4/4 = 100\% \checkmark$
$CFP \Rightarrow A$	4	$4/4 = 100\% \checkmark$
$A \Rightarrow CFP$	6	$4/6 = 66.7\% \times$
$P \Rightarrow ACF$	6	$4/6 = 66.7\% \times$

Reading the table

- **12 of the 14 rules are strong** at $\text{min_conf} = 80\%$.
- The only two failures are the ones with A or P alone on the left. Both occur in 6 of the 7 transactions, so they are **too common to predict anything**: a frequent antecedent divides the confidence down.
- **The general lesson**: confidence falls as the antecedent gets more frequent, and rises as it gets more specific. Longer left-hand sides almost always score higher.
- Seven rules here sit at **100%**. By §4.1 those are the *least* interesting of the twelve, because $\{A, F\}$ never appears without C and P anywhere in this database $\rightarrow$ nothing actionable. **Say so if the question asks you to comment**, do not just report the highest number.

Step by step: Apriori level by level, then the 14 rules



Problem 1.2 5

QUESTION AS PRINTED 2080 Baishakh · Q5 · [4+5]

Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2.

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: $C_1 \rightarrow L_1$

min count = 2

Item	In	Count	Keep
A	10, 30	2	✓
B	20, 30, 40	3	✓
C	10, 20, 30	3	✓
D	10	1	×
E	20, 30, 40	3	✓

 $L_1 = \{A(2), B(3), C(3), E(3)\}$ **Step 2:** $C_2 \rightarrow L_2$ all $\binom{4}{2} = 6$ pairs from L_1

Pair	In	Count	Keep
AB	30	1	×
AC	10, 30	2	✓
AE	30	1	×
BC	20, 30	2	✓
BE	20, 30, 40	3	✓
CE	20, 30	2	✓

 $L_2 = \{AC(2), BC(2), BE(3), CE(2)\}$ **Step 3:** $C_3 \rightarrow L_3$ join L_2 on a shared first item, prune, then count

Joined from	Candidate	Subset test	In	Count	Keep
BC + BE	BCE	BC ✓ BE ✓ CE ✓ $\rightarrow$ survives	20, 30	2	✓
AC + —	none	AC is the only pair starting with A, so it has nothing to join with	—	—	—

 $L_3 = \{BCE(2)\}$; C_4 needs two 3-itemsets sharing a prefix and there is only one $\rightarrow$ **stop**.**Final answer**

Frequent itemsets: A, B, C, E, AC, BC, BE, CE, BCE.

Step by step: Apriori level by level**1.** $C_1 \rightarrow L_1$, min count 2

count and keep

A 2 B 3 C 3 D 1 E 3

 L_1 : 4 kept of 5**2.** $C_2 \rightarrow L_2$

join count and keep

AB 1 AC 2 AE 1 BC 2 BE 3 CE 2

 L_2 : 4 kept of 6**3.** $C_3 \rightarrow L_3$

join count and keep

BCE 2

 L_3 : 1 kept of 1**4.** $C_4 \rightarrow L_4$

join stop

 $C_4 = \emptyset$: no two itemsets share a prefix.

KEY STEP

Problem 1.3 10

QUESTION AS PRINTED 2070 Chaitra · Q4 · [10]

Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database.

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

4 transactions, 50% $\rightarrow$ min count = 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1)× AC(2)✓ AD(1)× BC(1)× BD(2)✓ CD(1)×
- $L_2 = \{AC(2), BD(2)\}$
- C_3 : AC and BD share no common prefix $\rightarrow$ **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Step by step: Apriori level by level

1. Min support as a count

threshold to count

4 transactions $\times$ 50% = 2

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A 2 B 3 C 2 D 3 E 1 F 1

L_1 : 4 kept of 6

3. $C_2 \rightarrow L_2$

join count and keep

AB 1 AC 2 AD 1 BC 1 BD 2 CD 1

L_2 : 2 kept of 6

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$: no two itemsets share a prefix.

KEY STEP

Problems 1.4 and 1.5 5+3 8

QUESTION AS PRINTED 2076 Chaitra · Q7 · [5+3]

Consider the transaction data shown in the following table from a fast food restaurant.

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1–M5). Assume that the minimum support is $2/9$ and the minimum confidence is $7/9$.

- Apply the Apriori algorithm to the dataset of transactions and identify all frequent k -itemsets.
- Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

QUESTION AS PRINTED 2074 Ashwin · Q5 · [8]

Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm.

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

Take minimum support = 20%, minimum confidence 80%

Identical 9-transaction table, and both thresholds work out to a count of 2, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
M1	T1,4,5,7,8,9	6	✓
M2	T1,2,3,4,6,8,9	7	✓
M3	T3,5,6,7,8,9	6	✓
M4	T2,4	2	✓
M5	T1,8	2	✓

Nothing is dropped — all 5 items clear a count of 2.

Step 2: $C_2 \rightarrow L_2$

all $\binom{5}{2} = 10$ pairs

Pair	In	Count	Keep
M1M2	T1, T4, T8, T9	4	✓
M1M3	T5, T7, T8, T9	4	✓
M1M4	T4	1	×
M1M5	T1, T8	2	✓
M2M3	T3, T6, T8, T9	4	✓
M2M4	T2, T4	2	✓
M2M5	T1, T8	2	✓
M3M4	—	0	×
M3M5	T8	1	×
M4M5	—	0	×

$L_2 = \{M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)\}$

Step 3: $C_3 \rightarrow L_3$

join two L_2 pairs **only when both are in L_2 and they share their first item**

Joined from	Candidate	Subset test	In	Count	Keep
M1M2 + M1M3	M1M2M3	M1M2 ✓ M1M3 ✓ M2M3 ✓	T8, T9	2	✓
M1M2 + M1M5	M1M2M5	M1M2 ✓ M1M5 ✓ M2M5 ✓	T1, T8	2	✓
M1M3 + M1M5	M1M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M4	M2M3M4	M3M4 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M3 + M2M5	M2M3M5	M3M5 $\notin L_2 \rightarrow$ pruned	—	—	×
M2M4 + M2M5	M2M4M5	M4M5 $\notin L_2 \rightarrow$ pruned	—	—	×

$L_3 = \{M1M2M3(2), M1M2M5(2)\}$

- Note what is not in the table:** M1M2M4 never appears, because generating it would need M1M4, and M1M4 failed at step 2. **A candidate is only ever built from two surviving parents** — it is not built and then pruned.

Step 4: $C_4 \rightarrow L_4$

Joined from	Candidate	Subset test	Count	Keep
M1M2M3 + M1M2M5	M1M2M3M5	M1M3M5 $\notin L_3 \rightarrow$ pruned	—	×

$L_4 = \emptyset \rightarrow$ **stop**.

Step by step: Apriori level by level (both papers)**1. Min support as a count**

threshold to count

$2/9$ of $9 = 2$; 20% of $9 = 1.8 \rightarrow 2$

2. $C_1 \rightarrow L_1$, min count 2

count and keep

M1 6 M2 7 M3 6 M4 2 M5 2

L_1 : 5 kept of 5

3. $C_2 \rightarrow L_2$

join count and keep

M1M2 4 M1M3 4 M1M4 1 M1M5 2 M2M3 4
M2M4 2 M2M5 2 M3M4 0 M3M5 1 M4M5 0

L_2 : 6 kept of 10

4. $C_3 \rightarrow L_3$

join prune count and keep

M1M2M3 2 M1M2M5 2 M1M3M5
M2M3M4 M2M3M5 M2M4M5

pruned (dashed): M1M3M5 ($M3M5 \notin L_2$), M2M3M4 ($M3M4 \notin L_2$), M2M3M5 ($M3M5 \notin L_2$), M2M4M5 ($M4M5 \notin L_2$)
 C_3 : 2 kept of 6

KEY STEP

5. $C_4 \rightarrow L_4$

join prune count and keep stop

M1M2M3M5

pruned (dashed): M1M2M3M5 ($M1M3M5 \notin L_3$)

$L_4 = \emptyset$: nothing survives

The one idea that decides every rule below

- Every rule out of an itemset l has the *same* numerator: $conf(s \Rightarrow l - s) = \sigma(l)/\sigma(s)$. **Only the antecedent's count moves.**

- So a rule is strong exactly when **the antecedent is nearly as rare as the whole itemset** → when the antecedent hardly ever appears without the consequent.
- Here $\sigma(M4) = \sigma(M5) = 2$, and both *always* appear with their partners. **Expect every strong rule to have M4 or M5 on the left**, and check that prediction against the tables. Nothing else can win, because M1, M2 and M3 each occur 6–7 times while no itemset containing them has support above 4.

1.4 Rules of the form $X \wedge Y \Rightarrow Z$, $\min \text{conf} = 7/9 = 0.778$

- That form means a **2-item antecedent and a 1-item consequent**, so only the **3-itemsets** in L_3 can produce one. Two of them, three rules each → **6 candidates, and they are all of them**.

From	Rule	$\sigma(XY)$	$\text{Conf} = 2/\sigma(XY)$
M1M2M3 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 ×
	M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 ×
	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 ×
M1M2M5 $\sigma = 2$	M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 ×
	M1$\wedge$M5$\Rightarrow$M2	2	1.00 ✓
	M2$\wedge$M5$\Rightarrow$M1	2	1.00 ✓

Answer to 1.4

- **2 strong rules, both at 100 % confidence:**
M1 $\wedge$ M5 $\Rightarrow$ M2 and M2 $\wedge$ M5 $\Rightarrow$ M1.
- Every rule out of M1M2M3 fails, and all three fail *identically* at 0.50, because all three of its 2-subsets happen to have $\sigma = 4$.
- The two winners are exactly the ones predicted above: **M5 is in the antecedent**, and M5 occurs only in T1 and T8, both of which also hold M1 and M2.

1.5 All association rules, $\min \text{conf} = 0.80$

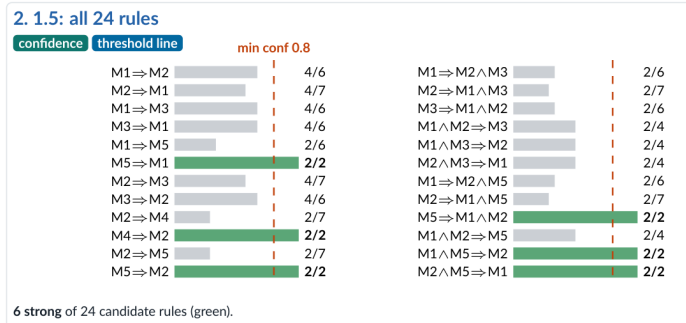
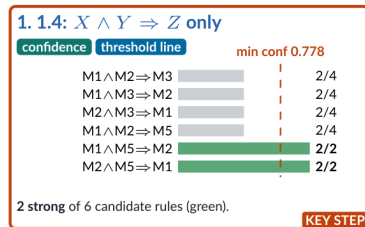
- Now every frequent itemset with $|I| \geq 2$ contributes: six 2-itemsets (2 rules each) and two 3-itemsets ($2^3 - 2 = 6$ rules each) → **12 + 12 = 24 candidate rules**. All 24 are listed, none is skipped.

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
M1M2 (4)	M1 $\Rightarrow$ M2	6	0.67 ×	M2M5 (2)	M2 $\Rightarrow$ M5	7	0.29 ×
	M2 $\Rightarrow$ M1	7	0.57 ×		M5$\Rightarrow$M2	2	1.00 ✓
M1M3 (4)	M1 $\Rightarrow$ M3	6	0.67 ×	M1M2M3 (2)	M1 $\Rightarrow$ M2M3	6	0.33 ×
	M3 $\Rightarrow$ M1	6	0.67 ×		M2 $\Rightarrow$ M1M3	7	0.29 ×
M1M5 (2)	M1 $\Rightarrow$ M5	6	0.33 ×		M3 $\Rightarrow$ M1M2	6	0.33 ×
	M5$\Rightarrow$M1	2	1.00 ✓		M1 $\wedge$ M2 $\Rightarrow$ M3	4	0.50 ×
M2M3 (4)	M2 $\Rightarrow$ M3	7	0.57 ×		M1 $\wedge$ M3 $\Rightarrow$ M2	4	0.50 ×
	M3 $\Rightarrow$ M2	6	0.67 ×	M1M2M5 (2)	M2 $\wedge$ M3 $\Rightarrow$ M1	4	0.50 ×
M2M4 (2)	M2 $\Rightarrow$ M4	7	0.29 ×		M1 $\Rightarrow$ M2M5	6	0.33 ×
	M4$\Rightarrow$M2	2	1.00 ✓		M2 $\Rightarrow$ M1M5	7	0.29 ×
					M5$\Rightarrow$M1$\wedge$M2	2	1.00 ✓
					M1 $\wedge$ M2 $\Rightarrow$ M5	4	0.50 ×
					M1$\wedge$M5$\Rightarrow$M2	2	1.00 ✓
					M2$\wedge$M5$\Rightarrow$M1	2	1.00 ✓

Answer to 1.5

- **6 strong rules, every one of them at 100 % confidence:**
M5 $\Rightarrow$ M1 M5 $\Rightarrow$ M2 M4 $\Rightarrow$ M2 M5 $\Rightarrow$ M1 $\wedge$ M2 M1 $\wedge$ M5 $\Rightarrow$ M2 M2 $\wedge$ M5 $\Rightarrow$ M1
- The other 18 fail, and **the prediction held**: every survivor has M4 or M5 on the left. M1, M2 and M3 are too frequent to be confident antecedents here, the best any of them manages is 0.67.
- **Cross-check with 1.4**: the two rules found there must reappear here, since 1.4's threshold (0.778) is the tighter of the two. They do, on the last two rows. ✓
- **Worth a closing sentence**: all six sit at 100 %, which by §4.1 makes them *reliable but not surprising* → M4 and M5 are simply never ordered alone. Confidence alone cannot see that; lift would.

Step by step: the rules for each paper's threshold



Problem 1.6 8

QUESTION AS PRINTED 2079 Bhadra · Q5 · [8]

Derive association rule for the following market basket transactions.

Minimum support = 50% Minimum confidence = 80%

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

This one has a surprise. 6 transactions, 50% $\rightarrow$ min count = 3.Step 1: $C_1 \rightarrow L_1$

Item	In	Count	Keep
A	1, 2, 3, 6	4	✓
B	1, 4, 5	3	✓
C	3, 6	2	×
D	2, 5	2	×
E	4, 5, 6	3	✓

 $L_1 = \{A(4), B(3), E(3)\}$

Final answer

- No frequent 2-itemset exists, so no association rule can be derived at 50% support. Say this explicitly – it is the correct answer, not a dead end.
- And it holds for every confidence threshold, including 0%: a rule needs a 2-itemset to split, and there is none. Lowering min_conf cannot rescue it, only lowering min_sup can (§4.2).
- Show you understand why: the data is spread thin, every pair occurs at most twice.
- If min support is relaxed to 33.3% (count 2), the picture changes:

Step 2: $C_2 \rightarrow L_2$ the 3 pairs from L_1

Pair	In	Count	Keep
AB	1	1	×
AE	6	1	×
BE	4, 5	2	×

 $L_2 = \emptyset$ – every pair falls short.

New L_2	Count	Rule	Confidence	Strong at 80 %?
AC	2	$A \Rightarrow C$	$2/4 = 0.50$	×
		$C \Rightarrow A$	$2/2 = 1.00$	✓
BE	2	$B \Rightarrow E$	$2/3 = 0.67$	×
		$E \Rightarrow B$	$2/3 = 0.67$	×

One rule survives: $C \Rightarrow A$ at 100%. Worth adding as a closing remark. verified/added

Step by step: nothing frequent at 50%, then the relaxed run

1. Min support as a count

threshold to count

6 transactions $\times$ 50% = 32. $C_1 \rightarrow L_1$, min count 3

count and keep

A 4 B 3 C 2 D 2 E 3

 L_1 : 3 kept of 53. $C_2 \rightarrow L_2$

join count and keep stop

AB 1 AE 1 BE 2

 $L_2 = \emptyset$: nothing survives

KEY STEP

4. Relaxed to count 2: the rules

confidence

threshold line

min conf 0.8

A $\Rightarrow$ C		2/4
C $\Rightarrow$ A		2/2
B $\Rightarrow$ E		2/3
E $\Rightarrow$ B		2/3

1 strong of 4 candidate rules (green).

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra • Q4 • [2+8]

Explain the term support and confidence in association analysis. Derive association rule for following transaction considering minimum support = 60% and confidence = 80% using Apriori Algorithm.

TID Transaction

T1	A, B, C, D, E, F
T2	B, C, D, E, F, G
T3	A, D, E, H
T4	A, D, F, I, J
T5	B, D, E, K

5 transactions, 60% $\rightarrow$ min count = 3.Step 1: $C_1 \rightarrow L_1$

11 distinct items

Item	In	Cnt	Keep
A	T1, T3, T4	3	✓
B	T1, T2, T5	3	✓
C	T1, T2	2	×
D	T1-T5	5	✓
E	T1, T2, T3, T5	4	✓
F	T1, T2, T4	3	✓
G	T2	1	×
H	T3	1	×
I	T4	1	×
J	T4	1	×
K	T5	1	×

 $L_1 = \{A(3), B(3), D(5), E(4), F(3)\}$ Step 2: $C_2 \rightarrow L_2$ all $\binom{5}{2} = 10$ pairs

Pair	In	Cnt	Keep
AB	T1	1	×
AD	T1, T3, T4	3	✓
AE	T1, T3	2	×
AF	T1, T4	2	×
BD	T1, T2, T5	3	✓
BE	T1, T2, T5	3	✓
BF	T1, T2	2	×
DE	T1, T2, T3, T5	4	✓
DF	T1, T2, T4	3	✓
EF	T1, T2	2	×

 $L_2 = \{AD(3), BD(3), BE(3), DE(4), DF(3)\}$ Step 3: $C_3 \rightarrow L_3$

Joined from	Cand.	Subset test	In	Cnt	Keep
BD + BE	BDE	BD ✓ BE ✓ DE ✓	T1, T2, T5	3	✓
AD + —	none	AD is the only pair starting with A	—	—	—
DE + DF	DEF	EF $\notin L_2 \rightarrow$ pruned	—	—	×

 $L_3 = \{BDE(3)\}$; $C_4 = \emptyset \rightarrow$ stop.

Step 4: the rules

 $5 \times 2 + (2^3 - 2) = 16$ candidates, min conf 80%

From	Rule	$\sigma(s)$	Conf	From	Rule	$\sigma(s)$	Conf
AD (3)	$A \Rightarrow D$	3	1.00 ✓	DF (3)	$D \Rightarrow F$	5	0.60 ×
	$D \Rightarrow A$	5	0.60 ×		$F \Rightarrow D$	3	1.00 ✓
BD (3)	$B \Rightarrow D$	3	1.00 ✓	BDE (3)	$B \Rightarrow DE$	3	1.00 ✓
	$D \Rightarrow B$	5	0.60 ×		$D \Rightarrow BE$	5	0.60 ×
BE (3)	$B \Rightarrow E$	3	1.00 ✓		$E \Rightarrow BD$	4	0.75 ×
	$E \Rightarrow B$	4	0.75 ×		$BD \Rightarrow E$	3	1.00 ✓
DE (4)	$D \Rightarrow E$	5	0.80 ✓		$BE \Rightarrow D$	3	1.00 ✓
	$E \Rightarrow D$	4	1.00 ✓		$DE \Rightarrow B$	4	0.75 ×

- **9 strong rules of 16.** $D \Rightarrow E$ scrapes in at exactly 0.80 – **$\geq$ the threshold counts as passing**, so keep it.
- Every failure has D or E alone on the left. D is in *all five* transactions, so it predicts nothing (§4.1); it is the consequent of four winning rules instead.

Step by step: Apriori level by level, then the 16 rules

1. Min support as a count

threshold to count

5 transactions $\times$ 60% = 3

2. $C_1 \rightarrow L_1$, min count 3

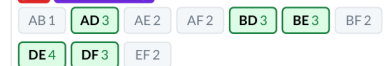
count and keep



L_1 : 5 kept of 11

3. $C_2 \rightarrow L_2$

join count and keep



L_2 : 5 kept of 10

4. $C_3 \rightarrow L_3$

join prune count and keep



pruned (dashed): DEF ($EF \notin L_2$)

L_3 : 1 kept of 2

KEY STEP

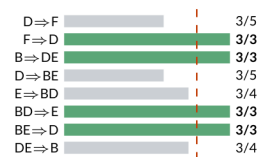
5. $C_4 \rightarrow L_4$

join stop

$C_4 = \emptyset$: no two itemsets share a prefix.

6. Every rule against min conf

confidence threshold line min conf 0.8



9 strong of 16 candidate rules (green).

Problem 1.8 2+6

QUESTION AS PRINTED 2081 Bhadra · Q7 · [2+6]

For the transaction given below, consider Confidence = 65% and minimum support = 50%. Identify frequent itemsets with possible association rules using Apriori Algorithm.

TID	Items
1	MILK, BREAD, CAKE
2	BUTTER, BREAD, EGG
3	MILK, BUTTER, BREAD, EGG
4	BUTTER, EGG

4 transactions, 50% $\rightarrow$ min count = 2. Note the unusually low confidence bar, 65%, which is why almost everything survives.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
MILK	1, 3	2	✓
BREAD	1, 2, 3	3	✓
CAKE	1	1	×
BUTTER	2, 3, 4	3	✓
EGG	2, 3, 4	3	✓

$L_1 = \{\text{MILK}(2), \text{BREAD}(3), \text{BUTTER}(3), \text{EGG}(3)\}$

Step 2: $C_2 \rightarrow L_2$

6 pairs

Pair	In	Cnt	Keep
BREAD, BUTTER	2, 3	2	✓
BREAD, EGG	2, 3	2	✓
BREAD, MILK	1, 3	2	✓
BUTTER, EGG	2, 3, 4	3	✓
BUTTER, MILK	3	1	×
EGG, MILK	3	1	×

Step 3: $C_3 \rightarrow L_3$

Joined from	Candidate	Subset test	Cnt	Keep
{BR,BU} + {BR,EG}	BREAD, BUTTER, EGG	all three 2-subsets in L_2 ✓	2	✓
{BR,BU} + {BR,MI}	BREAD, BUTTER, MILK	{BU,MI} $\notin L_2 \rightarrow$ pruned	—	×
{BR,EG} + {BR,MI}	BREAD, EGG, MILK	{EG,MI} $\notin L_2 \rightarrow$ pruned	—	×

$L_3 = \{\{\text{BREAD, BUTTER, EGG}\}(2)\}$, $C_4 = \emptyset \rightarrow$ stop.

Step 4: the rules

$4 \times 2 + 6 = 14$ candidates, min conf 65 %

Rule	$\sigma(s)$	Conf	Rule	$\sigma(s)$	Conf
BREAD $\Rightarrow$ BUTTER	3	0.667 ✓	BUTTER $\Rightarrow$ EGG	3	1.00 ✓
BUTTER $\Rightarrow$ BREAD	3	0.667 ✓	EGG $\Rightarrow$ BUTTER	3	1.00 ✓
BREAD $\Rightarrow$ EGG	3	0.667 ✓	BREAD $\Rightarrow$ BU,EG	3	0.667 ✓
EGG $\Rightarrow$ BREAD	3	0.667 ✓	BUTTER $\Rightarrow$ BR,EG	3	0.667 ✓
BREAD $\Rightarrow$ MILK	3	0.667 ✓	EGG $\Rightarrow$ BR,BU	3	0.667 ✓
MILK $\Rightarrow$ BREAD	2	1.00 ✓	BR,BU $\Rightarrow$ EGG	2	1.00 ✓
			BR,EG $\Rightarrow$ BUTTER	2	1.00 ✓
			BU,EG $\Rightarrow$ BREAD	3	0.667 ✓

- **All 14 rules are strong.** That is the point of the 65 % threshold: the lowest confidence anywhere in this database is $2/3 = 0.667$, which clears it.
- **Say so explicitly rather than listing 14 rules and stopping** — “every candidate rule passes, because min_conf sits below the minimum achievable confidence here” is the observation worth a mark. Raise the bar to 70 % and only the five 100 % rules survive.

Step by step: Apriori level by level, then the 14 rules

Problem 1.9 3+7

QUESTION AS PRINTED 2073 Chaitra · Q5 · [3+7]

Explain Apriori algorithm in market basket analysis? Derive association rule from the following market basket transactions with 50% of minimum support and confidence respectively.

Transaction	Itemsets
1	A, B, C
2	A, C
3	A, D
4	B, E, F

"50% of minimum support and confidence respectively" means **both** thresholds are 50%: min count = 2, min conf = 0.50.

Step 1: $C_1 \rightarrow L_1$

Item	In	Cnt	Keep
A	1, 2, 3	3	✓
B	1, 4	2	✓
C	1, 2	2	✓
D	3	1	×
E	4	1	×
F	4	1	×

$L_1 = \{A(3), B(2), C(2)\}$

Step 2: $C_2 \rightarrow L_2$

Pair	In	Cnt	Keep
AB	1	1	×
AC	1, 2	2	✓
BC	1	1	×

$L_2 = \{AC(2)\}$; only one pair survives, so $C_3 = \emptyset \rightarrow \text{stop}$.

Step 3: the rules

only $\{A, C\}$ can produce any, so $2^2 - 2 = 2$ candidates

Rule	$\sigma(s)$	Confidence	Strong at 50%?
$A \Rightarrow C$	3	$2/3 = 0.667$	✓
$C \Rightarrow A$	2	$2/2 = 1.000$	✓

- **Both are strong:** $A \Rightarrow C$ (67%) and $C \Rightarrow A$ (100%).
- **Transaction 4 contributes nothing.** $\{B, E, F\}$ shares no item pattern with the rest, which is why B survives as a 1-itemset but never appears in a rule. Worth a sentence.

Step by step: Apriori level by level, then the two rules

1. Min support as a count

threshold to count

50% of 4 = 2; min conf = 0.50

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A3 B2 C2 D1 E1 F1

L_1 : 3 kept of 6

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$; no two itemsets share a prefix.

5. Every rule against min conf

confidence threshold line min conf 0.5

A $\Rightarrow$ C 2/3
C $\Rightarrow$ A 2/2

2 strong of 2 candidate rules (green).

3. $C_2 \rightarrow L_2$

join count and keep

AB1 AC2 BC1

L_2 : 1 kept of 3

KEY STEP

Problem 1.10 5+3

QUESTION AS PRINTED 2079 Baishakh · Q5 · [5+3]

 a) Write Apriori algorithm and using the algorithm find all the frequent itemset for the following database. ($\text{min_sup} = 20\%$).

TID	A1	A2	A3	A4	A5	A6	A7	A8	A9
T2	0	1	0	1	0	0	0	1	0
T3	0	0	0	1	1	0	1	0	0
T4	0	0	1	0	0	0	0	0	0
T5	0	0	0	0	1	1	1	0	0
T6	0	1	1	1	0	0	0	0	0
T7	0	1	0	0	0	1	1	0	1
T8	0	0	0	0	1	0	0	0	0

b) Is it possible to generate any rules out of the frequent items, considering any value of confidence threshold?

The table really does start at T2, has seven rows, and its 8th column really is headed A8 (the scan clips the 8). 20% of 7 transactions = 1.4, and a count cannot be fractional, so **min count = 2**. **Column A1 is 0 in every row** — it can never be frequent, so drop it at once and say why.

(a) Step 1: $C_1 \rightarrow L_1$
Step 2: $C_2 \rightarrow L_2$
 $\binom{6}{2} = 15 \text{ pairs}$

Item	In	Cnt	Keep	Pair	Cnt	Keep	Pair	Cnt	Keep
A1	— (all zero)	0	×	A2,A3	1	×	A4,A5	1	×
A2	T2, T6, T7	3	✓	A2,A4	2	✓	A4,A6	0	×
A3	T4, T6	2	✓	A2,A5	0	×	A4,A7	1	×
A4	T2, T3, T6	3	✓	A2,A6	1	×	A5,A6	1	×
A5	T3, T5, T8	3	✓	A2,A7	1	×	A5,A7	2	✓
A6	T5, T7	2	✓	A3,A4	1	×	A6,A7	2	✓
A7	T3, T5, T7	3	✓	A3,A5	0	×			
A8	T2	1	×	A3,A6	0	×			
A9	T7	1	×	A3,A7	0	×			

Step 3: C_3

- $L_2 = \{A2A4(2), A5A7(2), A6A7(2)\}$.
- To build a 3-itemset two pairs must **share their first item**. A2A4, A5A7 and A6A7 all start with a different item, so **no candidate can be formed at all** $\rightarrow C_3 = \emptyset \rightarrow \text{stop}$.
- **Frequent itemsets:** A2, A3, A4, A5, A6, A7, {A2,A4}, {A5,A7}, {A6,A7}.

(b) Can any rule be generated, at any confidence threshold?

- **Yes**. The question is a trap only if you stop at L_1 : a rule needs a frequent itemset of size ≥ 2 to split into antecedent and consequent, and here **three 2-itemsets survive**, so rules exist.
- Each 2-itemset yields $2^2 - 2 = 2$ rules, so **6 rules in total**:

Rule	$\sigma(s)$	Confidence	Comment
A2 $\Rightarrow$ A4	3	$2/3 = 0.667$	
A4 $\Rightarrow$ A2	3	$2/3 = 0.667$	
A5 $\Rightarrow$ A7	3	$2/3 = 0.667$	
A7 $\Rightarrow$ A5	3	$2/3 = 0.667$	
A6 $\Rightarrow$ A7	2	$2/2 = 1.000$	A6 never occurs without A7
A7 $\Rightarrow$ A6	3	$2/3 = 0.667$	

- So the honest answer is: **rules exist for any threshold up to 0.667**, and above that only **A6 $\Rightarrow$ A7** survives. Above 1.0 nothing does.
- **The general principle** (§4.2): rules exist iff $L_2 \neq \emptyset$. **Check L_2 , not the confidence value** — if L_2 were empty no threshold, not even 0 %, could produce a rule. Compare Problem 1.6, where exactly that happens.

Step by step: Apriori level by level, then (b) the six rules

1. Min support as a count

threshold to count

20% of 7 = 1.4, a count cannot be fractional → 2

2. $C_1 \rightarrow L_1$, min count 2

count and keep

A2 3 A3 2 A4 3 A5 3 A6 2 A7 3 A8 1
A9 1

L_1 : 6 kept of 8

3. $C_2 \rightarrow L_2$

join count and keep

A2,A3 1 A2,A4 2 A2,A5 0 A2,A6 1
A2,A7 1 A3,A4 1 A3,A5 0 A3,A6 0
A3,A7 0 A4,A5 1 A4,A6 0 A4,A7 1
A5,A6 1 A5,A7 2 A6,A7 2

L_2 : 3 kept of 15

4. $C_3 \rightarrow L_3$

join stop

$C_3 = \emptyset$: no two itemsets share a prefix.

KEY STEP

5. (b) Every rule, at any threshold

confidence threshold line

A2 $\Rightarrow$ A4 2/3
A4 $\Rightarrow$ A2 2/3
A5 $\Rightarrow$ A7 2/3
A7 $\Rightarrow$ A5 2/3
A6 $\Rightarrow$ A7 2/2
A7 $\Rightarrow$ A6 2/3

6 strong of 6 candidate rules (green).

M2. Rule generation from frequent itemsets

Method

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.
- The support of every rule from l is the same ($= \text{sup}(l)$); only confidence varies.
- Shortcut: if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a subset of s also fails. Prune downward.

Worked example

PROBLEM not a PYQ · the rule half of 2081 Baishakh Q3 (b) verified/added

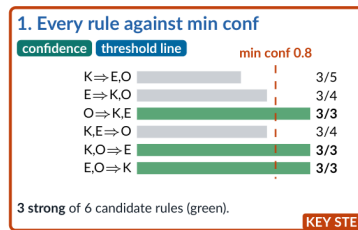
The frequent itemset $l = \{K, E, O\}$ has support count 3 in the 5-transaction database of Problem 3.1, where $\text{sup}(K) = 5$, $\text{sup}(E) = 4$, $\text{sup}(O) = 3$, $\text{sup}(K,E) = 4$, $\text{sup}(K,O) = 3$ and $\text{sup}(E,O) = 3$. List every association rule that l can generate and its confidence, and say which are strong at a minimum confidence of 80%.

From $l = \{K,E,O\}$ with $\text{sup} = 3$ (see Problem 3.1), $|l| = 3 \rightarrow 6$ candidate rules:

Rule	Confidence	Value	$\geq 80\%$?
$K \Rightarrow E,O$	3/5	0.60	×
$E \Rightarrow K,O$	3/4	0.75	×
$O \Rightarrow K,E$	3/3	1.00	✓
$K,E \Rightarrow O$	3/4	0.75	×
$K,O \Rightarrow E$	3/3	1.00	✓
$E,O \Rightarrow K$	3/3	1.00	✓

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

Step by step: the six rules from K, E, O



M3. FP-tree construction and FP-Growth mining

Method

- Scan 1:** count items, drop infrequent ones, sort the rest by descending support → **F-list**. **State your tie-break rule.**
- Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
- Mine bottom-up** (least frequent item first): conditional pattern base → conditional FP-tree → recurse.

Problem 3.1 2+7

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [2+7]

What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%.

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

- Find all frequent itemsets using FP-growth.
- List all of the strong association rules (with support and confidence).

- T500 prints O twice** — a transaction is a set, so it is {C,O,K,I,E}.
- 5 transactions, 60% → **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 ✓ N:2, C:2, D:1, A:1, U:1, I:1 ✗
- F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

Drawn below, after every transaction; the last panel is the finished tree.

- Check:** counts on each level sum correctly — K:5 = E:4 + M:1. ✓

The node-link (header) table

- One entry per F-list item, pointing at every node carrying that label:

K → K:5

E → E:4

M → M:2 (under E), M:1 (under K)

O → O:1 (under M), O:2 (under E)

Y → Y:1 × 3 (three different branches)

- This table is what makes mining possible** — it is how you find all occurrences of an item without walking the tree.

Step by step: the FP-tree after each transaction



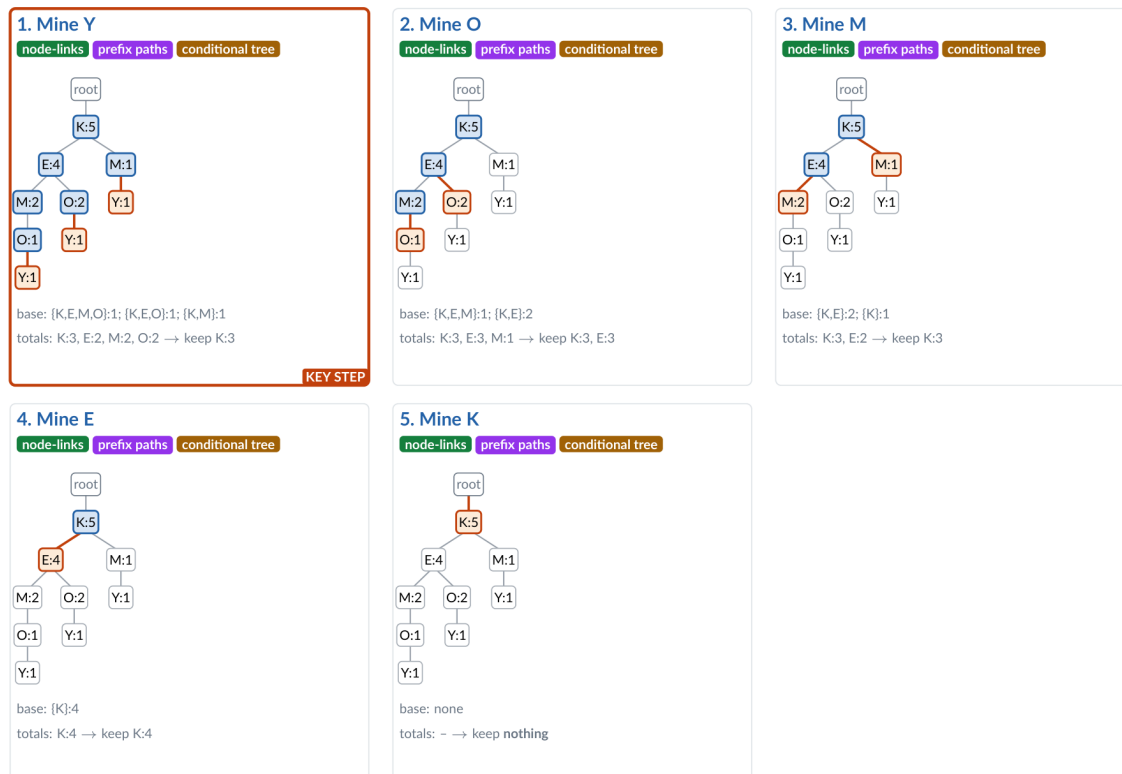
Step 4: mine the tree bottom-up (FP-Growth proper — this is where the marks are)

- The procedure, for one item i at a time, taken in **reverse F-list order** (least frequent first, so $Y \rightarrow O \rightarrow M \rightarrow E \rightarrow K$):
 - Follow i 's node-links to **every** node labelled i .
 - For each such node, write down the **prefix path**: the labels from the root down to it, **excluding i itself**, and tag that whole path with **the count of the i -node**, not the counts written on the path. Together these are i 's **conditional pattern base**.
 - Add up each label's tag totals across the base. Drop labels below min count. What remains is i 's **conditional FP-tree**.
 - Every itemset in that conditional tree, with i appended, is frequent. Its support is the count in the conditional tree.
 - If the conditional tree branches, recurse on it the same way.
- The **single most common mistake** is truncating a prefix path. A path is the *whole* route from the root, so an item sitting between the root and i stays in even when it is not adjacent to i .

Item	Conditional pattern base	Totals $\rightarrow$ cond. FP-tree	Frequent patterns
Y	{K,E,M,O}:1 {K,E,O}:1 {K,M}:1	K:3, E:2, M:2, O:2 $\rightarrow$ only K:3	{K,Y}:3
O	{K,E,M}:1 {K,E}:2	K:3, E:3, M:1 $\rightarrow$ K:3 $\rightarrow$ E:3	{K,O}:3, {E,O}:3, {K,E,O}:3
M	{K,E}:2 {K}:1	K:3, E:2 $\rightarrow$ only K:3	{K,M}:3
E	{K}:4	K:4 $\rightarrow$ K:4	{K,E}:4
K	— (root child, no prefix)	—	{K}:5

- Worked in full for Y**, since it is the one people get wrong. Y sits at the end of three different branches: $K \rightarrow E \rightarrow M \rightarrow O \rightarrow Y:1$, $K \rightarrow E \rightarrow O \rightarrow Y:1$ and $K \rightarrow M \rightarrow Y:1$. Strip the Y and keep **the whole remaining path each time**, so O belongs in the first two entries. Totals give K:3 only, hence the one pattern {K,Y}:3.
- Why only O recurses**: its conditional tree still has two items (K and E), so the single branch $K:3 \rightarrow E:3$ yields {K,O}, {E,O} and the combination {K,E,O}. Every other conditional tree is a single node, so it yields exactly one pattern.

Step by step: mining each item, least frequent first



All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)

- $K \Rightarrow E = 4/5 = 0.80 \checkmark$ $E \Rightarrow K = 4/4 = 1.0 \checkmark$
- $M \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow K = 3/3 = 1.0 \checkmark$ $Y \Rightarrow K = 3/3 = 1.0 \checkmark$ $O \Rightarrow E = 3/3 = 1.0 \checkmark$
- $K, O \Rightarrow E = 1.0 \checkmark$ $E, O \Rightarrow K = 1.0 \checkmark$ $O \Rightarrow K, E = 1.0 \checkmark$
- Failing, all 7 of them: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ and $K \Rightarrow E, O$ (all $3/5 = 0.6$), $E \Rightarrow O$, $K, E \Rightarrow O$ and $E \Rightarrow K, O$ (all $3/4 = 0.75$).
- Count check:** five 2-itemsets give $5 \times 2 = 10$ rules and {K,E,O} gives $2^3 - 2 = 6$, so **16 candidates = 9 strong + 7 failing.** ✓
- Note K is on the right of five of the nine.** K is in every transaction, so anything implies it → high confidence, no information (§4.1).

Step by step: the 16 rules against 80%



Problem 3.2 4+4

QUESTION AS PRINTED 2075 Chaitra · Q4 · [4+4]

What do you mean by frequent Pattern growth, draw FP-tree with given tabular data.

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

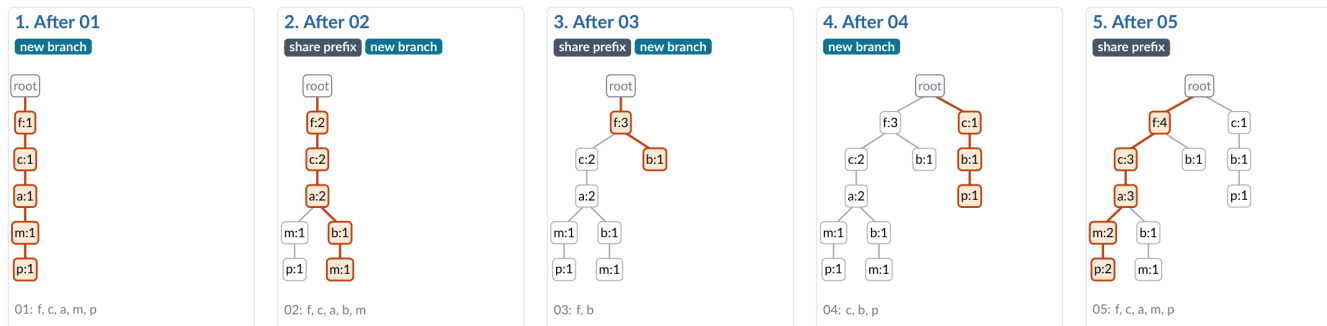
Min support is not printed; use 3, which is the standard for this dataset. verified/added

Counts and F-list

- f:4, c:4, a:3, b:3, m:3, p:3 ✓ l:2, o:2, and d,g,i,h,j,k,s,e,n,w all 1 ×
- **F-list = f, c, a, b, m, p**
- **T05 prints a capital "A"** — same item as a.

Reordered transactions

- 01 → f, c, a, m, p 02 → f, c, a, b, m
- 03 → f, b 04 → c, b, p
- 05 → f, c, a, m, p

Step by step: the FP-tree after each transaction**The FP-tree**

Drawn below, after every transaction; the last panel is the finished tree.

- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

QUESTION AS PRINTED 2078 Bhadra · Q5 · [4+4]

Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following:

- Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
- Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- 6 transactions, 33.34% → **min count = 2**.
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent**.
- **F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2)** (4-way tie broken alphabetically).

Tree, built transaction by transaction

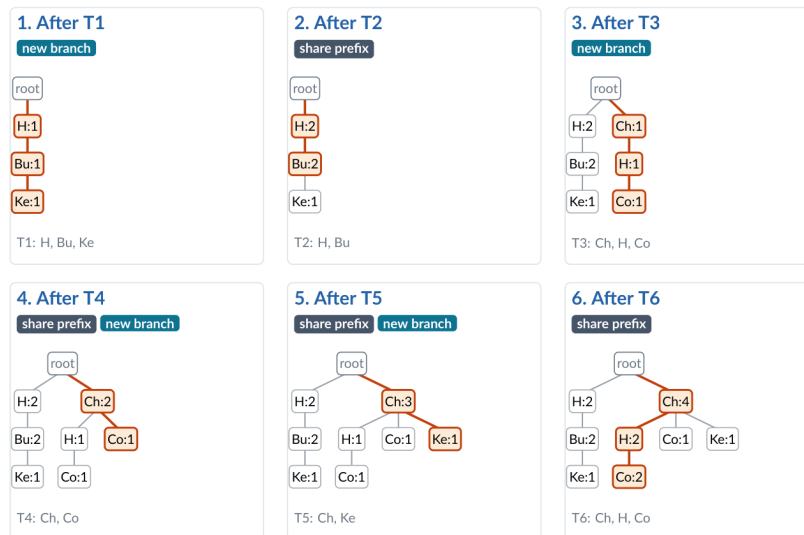
Final tree

Drawn below, after every transaction; the last panel is the finished tree.

T	Reordered	Effect on the tree
T1	H, Bu, Ke	new branch $H:1 \rightarrow Bu:1 \rightarrow Ke:1$
T2	H, Bu	prefix shared $\rightarrow H:2, Bu:2$
T3	Ch, H, Co	new root child $Ch:1 \rightarrow H:1 \rightarrow Co:1$
T4	Ch, Co	$Ch:2$, new child $Co:1$ under Ch
T5	Ch, Ke	$Ch:3$, new child $Ke:1$ under Ch
T6	Ch, H, Co	fully shared $\rightarrow Ch:4, H:2, Co:2$

- **Two root children**, because T1 and T2 do not contain Chips and so cannot join the Chips prefix.

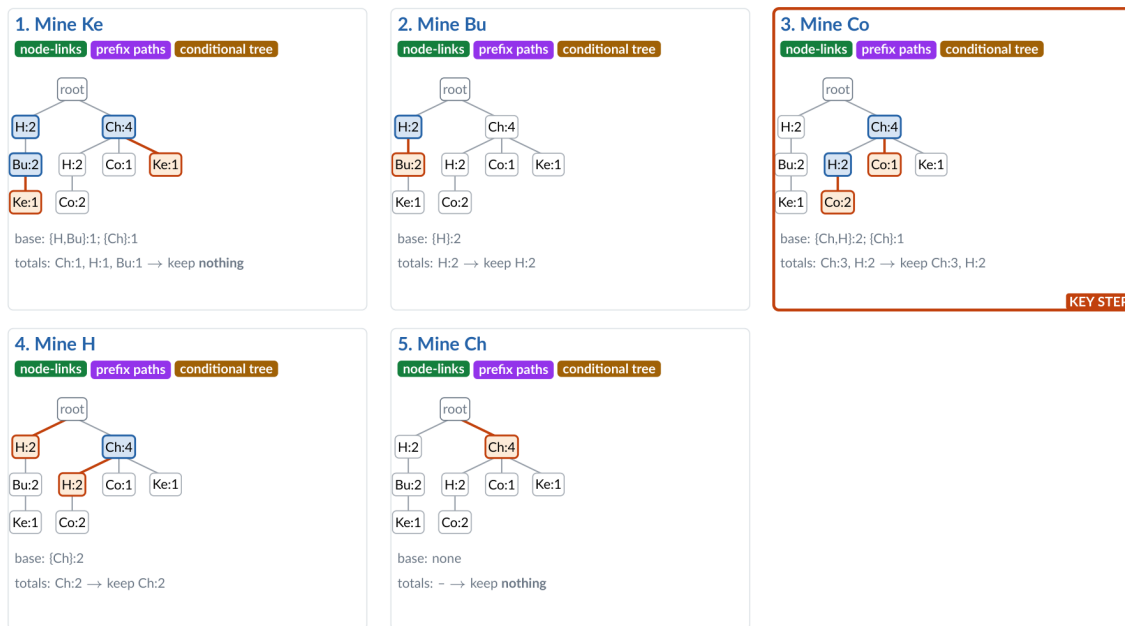
Step by step: the FP-tree after each transaction (orange = the path it touches)



Mining, least-frequent item first

Item	Conditional pattern base	Totals $\rightarrow$ patterns
Ketchup	{H, Bu}:1, {Ch}:1	$H:1, Bu:1, Ch:1 \rightarrow$ all below 2, no pattern
Buns	{H}:2	$H:2 \rightarrow \{H, Bu\}:2$
Coke	{Ch, H}:2, {Ch}:1	$Ch:3, H:2 \rightarrow \{Ch, Co\}:3, \{H, Co\}:2, \{Ch, H, Co\}:2$
HotDogs	{ }:2, {Ch}:2	$Ch:2 \rightarrow \{Ch, H\}:2$
Chips	– (root child)	$- \rightarrow \{Ch\}:4$

Step by step: mining each item, least frequent first



Frequent itemsets

- **1-itemsets:** Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- **2-itemsets:** {Chips,Coke} 3, {Chips,HotDogs} 2, {HotDogs,Coke} 2, {HotDogs,Buns} 2
- **3-itemsets:** {Chips,HotDogs,Coke} 2
- **HotDogs has an empty prefix once** (it is a root child from T1/T2), which contributes nothing to its conditional base. Write the empty set rather than omitting the row.

Problem 3.4 2+8

QUESTION AS PRINTED 2074 Chaitra · Q6 · [2+8]

How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree.

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

Min support is not printed; use 2. Abbreviations: E = Earphone, L = Laptop, Mo = Mobile, Ca = Camera, Pd = Pen drive.

Step 1: counts and F-list

Item	In	Count	Keep
Earphone	T3, T4, T5, T6	4	✓
Laptop	T1, T2, T3, T6	4	✓
Mobile	T3, T4, T6	3	✓
Camera	T1, T5	2	✓
Pen drive	T1, T2	2	✓

F-list = E(4), L(4), Mo(3), Ca(2), Pd(2) – **all five are frequent**, and the E/L tie at 4 is broken alphabetically.

Step 2: reorder each transaction by the F-list

T	As printed	Reordered
T1	Camera, Laptop, Pen drive	L, Ca, Pd
T2	Laptop, Pen drive	L, Pd
T3	Laptop, Mobile, Earphone	E, L, Mo
T4	Earphone, Mobile	E, Mo
T5	Camera, Earphone	E, Ca
T6	Laptop, Mobile, Earphone	E, L, Mo

Step 3: insert them one at a time

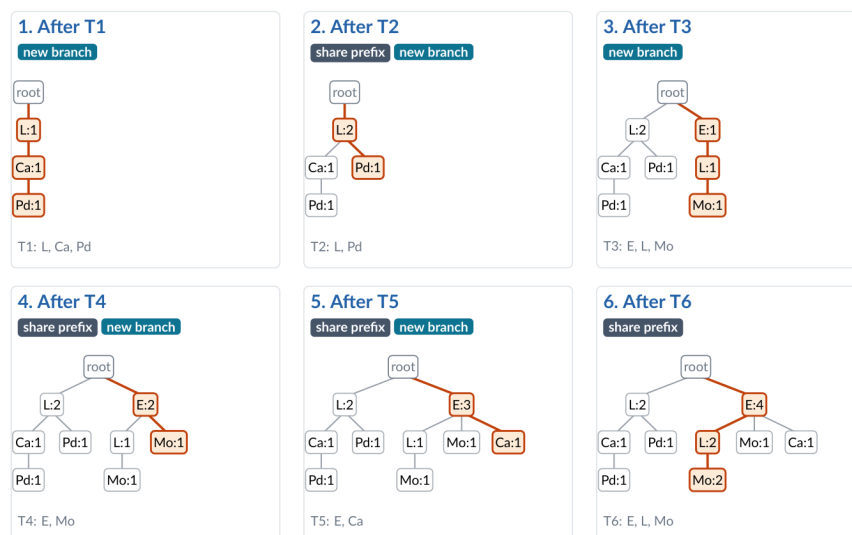
T	Effect on the tree
T1	new branch L:1→Ca:1→Pd:1
T2	L:2, new child Pd:1 under L
T3	new root child E:1→L:1→Mo:1
T4	E:2, new child Mo:1 under E
T5	E:3, new child Ca:1 under E
T6	fully shared → E:4, L:2, Mo:2

- **Two root children:** T1 and T2 hold no Earphone, so they cannot join the E prefix.

Final FP-tree

Drawn below, after every transaction; the last panel is the finished tree.

- **Check:** L appears 4 times in D and the tree shows 2 under the root plus 2 under E. ✓ Do this for one item, it catches most construction errors.
- Mining it gives the key pattern {Earphone, Laptop, Mobile}:2 (T3 and T6). The question only asks for the tree, so this is a bonus line.

Step by step: the FP-tree after each transaction (orange = the path it touches)

M4. Pattern evaluation: lift and correlation

Method

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{conf(A \Rightarrow B)}{sup(B)} \quad \chi^2 = \sum \frac{(obs - exp)^2}{exp}$$

- $lift > 1$ positive · $= 1$ independent · < 1 negative.

Problem 4.1 8

QUESTION AS PRINTED 2070 Chaitra · Q5 · [8]

Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules.

The paper prints no data — the example is yours to supply. The standard *game* / *video* contingency table below is the one every textbook uses. verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $sup(game \Rightarrow video) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $conf(game \Rightarrow video) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(video) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$lift = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<i>game</i>	row
video	4000 (4500)	3500 (3000)	7500
<i>video</i>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer **all_conf**, **max_conf**, **Kulczynski** or **cosine**.

Step by step: why a 66.7% rule is misleading

1. Support and confidence pass

confidence

sup = 4000/10000 = 40% (over 30%)

conf = 4000/6000 = 66.7% (over 60%)

By these two measures the rule *game* $\Rightarrow$ *video* is strong.

2. Compare with the base rate

confidence

P(video) 75%

P(video|game) 66.7%

Buying a game lowers the chance of a video.

3. Lift

lift

$\frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89$

< 1: **negatively** correlated, despite 66.7% confidence.

KEY STEP

4. Chi-square

chi-square

	game	<i>game</i>
video	4000 (4500)	3500 (3000)
<i>video</i>	2000 (1500)	500 (1000)

obs (exp):

$\chi^2 = 55.6 + 83.3 + 166.7 + 250 = 555.6$;

obs < exp for both: negative.

M5. Subsequence enumeration

Method

- A k -**subsequence** contains k items in total (not k elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (a, b, c, d) of items taken per element, then remove duplicates.

Problem 5.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (a) · [3]

List all the 4-subsequences contained in the data sequence: $\langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

What counts as a subsequence (get this right first)

- Take a subset from each element, **possibly empty**. Elements chosen empty simply disappear.
- **Element order must be preserved**, and items from two different elements can never be merged into one element.
- A k -**subsequence** has k items in total, counted across all surviving elements. It is **not** k elements. This is the single most common misreading.
- So $\langle \{1, 3\} \{2\} \rangle$ is a 3-subsequence, not a 2-subsequence.

Why enumerate by splits

- A choice is fully described by **how many items are taken from each element**, written (a, b, c, d) for $e_1 \dots e_4$.
- The bounds are the element sizes: $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, and we need $a + b + c + d = 4$.
- For a fixed split the number of ways is a product of binomials, $\binom{2}{a} \binom{1}{b} \binom{2}{c} \binom{1}{d}$, because the elements are chosen independently.
- **Enumerating splits first guarantees nothing is missed**, which listing by hand does not. Then remove duplicates at the end.

Step 1: enumerate the item-count splits

(a, b, c, d) with $a \leq 2$, $b \leq 1$, $c \leq 2$, $d \leq 1$, sum = 4

(a, b, c, d)	Ways	Subsequences
(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Step 2: check the count, then remove duplicates

- Ways per split: $2 + 1 + 1 + 2 + 2 + 4 + 2 + 1 = 15$ selections. **Do this arithmetic** — it is how you know no split was dropped.
- **A duplicate appears when the same items can be taken from different elements.** Here $e_2 = \{2\}$ and $e_3 = \{2, 3\}$ share the item 2, so $\langle \{1, 3\} \{2\} \{4\} \rangle$ is produced twice: once as $(2, 1, 0, 1)$ taking the 2 from e_2 , once as $(2, 0, 1, 1)$ taking it from e_3 . **Same sequence, so it counts once.**
- No other pair of splits can collide, because every other split keeps a different number of surviving elements or a different element content.

Final answer

- 15 selections — 1 duplicate $\rightarrow$ **14 distinct 4-subsequences.**
- **The exam usually only wants “at least four”** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Step by step: enumerate by split, then remove the duplicate

1. Splits (a, b, c, d) summing to 4

item splits

e_1	$\{1, 3\}$	e_2	$\{2\}$	e_3	$\{2, 3\}$	e_4	$\{4\}$	ways
2	1	1	0	2				$\binom{2}{a} \binom{1}{b} \binom{2}{c} \binom{1}{d}$ per split, total 15
2	1	0	1	1				
2	0	2	0	1				
2	0	1	1	2				
1	1	2	0	2				
1	1	1	1	1				
1	0	2	1	2				
0	1	2	1	1				

2. Remove the duplicate

remove duplicates

$\langle \{1, 3\} \{2\} \{4\} \rangle$ arises twice:
 $(2, 1, 0, 1)$: the 2 from e_2 $(2, 0, 1, 1)$: the 2 from e_3
 e_2 and e_3 share the item 2, so the same sequence is reached twice.

$15 - 1 = 14$ distinct 4-subsequences.

KEY STEP

Problem 5.2 3

QUESTION AS PRINTED 2079 Bhadra · Q6 (a) · [3+3]

How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$.

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

Step by step: four order-preserving picks

1. Pick within each element

keep element order

2	3	5	6	7	8	9	1	7	4	$\langle \{2\} \{6\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{3, 5\} \{7\} \{9\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$
2	3	5	6	7	8	9	1	7	4	$\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$

Green = kept. Each pick is a subset of its own element, in element order.

KEY STEP

M6. Subgraph candidate generation by edge growing

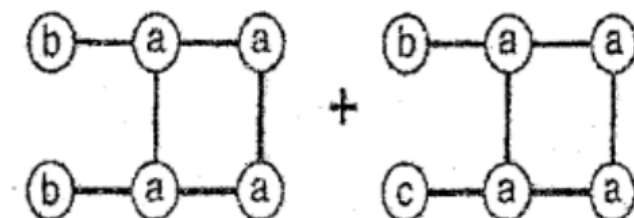
Method

- Two frequent subgraphs are joined only if they share a **common core**: the subgraph left when one edge is removed from each.
- Edge growing** joins two k -edge graphs sharing a $(k - 1)$ -edge core and produces a $(k + 1)$ -**edge candidate** = core + the edge unique to each parent.
- Vertex growing** instead adds a vertex, and generates far more candidates. **Edge growing is preferred for exactly that reason.**
- The trap, and the whole point of this question:** when the core carries **repeated vertex labels**, the two extra edges can be attached in more than one non-isomorphic way, so **one join yields several candidates**. Enumerate them, do not assume there is one.

Problem 6.1 3

QUESTION AS PRINTED 2076 Chaitra · Q8 (b) · [3]

Draw all candidate sub-graphs obtained from joining the pair of graphs shown below using edge-growing method to expand the sub-graphs.



Step 1: read the two parents

- Both are a **4-cycle of a vertices** with two pendant vertices hanging off two *adjacent* corners of that cycle.
- G_1 : pendants are b and b . G_2 : pendants are b and c .
- Each has **6 vertices and 6 edges**, so $k = 6$.

Step 2: find the core

- Delete one edge from each so that what is left is the same graph: drop the second pendant edge from both.
- Core = 4-cycle of a 's + one pendant b** , 5 vertices and 5 edges = $k - 1$. ✓ the graphs are joinable.
- Edge unique to G_1 : $a - b$. Edge unique to G_2 : $a - c$.

Step 3: the candidate size

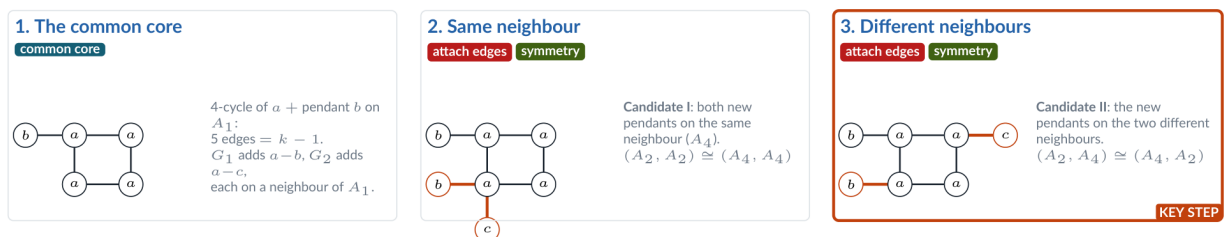
- Candidate = core + both unique edges $\rightarrow 5 + 2 = 7$ edges, 7 vertices.

Final answer: 2 candidate sub-graphs, each with 7 vertices and 7 edges

Step by step: the core, then the two ways to attach (7 vertices, 7 edges each)

Step 4: where can the two new edges attach?

- Call the core's pendant-carrying a vertex A_1 . In both parents the second pendant sits on a cycle vertex **adjacent to A_1** , and A_1 has two neighbours in the cycle, A_2 and A_4 .
- All four placements (b, c) are legal: (A_2, A_2) , (A_4, A_4) , (A_2, A_4) , (A_4, A_2) .
- The core has a **mirror symmetry swapping A_2 and A_4** , so these collapse to **two non-isomorphic candidates**:
 $(A_2, A_2) \cong (A_4, A_4) \rightarrow$ **candidate I**
 $(A_2, A_4) \cong (A_4, A_2) \rightarrow$ **candidate II**
- This multiplicity comes entirely from the four a 's sharing one label.** Had the cycle been a, b, c, d the join would have given a single candidate.



- Marking note:** the marks are for showing the *core* and then arguing the placements. A single drawing with no core and no count of candidates scores badly even if the drawing itself is one of the two right answers.
- Cross-check:** every candidate must contain both parents as subgraphs, and must have exactly $k + 1 = 7$ edges. Both do.

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

135 numerical marks, 7.0% of the archive, 37% of this chapter · 3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
- **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
- **M3** DBSCAN point classification *practice*

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-5 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
seed rule	State the seeds; with one given, take the point farthest from it.	1.1/1.2 1; 1.3 1; 1.4 1; 1.7 1; 1.8 1
assign nearest	Each point joins the centroid it is nearest to.	1.1/1.2 2, 3 ; 1.3 2, 3 ; 1.4 2,3 ; 1.5 1,2, 3 ; 1.7 2, 3 ; 1.8 2,3; 1.9 1,2,3,4; 1.10 1,2, 3
tie-break	Equal distances: name a rule and apply it; say which way the point went.	1.4 2
move to mean	Each centroid moves to the mean of its members.	1.1/1.2 2; 1.3 2; 1.4 2 ; 1.5 1,2; 1.7 2; 1.8 2; 1.9 1,2,3; 1.10 1,2, 3
unchanged = done	An iteration that reassigns nothing proves convergence.	1.1/1.2 3 ; 1.3 3 ; 1.4 3; 1.5 3 ; 1.7 3 ; 1.8 3; 1.9 4
Manhattan	$ dx + dy $ instead of the straight line.	1.10 1,2, 3
SSE	Sum of squared distances to each point's own centroid: what K-means minimises.	1.9 5
smallest entry	Merge the two clusters with the smallest distance; that value is the bar height.	2.1 1,2, 3 ,4,5; 2.2 1,2, 3 ,4,5; 2.3a 1,2, 3 ,4,5; 2.3b 1,2, 3 ,4,5
MIN update	Single link: the merged row takes the smaller of the two old rows.	2.1 2, 3 ,4,5; 2.3a 2, 3 ,4,5
MAX update	Complete link: the merged row takes the larger of the two old rows.	2.2 2, 3 ,4,5; 2.3b 2, 3 ,4,5
ε-neighbourhood	Count the points within ε , the point itself included, $d \leq \varepsilon$.	3.1 1; Practice 1
core / noise	Count $\geq$ MinPts is core; not core and near no core point is noise.	3.1 2 ; Practice 2
connect cores	Cores within ε of each other form one cluster; borders attach.	3.1 3; Practice 3

M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

QUESTION AS PRINTED 2078 Bhadra · Q8 · [2+6]

Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

QUESTION AS PRINTED 2074 Ashwin · Q7 · [2+8]

Write K-means algorithm and find clusters for following data set.

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

(Take $K = 2$)

The same 6 instances, only the column names differ.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	$d \text{ to } c_1(1, 2)$	$d \text{ to } c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow \text{new centroid } \left(\frac{1+2.5}{2}, \frac{2+1}{2}\right) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow \text{new centroid } \left(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}\right) = (4, 2)$

Iteration 2

Point	$d \text{ to } (1.75, 1.5)$	$d \text{ to } (4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged $\rightarrow$ converged.**

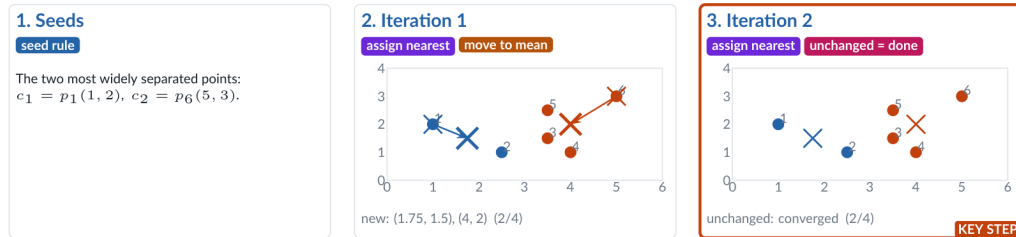
Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity — worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, p_3, p_4, p_5, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**.

Same data, same K , **different answer** — this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. **verified/added**

Step by step: seeds, then each iteration (x = centroid; hollow = before, filled = after)



Problem 1.3 4+6

QUESTION AS PRINTED 2075 Ashwin · Q7 · [4+6]

Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering.

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

Seeds

$c_1 = (1, 2)$ (clearly isolated) and $c_2 = (4, 6)$. (State the choice.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(4, 6)$	Cl.
(1, 2)	0	5.000	C1
(2.5, 4.5)	2.915	2.121	C2
(4, 6)	5.000	0	C2
(3.5, 4)	3.202	2.062	C2
(4, 5.5)	4.610	0.500	C2
(3, 6)	4.472	1.000	C2

- $C1 = \{(1, 2)\} \rightarrow$ centroid stays (1, 2)
- $C2 \rightarrow$ centroid

$$\left(\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5} \right) = (3.4, 5.2)$$

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid (3.4, 5.2).

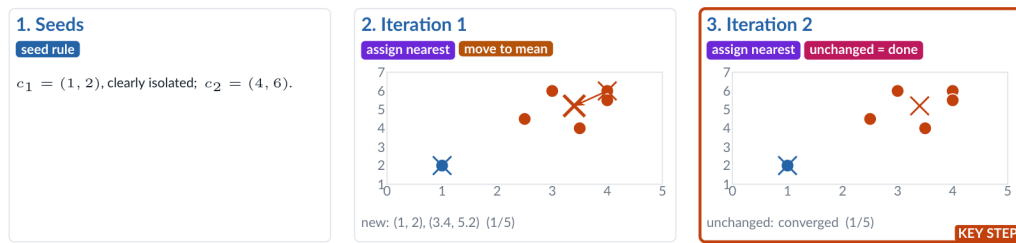
- **Note:** (1, 2) is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Iteration 2

Point	d to (1, 2)	d to (3.4, 5.2)	Cl.
(1, 2)	0	4.000	C1
(2.5, 4.5)	2.915	1.140	C2
(4, 6)	5.000	1.000	C2
(3.5, 4)	3.202	1.204	C2
(4, 5.5)	4.610	0.671	C2
(3, 6)	4.472	0.894	C2

- **Assignment unchanged $\rightarrow$ converged** after 2 iterations.

Step by step: seeds, then each iteration



Problem 1.4 10

QUESTION AS PRINTED 2071 Shrawan · Q8 · [10]

Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset.

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

The paper prints no K ; take $K = 2$, the value its sister questions use.

Seeds

 $c_1 = (1, 1)$, $c_2 = (5, 7)$ – the two most widely separated points.

Iteration 1

Point	d to $c_1(1, 1)$	d to $c_2(5, 7)$	Cl.
(1, 1)	0	7.211	C1
(1.5, 2)	1.118	6.104	C1
(3, 4)	3.606	3.606	tie
(5, 7)	7.211	0	C2
(3.5, 5)	4.717	2.500	C2
(4.5, 5)	5.315	2.062	C2
(3.5, 4.5)	4.301	2.915	C2

Iteration 2

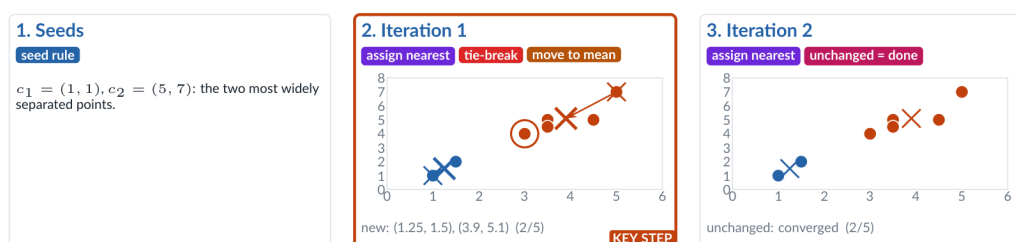
Point	d to (1.25, 1.5)	d to (3.9, 5.1)	Cl.
(1, 1)	0.559	5.022	C1
(1.5, 2)	0.559	3.921	C1
(3, 4)	3.052	1.421	C2
(5, 7)	6.657	2.195	C2
(3.5, 5)	4.161	0.412	C2
(4.5, 5)	4.776	0.608	C2
(3.5, 4.5)	3.750	0.721	C2

- (3, 4) is an exact tie: $\sqrt{4+9} = \sqrt{4+9} = 3.606$ both ways. **State a tie-break rule and use it** – here, send it to C2. Examiners give the mark for naming the rule, not for the direction chosen.
- C1 → centroid (1.25, 1.5)
- C2 → $\left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = (3.9, 5.1)$
- **Assignment unchanged → converged.** The tie at (3, 4) is now decisive (1.421 against 3.052), so the arbitrary choice in iteration 1 did not matter here.

Final answer

C1 = {(1, 1), (1.5, 2)} centroid (1.25, 1.5); C2 = {(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)} centroid (3.9, 5.1).

Step by step: the exact tie at (3,4), ringed in orange, sent to C2



Problem 1.5 8

QUESTION AS PRINTED 2076 Chaitra · Q9 · [8]

Given the matrix X whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here K is chosen as 3. The center of the 3 clusters are initialized as red (6.2, 3.2), green (6.6, 3.7) and blue (6.5, 3.0). Provide the final cluster centers and comment on the number of iterations required for the clusters to converge.

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red (5.171, 3.171), green (5.5, 4.2), blue (6.45, 2.95)

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red (4.8, 3.05), green (5.3, 4.0), blue (6.2, 3.025)

Iteration 3

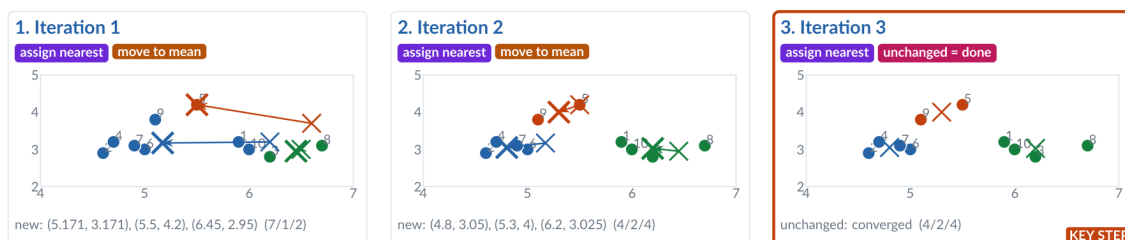
- Every point stays where it was → **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.
- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2-6.6$), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Step by step: red, green, blue centres (blue, orange, green here) over three passes



Problem 1.6 – the defective question 5+3

QUESTION AS PRINTED 2081 Baishakh · Q4 (b) · [5+3]

Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.

(nothing follows on the paper – the next line is Q5)

The paper prints NO data table for this question — it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (\$5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points — it is the dataset this paper's sister questions use. verified/added

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra · Q6 · [2+8]

Describe the K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	1.5	1
3	3.5	1.5
4	4	3
5	3.5	2.5
6	6	4

Seeds

the two most widely separated points, $c_1 = (1, 2)$ and $c_2 = (6, 4)$.

Iteration 1

Point	d to $(1, 2)$	d to $(6, 4)$	Cl.
(1, 2)	0	5.385	C1
(1.5, 1)	1.118	5.408	C1
(3.5, 1.5)	2.550	3.536	C1
(4, 3)	3.162	2.236	C2
(3.5, 2.5)	2.550	2.915	C1
(6, 4)	5.385	0	C2
- C1 $\rightarrow \left(\frac{1+1.5+3.5+3.5}{4}, \frac{2+1+1.5+2.5}{4} \right) = (2.375, 1.75)$			
- C2 $\rightarrow \left(\frac{4+6}{2}, \frac{3+4}{2} \right) = (5, 3.5)$			

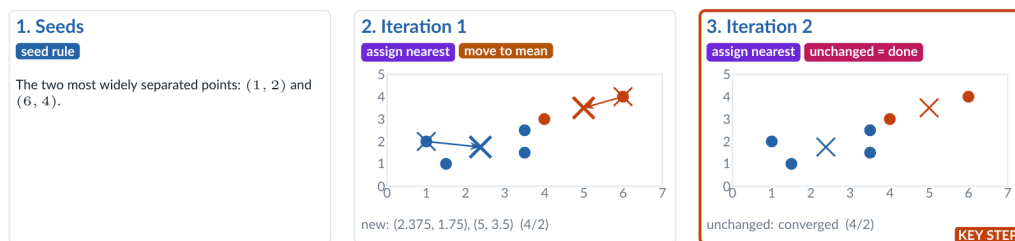
Iteration 2

Point	d to $(2.375, 1.75)$	d to $(5, 3.5)$	Cl.
(1, 2)	1.398	4.272	C1
(1.5, 1)	1.152	4.301	C1
(3.5, 1.5)	1.152	2.500	C1
(4, 3)	2.050	1.118	C2
(3.5, 2.5)	1.352	1.803	C1
(6, 4)	4.267	1.118	C2
- Assignment unchanged $\rightarrow$ converged after 2 iterations.			

Final answer

- **Cluster 1** = $\{(1, 2), (1.5, 1), (3.5, 1.5), (3.5, 2.5)\}$, centroid $(2.375, 1.75)$
- **Cluster 2** = $\{(4, 3), (6, 4)\}$, centroid $(5, 3.5)$

Step by step: seeds, then each iteration



Problem 1.8 5+3

QUESTION AS PRINTED 2081 Bhadra · Q8 · [5+3]

Consider the data points $\{(2, 3), (3, 3), (6, 8), (8, 8), (7, 5)\}$ with initial centroid $(2, 3)$. Apply K-means algorithm to choose the next centroid and cluster these data points into $k = 2$. Discuss the problems that arise with K-means clustering algorithm and ways to solve it.

Only ONE centroid is given, and "choose the next centroid" is the first thing asked. Take the **farthest point** from the given one — the standard max-separation seeding rule — and say so.

Step 1: choose the second centroid

Distance from (2, 3)	(3, 3)	(6, 8)	(8, 8)	(7, 5)
Euclidean	1.000	6.403	7.810	5.385

(8, 8) is farthest $\rightarrow c_2 = (8, 8)$.

Iteration 1

Point	d to (2, 3)	d to (8, 8)	Cl.
(2, 3)	0	7.810	C1
(3, 3)	1.000	7.071	C1
(6, 8)	6.403	2.000	C2
(8, 8)	7.810	0	C2
(7, 5)	5.385	3.162	C2

- C1 $\rightarrow (2.5, 3)$ C2 $\rightarrow (\frac{6+8+7}{3}, \frac{8+8+5}{3}) = (7, 7)$

Iteration 2

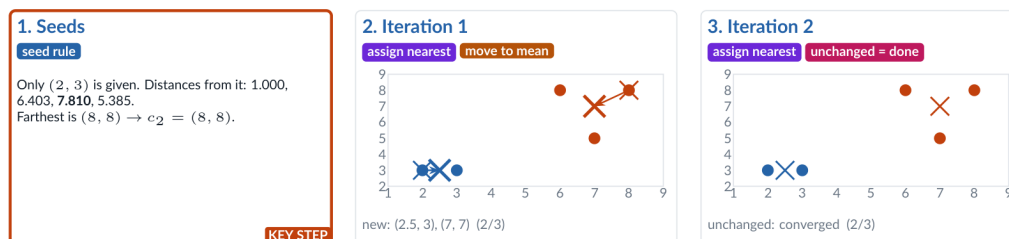
Point	d to (2.5, 3)	d to (7, 7)	Cl.
(2, 3)	0.500	6.403	C1
(3, 3)	0.500	5.657	C1
(6, 8)	6.103	1.414	C2
(8, 8)	7.433	1.414	C2
(7, 5)	4.924	2.000	C2

- **Unchanged $\rightarrow$ converged.**

Final answer

C1 = {(2, 3), (3, 3)} centroid (2.5, 3); C2 = {(6, 8), (8, 8), (7, 5)} centroid (7, 7). The 3-mark “problems and ways to solve” half is the limitations list in §5.2.

Step by step: choose the second seed, then iterate



Problem 1.9 4+8

QUESTION AS PRINTED 2073 Chaitra · Q7 · [4+8]

How clustering differ from classification? Given the one-dimensional points {5, 12, 18, 24, 30, 42, 48} with initial centroids {5, 12, 18}, create three clusters by K-Means algorithm and calculate SSE for this clustering result.

One dimension, so “distance” is just $|x - c|$. This is the only PYQ that asks for SSE, and it is the only one that runs past two iterations.

Iter.	Assignment (nearest centroid)	Clusters	New centroids
1	centroids 5, 12, 18	{5}, {12}, {18, 24, 30, 42, 48}	5, 12, $\frac{162}{5} = 32.4$
2	centroids 5, 12, 32.4	{5}, {12, 18}, {24, 30, 42, 48}	5, 15, $\frac{144}{4} = 36$
3	centroids 5, 15, 36	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40
4	centroids 5, 18, 40	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40 unchanged

- **Converged at iteration 4.** Note the cluster boundaries kept moving for three rounds — do not stop after two just because the earlier problems did.

SSE

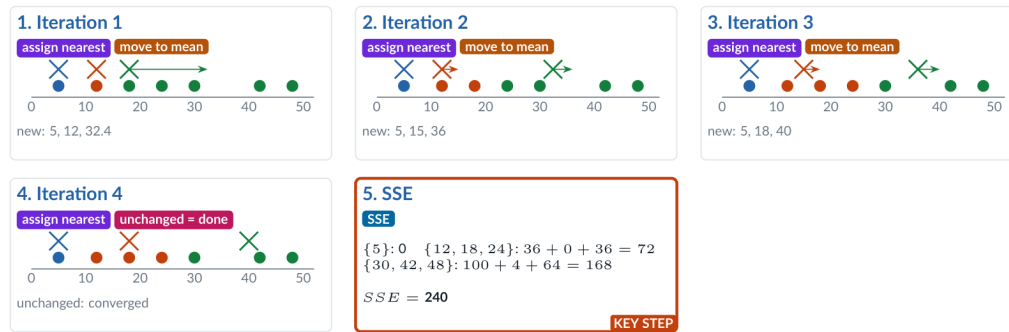
$$SSE = \sum_i \sum_{x \in C_i} (x - c_i)^2$$

Cluster	Centroid	Squared errors	Subtotal
{5}	5	$(5 - 5)^2 = 0$	0
{12, 18, 24}	18	$(12 - 18)^2 + (18 - 18)^2 + (24 - 18)^2 = 36 + 0 + 36$	72
{30, 42, 48}	40	$(30 - 40)^2 + (42 - 40)^2 + (48 - 40)^2 = 100 + 4 + 64$	168

$$SSE = 0 + 72 + 168 = 240$$

- **SSE is the objective K-means minimises**, so quoting it is how you show the clustering is the algorithm’s answer and not just a guess. A different seeding that gave a larger SSE would be a worse local optimum.

Step by step: four passes along the line (the boundaries move three times), then SSE



Problem 1.10 – the Manhattan one 8

QUESTION AS PRINTED 2079 Baishakh · Q8 · [8]

Define clustering. Given the matrix whose column represent different data points, perform a K-means clustering on this dataset using the **Manhattan** as the distance function. The center of the 3 clusters are initiated as $A(6.2, 3.2)$, $B(6.6, 3.7)$ and $C(6.5, 3.0)$. Provide the final cluster centers after 3 iteration.

x	5.9	4.6	6.2	4.7	5.5	5.0	4.9	6.7	5.1	6.0
y	3.2	2.9	2.8	3.2	4.2	3.0	3.1	3.1	3.8	3.0

Manhattan, not Euclidean: $d = |x_1 - x_2| + |y_1 - y_2|$, no squares and no square root. **Stop after 3 iterations** – the question says so, and it has *not* converged by then. Say that in the answer.

Iteration 1

centres $A(6.2, 3.2)$, $B(6.6, 3.7)$, $C(6.5, 3.0)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.30	1.20	0.80	A
(4.6, 2.9)	1.90	2.80	2.00	A
(6.2, 2.8)	0.40	1.30	0.50	A
(4.7, 3.2)	1.50	2.40	2.00	A
(5.5, 4.2)	1.70	1.60	2.20	B
(5.0, 3.0)	1.40	2.30	1.50	A
(4.9, 3.1)	1.40	2.30	1.70	A
(6.7, 3.1)	0.60	0.70	0.30	C
(5.1, 3.8)	1.70	1.60	2.20	B
(6.0, 3.0)	0.40	1.30	0.50	A

New centres: $A = (5.329, 3.029)$ (7 points), $B = (5.3, 4.0)$ (2), $C = (6.7, 3.1)$ (1).

Iteration 2

centres $A(5.329, 3.029)$, $B(5.3, 4.0)$, $C(6.7, 3.1)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.74	1.40	0.90	A
(4.6, 2.9)	0.86	1.80	2.30	A
(6.2, 2.8)	1.10	2.10	0.80	C
(4.7, 3.2)	0.80	1.40	2.10	A
(5.5, 4.2)	1.34	0.40	2.30	B
(5.0, 3.0)	0.36	1.30	1.80	A
(4.9, 3.1)	0.50	1.30	1.80	A
(6.7, 3.1)	1.44	2.30	0.00	C
(5.1, 3.8)	1.00	0.40	2.30	B
(6.0, 3.0)	0.70	1.70	0.80	A

New centres: $A = (5.183, 3.067)$ (6), $B = (5.3, 4.0)$ (2), $C = (6.45, 2.95)$ (2).

Iteration 3

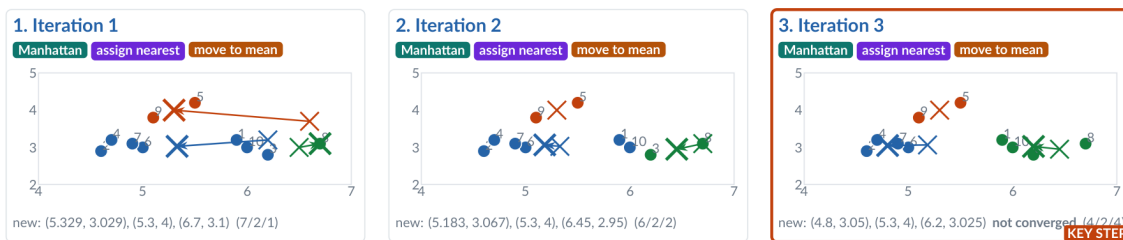
centres $A(5.183, 3.067)$, $B(5.3, 4.0)$, $C(6.45, 2.95)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.85	1.40	0.80	C
(4.6, 2.9)	0.75	1.80	1.90	A
(6.2, 2.8)	1.28	2.10	0.40	C
(4.7, 3.2)	0.62	1.40	2.00	A
(5.5, 4.2)	1.45	0.40	2.20	B
(5.0, 3.0)	0.25	1.30	1.50	A
(4.9, 3.1)	0.32	1.30	1.70	A
(6.7, 3.1)	1.55	2.30	0.40	C
(5.1, 3.8)	0.82	0.40	2.20	B
(6.0, 3.0)	0.88	1.70	0.50	C

Final answer after 3 iterations

- $A = (4.8, 3.05)$ from $\{(4.6, 2.9), (4.7, 3.2), (5.0, 3.0), (4.9, 3.1)\}$
- $B = (5.3, 4.0)$ from $\{(5.5, 4.2), (5.1, 3.8)\}$
- $C = (6.2, 3.025)$ from $\{(5.9, 3.2), (6.2, 2.8), (6.7, 3.1), (6.0, 3.0)\}$
- **The assignment was still changing at iteration 3**, so this is *not* a converged clustering. The question asked for 3 iterations, so stop there and **state that it has not converged** – that sentence is worth a mark.
- **Note how the three seeds were all bunched near (6.5, 3)**, which is why cluster C started with a single point and had to grow every round. A textbook illustration of **poor seeding**.

Step by step: Manhattan distance, three passes, still moving



M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
 2. Find the **smallest** entry → merge those 2 clusters. **Record the merge height** – that value is the dendrogram bar.
 3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
 4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
- **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q4 (b) · [2+7]

What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points.

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	0.00	0.24	0.22	0.37	0.34	0.23
p_2	0.24	0.00	0.15	0.20	0.14	0.25
p_3	0.22	0.15	0.00	0.15	0.28	0.11
p_4	0.37	0.20	0.15	0.00	0.29	0.22
p_5	0.34	0.14	0.28	0.29	0.00	0.39
p_6	0.23	0.25	0.11	0.22	0.39	0.00

Distances are given directly, so no coordinate work. The paper does not name a linkage — use **single link (MIN)** and say so. **Recompute and rewrite the whole matrix after every merge** — that redrawing is the method, and it is where the marks are.

Step 1

smallest entry in the matrix above is $d(p_3, p_6) = 0.11 \rightarrow$ merge at height **0.11**

Matrix after merge 1

5 clusters

	p_1	p_2	p_4	p_5
p_1	0.00	0.24	0.37	0.34
p_2	0.24	0.00	0.20	0.14
p_4	0.37	0.20	0.00	0.29
p_5	0.34	0.14	0.29	0.00
$\{p_3p_6\}$	0.22	0.15	0.15	0.28

How the new row was computed

MIN of the two old rows

- $d(\{p_3p_6\}, p_1) = \min(0.22, 0.23) = 0.22$
- $d(\{p_3p_6\}, p_2) = \min(0.15, 0.25) = 0.15$
- $d(\{p_3p_6\}, p_4) = \min(0.15, 0.22) = 0.15$
- $d(\{p_3p_6\}, p_5) = \min(0.28, 0.39) = 0.28$
- Every other entry is copied over unchanged.

Step 2

smallest is now $d(p_2, p_5) = 0.14 \rightarrow$ merge at height **0.14**

Matrix after merge 2

4 clusters

	p_1	p_4	$\{p_3p_6\}$
p_1	0.00	0.37	0.22
p_4	0.37	0.00	0.15
$\{p_3p_6\}$	0.22	0.15	0.00
$\{p_2p_5\}$	0.24	0.20	0.15

New row

- $d(\{p_2p_5\}, p_1) = \min(0.24, 0.34) = 0.24$
- $d(\{p_2p_5\}, p_4) = \min(0.20, 0.29) = 0.20$
- $d(\{p_2p_5\}, \{p_3p_6\}) = \min(0.15, 0.28) = 0.15$
- **A tie at 0.15**, between $(\{p_3p_6\}, p_4)$ and $(\{p_3p_6\}, \{p_2p_5\})$. **Either may be merged first** and the dendrogram heights come out the same; state the rule you use. Taken here in the order the matrix is read.

Step 3

take $d(\{p_3p_6\}, p_4) = 0.15 \rightarrow$ merge at height **0.15**

Matrix after merge 3

3 clusters

	p_1	$\{p_2p_5\}$
p_1	0.00	0.24
$\{p_2p_5\}$	0.24	0.00
$\{p_3p_4p_6\}$	0.22	0.15

New row

- $d(\{p_3p_4p_6\}, p_1) = \min(0.22, 0.37) = 0.22$
- $d(\{p_3p_4p_6\}, \{p_2p_5\}) = \min(0.15, 0.20) = 0.15$

Step 4

smallest is **0.15** $\rightarrow$ merge $\{p_3p_4p_6\}$ with $\{p_2p_5\}$ at height **0.15**

Matrix after merge 4

2 clusters

	p_1
p_1	0.00
$\{p_2p_3p_4p_5p_6\}$	0.22

New row

- $d(\{p_2p_3p_4p_5p_6\}, p_1) = \min(0.24, 0.22) = 0.22$, the smaller of the two surviving links to p_1 .

Step 5

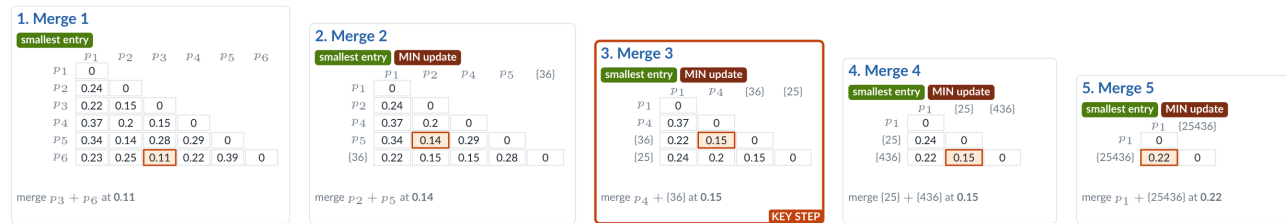
only one distance left $\rightarrow$ merge everything at height **0.22**

Final answer

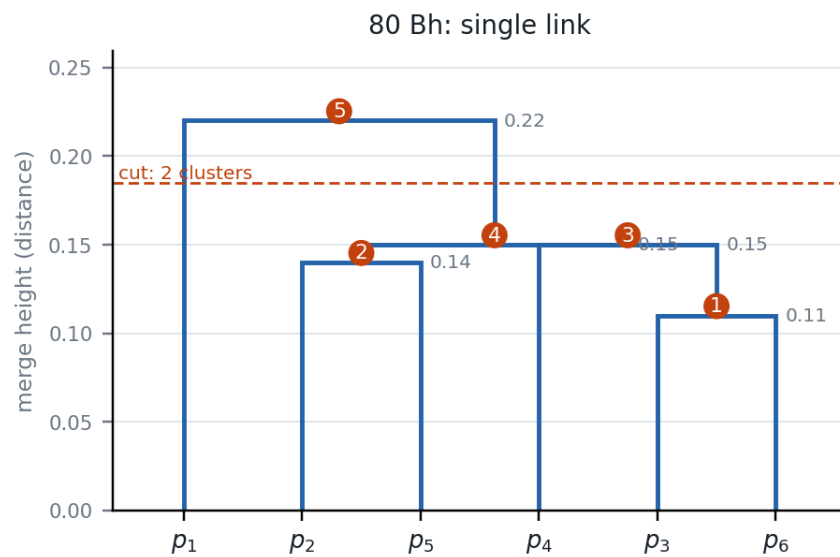
Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22**.

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

Step by step: the matrix before each merge, smallest entry lit (single link)



The dendrogram



Bars at the five merge heights, numbered in merge order (Steps 1 to 5 above); merges 3 and 4 share the height 0.15, the tie. The dashed line is the 2-cluster cut. Drawn by `dendfig.py`, which replays the merges and checks them against this page.

Problem 2.2 8

QUESTION AS PRINTED 2080 Baishakh · Q7 · [8]

Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance.

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Step 1: Euclidean distance matrix

$p_1(0.40, 0.53)$ $p_2(0.22, 0.38)$ $p_3(0.35, 0.32)$ $p_4(0.26, 0.19)$ $p_5(0.08, 0.41)$ $p_6(0.45, 0.30)$

	p_1	p_2	p_3	p_4	p_5
p_2	0.234				
p_3	0.216	0.143			
p_4	0.368	0.194	0.158		
p_5	0.342	0.143	0.285	0.284	
p_6	0.235	0.244	0.102	0.219	0.386

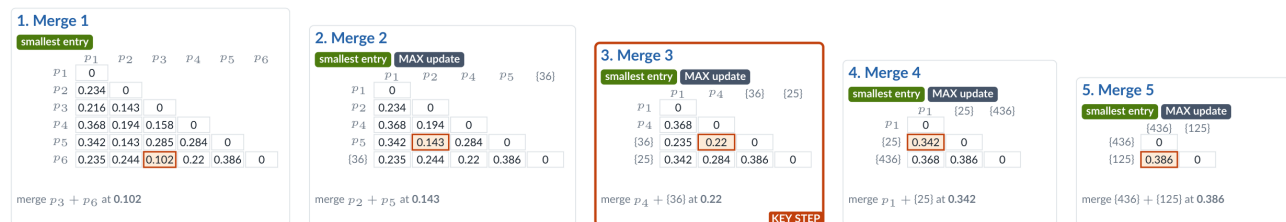
Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3p_6\}$ 0.386
3	$\{p_3p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2p_5\}$ 0.386
4	$p_1, \{p_2p_5\}$	0.342	to $\{p_3p_4p_6\}$ 0.386
5	the two remaining	0.386	—

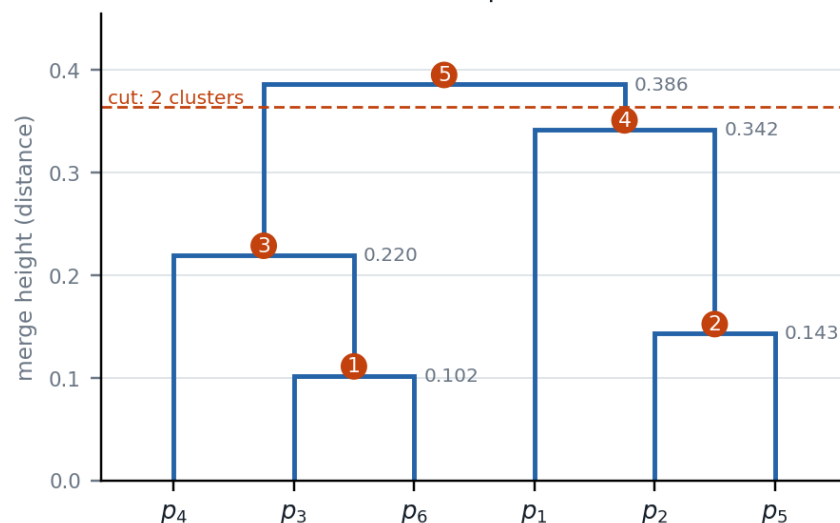
Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.

Step by step: the matrix before each merge (complete link: the merged row takes the MAX)



80 Ba: complete link



The dendrogram the question asks for, merges numbered as in the Step 2 table; the cut below 0.386 leaves the two groups. From `dendfig.py`.

Problem 2.3 4+4

QUESTION AS PRINTED 2076 Chaitra · Q10 · [4+4]

The table below is a distance matrix for six objects:

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.

The ideal question for showing you understand the difference — one matrix, both linkages.

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → **chaining**.

(b) Complete link (MAX)

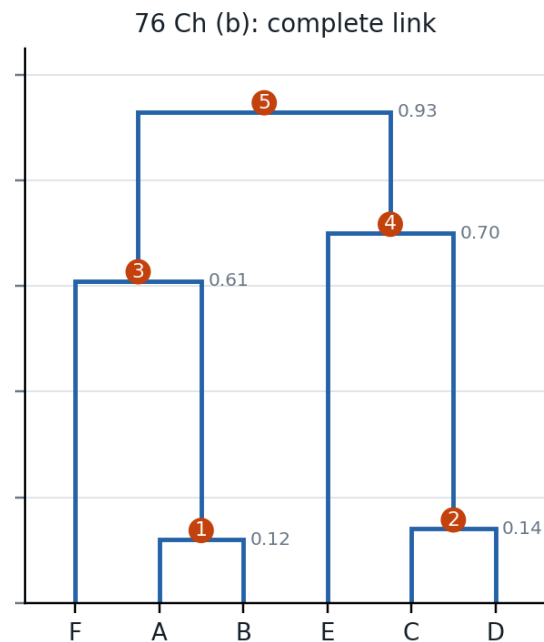
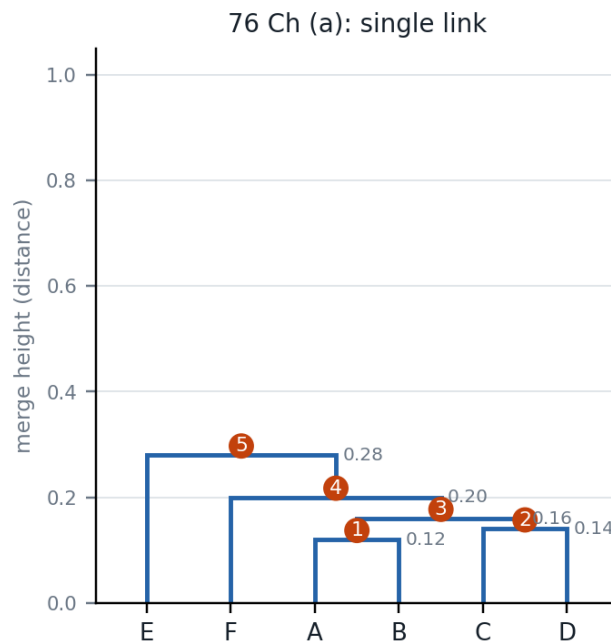
	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** – a balanced split.

The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A-B-C-D chain and leaves E as a straggler, while complete link produces two clean balanced groups.

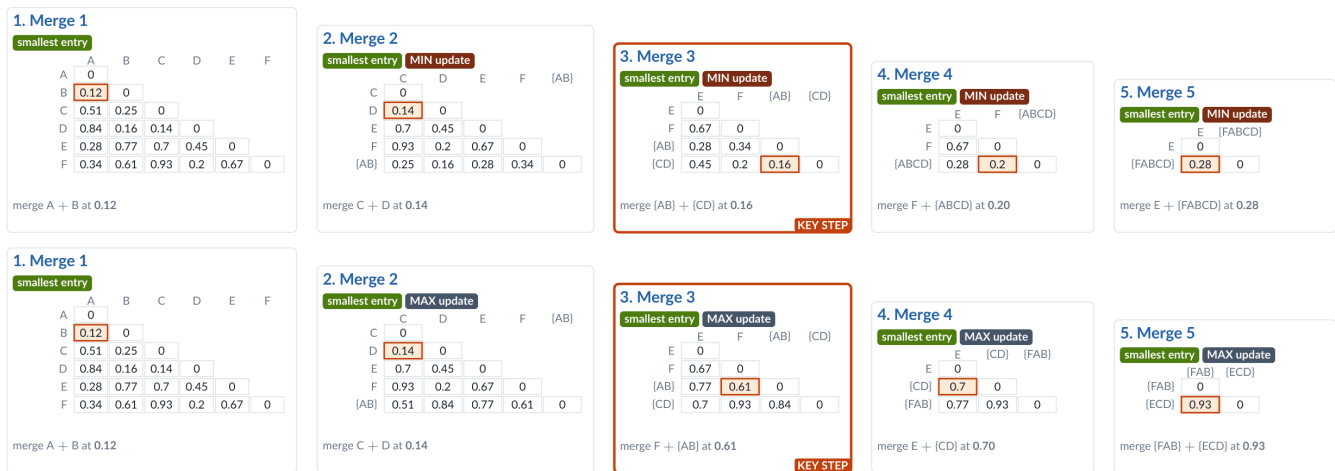
Single link chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

Both dendrograms, to one scale



(a) single link: everything joins below 0.3, the chaining in one look. (b) complete link: two compact groups that only meet at 0.93. Merges numbered as in the two tables. From `dendfig.py`.

Step by step: (a) single link, top row; (b) complete link, bottom row



M3. DBSCAN point classification

Method

- For each point, count how many points lie within ε (include the point itself).
- Count $\geq \text{MinPts}$ → **core**.
- Otherwise, within ε of a core point → **border**.
- Otherwise → **noise**.
- Join core points within ε of each other into clusters; attach the border points.

Problem 3.1 3+5

QUESTION AS PRINTED 2082 Bhadra · Q5 · [3+5]

What are core, border and noise points? Consider the following points (2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4), (1, 2), (4, 9). Use DBSCAN algorithm to find cluster with $\varepsilon = 2$ and $\text{minPts} = 2$.

The paper prints the radius as a plain ε ; it is ε . Label the points $A \dots H$ in the order given, so the working can name them.

Step 1: neighbourhood of every point

$\varepsilon = 2$, and **the point counts itself**, so $\text{MinPts} = 2$ means “at least one other point within 2”.

Point	Others within $\varepsilon = 2$	$ N $	Type	Why
A(2, 10)	— (nearest is H at 2.236)	1	noise	no neighbour
B(2, 5)	— (nearest is G at 3.162)	1	noise	no neighbour
C(8, 4)	E (1.414), F (2.000)	3	core	≥ 2
D(5, 8)	H (1.414)	2	core	≥ 2
E(7, 5)	C (1.414), F (1.414)	3	core	≥ 2
F(6, 4)	C (2.000), E (1.414)	3	core	≥ 2
G(1, 2)	— (nearest is B at 3.162)	1	noise	no neighbour
H(4, 9)	D (1.414)	2	core	≥ 2

- C–F is exactly 2.000**, which is **inside** the neighbourhood: the definition is $d \leq \varepsilon$, not $d < \varepsilon$. Getting this wrong splits the cluster.
- There are no border points here.** A, B and G are not core, and none of them is within ε of a core point either, so all three are noise. Say that explicitly – the question names all three categories.

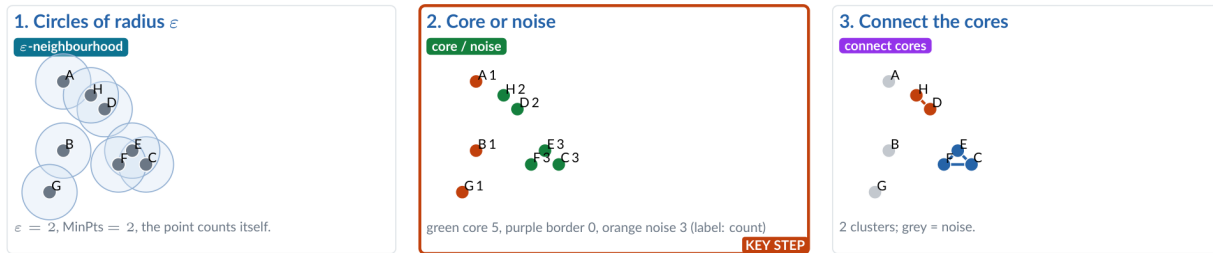
Step 2: join the core points

- C, E, F are mutually within ε → **Cluster 1** = {(8, 4), (7, 5), (6, 4)}.
- D and H are within ε of each other and of nothing else → **Cluster 2** = {(5, 8), (4, 9)}.
- No border point is left to attach.

Final answer

- **2 clusters:** $\{(8, 4), (7, 5), (6, 4)\}$ and $\{(5, 8), (4, 9)\}$.
- **Core:** all 5 of those points. **Border:** none. **Noise:** $(2, 10), (2, 5), (1, 2)$.
- **Check:** $5 + 0 + 3 = 8$ points. ✓

Step by step: neighbourhoods, point types, clusters



Practice – the same method with MinPts = 3

PROBLEM not a PYQ · written to show the MinPts effect verified/added

Classify each of the points $A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(8, 8)$, $E(8, 9)$, $F(25, 25)$ as a **core**, **border** or **noise** point under DBSCAN with $\varepsilon = 2$ and MinPts = 3 (Euclidean distance, the point itself counted in its own neighbourhood). Then give the clusters DBSCAN produces.

Kept because it isolates **what MinPts alone changes**: the same shaped data gives a different answer at MinPts 3 than at MinPts 2. verified/added

Step 1: neighbourhood counts

(including the point itself)

Point	Points within $\varepsilon = 2$	Count	Type
$A(1, 1)$	A, B (1.00), C (1.00)	3	core
$B(1, 2)$	B, A (1.00), C (1.41)	3	core
$C(2, 1)$	C, A (1.00), B (1.41)	3	core
$D(8, 8)$	D, E (1.00)	2	not core
$E(8, 9)$	E, D (1.00)	2	not core
$F(25, 25)$	F only	1	noise

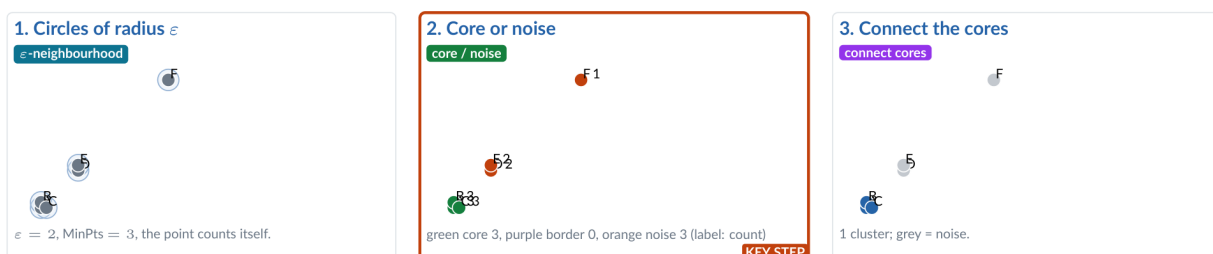
Step 2: form clusters

- A, B, C are all core and all within ε of each other → **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** → they are **not border points** either, they are **noise**.
- F is isolated → **noise**.

Final answer

- **1 cluster:** $\{A, B, C\}$. **Noise:** D, E, F .
- **The trap:** D and E look like a cluster to the eye, but with MinPts = 3 a pair can never form one. **A border point must be within ε of an actual core point** – being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Step by step: the same method with MinPts 3 (D and E are noise)



Ch 6 – Anomaly / Fraud Detection – Numerical Problems & Solutions

3 numerical marks, 0.2% of the archive, 2% of this chapter · This chapter is **almost entirely theory**. Across all 23 papers there is exactly **one** numerical 76 Ch, on the base rate fallacy. It is solved below, plus 2 practice problems covering the patterns a future numerical would most likely take.

Methods

- **M1** Base rate fallacy: detection rate, false alarm rate, Bayesian detection rate
- **M2** Distance-based outlier: the $DB(r, \pi)$ test

76 Ch
practice

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-6 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
round population	Invent a round population and count objects; no formula to misremember.	1.1 1; Practice A 1
apply both rates	Detection rate hits the anomalies, false alarm rate hits the much larger normal group.	1.1 2; Practice A 2
read the alarm bar	$P(I A)$ = true alarms / all alarms, which is what an analyst sees.	1.1 3 ; Practice A 3
base rate moved	Same detector, rarer anomalies: only the prior changed, and $P(I A)$ collapses.	1.1 3 ; Practice A 3
count within r	Count the objects within r , the point itself included.	Practice B 1
compare with π	Outlier when that count over $ D $ is at most π .	Practice B 2
k -NN cross-check	With $k = \lceil \pi D \rceil$, outlier iff the k -th neighbour is farther than r .	Practice B 3

M1. Base rate fallacy

Method

- Let I = anomalous, $\neg I$ = normal, A = alarm raised.
- **Detection rate** = $P(A|I)$ **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** (what an analyst actually cares about):

$$P(I|A) = \frac{P(A|I) P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

- **Fastest safe route**: invent a round population (10,000), build the confusion matrix, read every rate off it. No formula to misremember.

Problem 1.1 3

QUESTION AS PRINTED 2076 Chaitra · Q11 (b) · [3]

If the probability that a normal object is classified as an anomaly is 0.01 and the probability that an anomalous object is classified as anomalous is 0.99, then what is the false alarm rate and detection rate if 99% of the objects are normal?

Given

- $P(A|\neg I) = 0.01$ (normal object raises an alarm)
- $P(A|I) = 0.99$ (anomalous object raises an alarm)
- $P(\neg I) = 0.99$, so $P(I) = 0.01$

Step 1: the two rates asked for, read straight off the givens

- **False alarm rate** = $P(A|\neg I) = 0.01$ (1%)
- **Detection rate** = $P(A|I) = 0.99$ (99%)

Step 2: why the 99% prior was given – build the matrix on 10,000 objects

	Alarm	No alarm	Total
Anomalous (I)	$0.99 \times 100 = 99$ (TP)	1 (FN)	100
Normal ($\neg I$)	$0.01 \times 9900 = 99$ (FP)	9801 (TN)	9,900
Total	198	9,802	10,000

– Check: detection rate = $99/100 = 0.99 \checkmark$ false alarm rate = $99/9900 = 0.01 \checkmark$

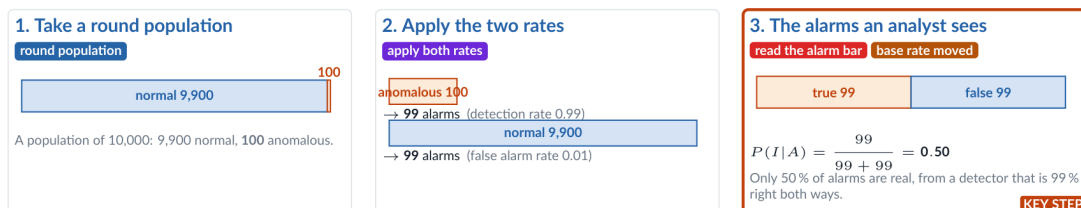
Step 3: Bayesian detection rate

$$P(I|A) = \frac{(0.99)(0.01)}{(0.99)(0.01) + (0.01)(0.99)} = \frac{0.0099}{0.0099 + 0.0099} = \frac{99}{198} = 0.50$$

Final answer

- False alarm rate = **0.01**, detection rate = **0.99**.
- **But the Bayesian detection rate is only 0.50** – half of every alarm raised is false, despite a detector that is 99% accurate in both directions.
- **This is the base rate fallacy.** The normal population (9,900) is so much larger than the anomalous one (100) that its 1% error rate manufactures exactly as many false alarms (99) as there are true detections (99).
- **Say this in the exam:** it is the whole point of the question and is worth more than the two arithmetic lines.

If the examiner wanted only the two conditional rates, they are given directly and the question is trivial – which is the clue that the **99% prior** must be used for something. Answer both readings: state the two rates, then compute $P(I|A)$ and name the fallacy. verified/added

Step by step: why a 99% detector leaves half its alarms false**Practice A – rarer anomalies**

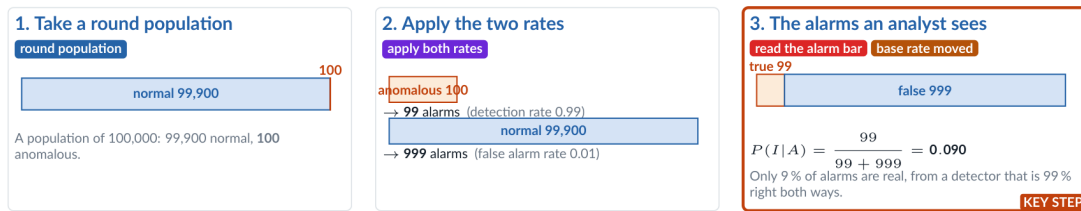
PROBLEM not a PYQ · same detector, rarer anomalies verified/added

The detector of Problem 1.1 is unchanged: a normal object is classified as an anomaly with probability 0.01, and an anomalous object is classified as anomalous with probability 0.99. Now only **1 object in 1,000** is anomalous instead of 1 in 100. Find the detection rate, the false alarm rate, and the probability that an object raising an alarm really is an anomaly.

Shows how fast the Bayesian rate collapses. verified/added

- On 100,000 objects: 100 anomalous, 99,900 normal.
- TP = $0.99 \times 100 = 99$ FP = $0.01 \times 99,900 = 999$
- $P(I|A) = \frac{99}{99 + 999} = \frac{99}{1098} = 0.090$
- **Only 9% of alarms are real.** Detection rate and false alarm rate are unchanged at 0.99 and 0.01 – **only the base rate moved.**
- **Conclusion (Axelsson):** to keep $P(I|A)$ usable when $P(I)$ is tiny you must drive the *false alarm rate* down, not push the detection rate up.

Step by step: the same detector, anomalies ten times rarer



M2. Distance-based outlier: the $DB(r, \pi)$ test

Method

- Object o is a $DB(r, \pi)$ -outlier if

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{|D|} \leq \pi$$

- In words: count how many objects (including o itself, per Han & Kamber) lie within distance r ; if that count as a fraction of $|D|$ is at most π , it is an outlier.
- Equivalent form: with $k = \lceil \pi |D| \rceil$, o is an outlier if $\text{dist}(o, o_k) > r$.

Practice B

PROBLEM not a PYQ · written to fill the gap **verified/added**

A dataset contains the points $P_1(1, 1)$, $P_2(1, 2)$, $P_3(2, 1)$, $P_4(2, 2)$ and $P_5(8, 8)$. Taking $r = 2$ and $\pi = 0.4$, and using Euclidean distance, determine which points are $DB(2, 0.4)$ -outliers.

No PYQ has set this yet, but it is the only computable form of the distance-based question that is asked in 6 papers, so it is worth one run-through.

verified/added

Step 1: distance matrix

	P_1	P_2	P_3	P_4	P_5
P_1	0	1	1	1.414	9.899
P_2	1	0	1.414	1	9.220
P_3	1	1.414	0	1	9.220
P_4	1.414	1	1	0	8.485
P_5	9.899	9.220	9.220	8.485	0

Step 2: count neighbours within $r = 2$

(threshold $\pi|D| = 0.4 \times 5 = 2$ objects)

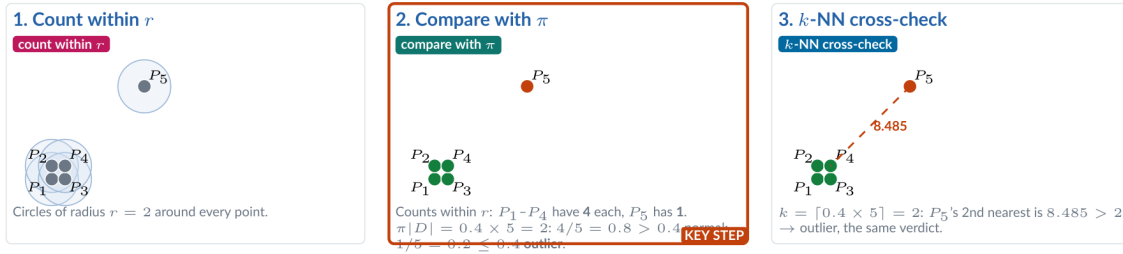
- P_1 : itself, $P_2, P_3, P_4 = 4$ $4/5 = 0.8 > 0.4 \rightarrow$ normal
- P_2, P_3, P_4 : 4 each by symmetry $\rightarrow$ normal
- P_5 : only itself = 1 $1/5 = 0.2 \leq 0.4 \rightarrow$ **outlier**

Final answer

P_5 is the only $DB(2, 0.4)$ -outlier.

- Cross-check with the k -NN form:** $k = \lceil 0.4 \times 5 \rceil = 2$. For P_5 , its 2nd nearest neighbour is at $8.485 > 2 \rightarrow$ outlier $\checkmark$. For P_1 , its 2nd nearest is at $1 \leq 2 \rightarrow$ normal $\checkmark$.
- Note the sensitivity to r :** raise r to 9 and P_5 stops being an outlier altogether. Quoting the chosen r and π is part of the answer.

Step by step: the DB(2, 0.4) test, and the k -NN form of the same test



Ch 7 – Advanced Applications – Numerical Problems & Solutions

8 numerical marks, 0.4% of the archive, 6% of this chapter · This chapter is almost entirely short notes. Across all 23 papers there is exactly one numerical 76 Ch, on PageRank. It is solved in full below, plus 1 practice problem on the variant most likely to be set next.

Both end with the solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**.

The moves the chapter-7 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
outlinks vs inlinks	Outlinks give the divisor C ; inlinks say which terms to add.	1.1 1
split a rank	A page's rank is shared equally among the pages it links to.	1.1 1,2
use the old values	Compute every page from the PREVIOUS iteration, then substitute together.	1.1 3
no inlinks	A page nothing points at stays at $1 - d$ for ever.	1.1 2
order before value	The ranking settles long before the numbers stop moving.	1.1 4
sum check	In this form the converged ranks add up to N .	1.1 5
normalised form	$\frac{1-d}{N}$ instead of $1 - d$: the same ordering, ranks summing to 1.	Practice 1

M1. PageRank

Method

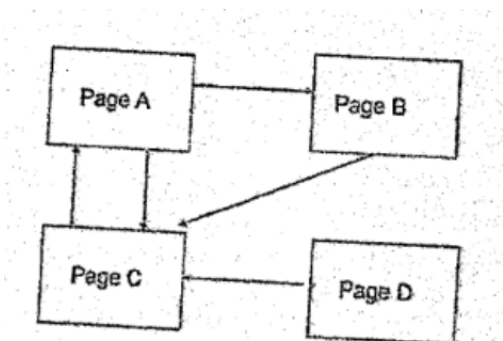
$$PR(A) = (1 - d) + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

- 1. Write down the link table first:** for every page, its **outlinks** (to get C) and its **inlinks** (to know which terms to sum). **Nearly every error comes from confusing the two.**
2. Initialise all ranks (here 0.25).
3. Each iteration, compute all pages using the **previous** iteration's values, then substitute in together.
4. Repeat for the required number of iterations, or until the values settle.
 - **A page's own rank is split equally among its outlinks** – that is the $/C(T_i)$.
 - A page with **no inlinks** sits permanently at $(1 - d) = 0.15$.

Problem 1.1 8

QUESTION AS PRINTED 2076 Chaitra · Q12 · [8]

Consider the following subset of pages and their links. Apply the PageRank algorithm using a damping factor of 0.85. A minimum of five iterations are required. Assume initial page rank of all pages is 0.25.



Step 1: read the graph

Page	Outlinks	C	Inlinks
A	B, C	2	C
B	C	1	A
C	A	1	A, B, D
D	C	1	none

Step 2: write the 4 equations

($d = 0.85$, $1 - d = 0.15$)

- $PR(A) = 0.15 + 0.85 PR(C)$
- $PR(B) = 0.15 + 0.85 \frac{PR(A)}{2}$
- $PR(C) = 0.15 + 0.85 \left(\frac{PR(A)}{2} + PR(B) + PR(D) \right)$
- $PR(D) = 0.15 + 0.85(0) = \mathbf{0.15}$ **constant, D has no inlinks**
- Note A's rank is divided by 2 because A links to *both* B and C.

Final answer after 5 iterations

- $PR(A) = 0.972$, $PR(B) = 0.519$, $PR(C) = 1.028$, $PR(D) = 0.150$
- **Ranking: C > A > B > D**

Step 3: iterate

Iter	PR(A)	PR(B)	PR(C)	PR(D)
0	0.2500	0.2500	0.2500	0.2500
1	0.3625	0.2563	0.6813	0.1500
2	0.7291	0.3041	0.6494	0.1500
3	0.7020	0.4599	0.8459	0.1500
4	0.8690	0.4484	0.9668	0.1500
5	0.9718	0.5193	1.0280	0.1500

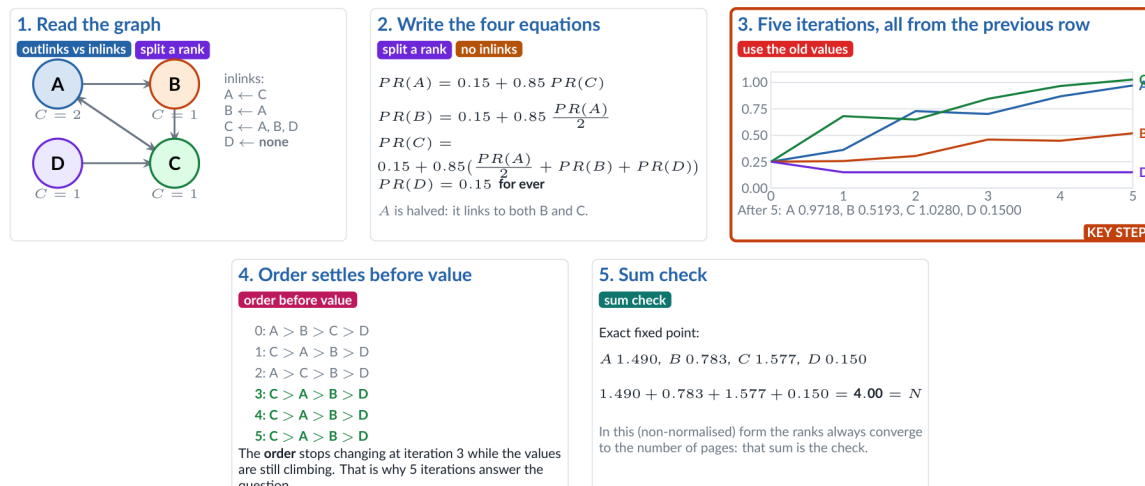
Row 0 is the given start value.

Sample working (iteration 1)

- $PR(A) = 0.15 + 0.85(0.25) = 0.3625$
- $PR(B) = 0.15 + 0.85(0.25/2) = 0.2563$
- $PR(C) = 0.15 + 0.85(0.125 + 0.25 + 0.25) = 0.6813$
- $PR(D) = 0.15$

Two comments worth writing, they show understanding. verified/added

1. The values are still climbing at iteration 5 but the **ranking has been stable since iteration 3**. PageRank converges on *order* much faster than on *value*, which is why 5 iterations is enough for the question.
2. Solving the equations exactly gives the fixed point $PR(A) = 1.490$, $PR(B) = 0.783$, $PR(C) = 1.577$, $PR(D) = 0.150$, and these sum to $4.00 = N$. **That sum is your check** – in this (non-normalised) form the ranks always converge to the number of pages.
3. **C wins** because it collects links from three pages while A collects only one, and **D is stuck at 0.15** because nothing points to it. Linking out never raises your own rank.

Step by step: the graph, the equations, five iterations, and the two checks**Practice – the normalised form**

PROBLEM not a PYQ · the same graph, normalised form verified/added

Repeat Problem 1.1 on the same four-page graph, but using the *normalised* PageRank formula

$$PR(A) = \frac{1-d}{N} + d \sum_i \frac{PR(T_i)}{C(T_i)}$$

with $d = 0.85$, $N = 4$ and an initial rank of 0.25 for every page. Compare the converged ranks and the ordering with those of Problem 1.1.

This form makes the ranks sum to 1 instead of N . Know both. verified/added

- With $N = 4$, $\frac{1-d}{N} = \frac{0.15}{4} = 0.0375$.
- Iteration 1 from the same 0.25 start:
 - $PR(A) = 0.0375 + 0.85(0.25) = 0.2500$
 - $PR(B) = 0.0375 + 0.85(0.125) = 0.1438$
 - $PR(C) = 0.0375 + 0.85(0.625) = 0.5688$
 - $PR(D) = 0.0375$
- Sum = 1.0001 ≈ 1 ✓ — **that is the check for this form.**
- **The ranking is identical** ($C > A > B > D$); only the scale differs.
- **In the exam:** state which form you are using in one line. Either is accepted; mixing them is not.

Step by step: the same graph in the normalised form (ranks sum to 1)

1. The normalised form

normalised form

$$\frac{1-d}{N} = \frac{0.15}{4} = 0.0375 \text{ replaces } 0.15$$

Iteration 1: A 0.2500, B 0.1438, C 0.5688, D 0.0375

Sum = 1.0001 ≈ 1 : the check for this form.

Same ordering $C > A > B > D$, only the scale differs.
State which form you are using.

KEY STEP

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.